

# GENERAL NOTES

1. LCG POSITIVE AFT OF FP. TCG POSITIVE PORT OF CL. VCG POSITIVE ABOVE BL.
- 2.
- 3.

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| 1   |       |             |
| NO. | TITLE | DRAWING NO. |

## REFERENCE PLANS

## SPECIAL NOTATIONS

PRELIMINARY DOCUMENT ISSUED TO SUPPORT DRAFT RFP. DOCUMENT IS UNDER DEVELOPMENT, AND NOT APPROVED.

## STINFO DESIGNATION

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|                                                                                     |                             |                                                   |                        |      |
|-------------------------------------------------------------------------------------|-----------------------------|---------------------------------------------------|------------------------|------|
|  | UNITED STATES COAST GUARD   |                                                   | WASHINGTON, D.C. 20593 |      |
|                                                                                     | OFFICE OF NAVAL ENGINEERING |                                                   |                        |      |
|                                                                                     | 69 FT                       | HOMELAND SECURITY CUTTER LIGHT<br>WEIGHT ESTIMATE |                        | HSCL |
|                                                                                     | SIZE                        | DATE                                              | USCG DRAWING NO.       | REV  |
|                                                                                     | A                           | 3/11/2026                                         | 69-HSCL-096-001        | 0    |

| REVISIONS |      |      |                             |           |         |
|-----------|------|------|-----------------------------|-----------|---------|
| REV       | ITEM | SWBS | DESCRIPTION                 | DATE      | INITIAL |
| -         | ALL  | ALL  | PRELIMINARY RELEASE FOR RFP | 3/13/2026 | SPS     |
|           |      |      |                             |           |         |
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|----------------------------------------------------------------------------------------------------------|-------------|---------------|-----|
| <p align="center"><u>United States Coast Guard</u></p> <p align="center"><u>HSC-L Design Support</u></p> |             |               |     |
| <p align="center">SHEET INDEX</p>                                                                        |             |               |     |
| SIZE                                                                                                     | PROJECT NO. | DWG. NO.      | REV |
| A                                                                                                        | N/A         | N/A           | -   |
| SCALE: NTS                                                                                               |             | SHEET 1 OF 57 |     |

## HOMELAND SECURITY CUTTER - LIGHT ICEBREAKER

## 1-DIGIT WEIGHT AND MOMENT ESTIMATE

WEIGHT ESTIMATE

11-Mar

IMPERIAL UNITS - POUNDS, LONG TONS, FEET, LB-FT

SHIP: HSC - L

REFERENCED FROM THE FORWARD PERPENDICULAR, CENTERLINE AND BASELINE

| SWBS<br>GROUP | DESCRIPTION                                 | CURRENT<br>WEIGHT | VCG | VERTICAL<br>MOMENT | LCG               | LONGITUDINAL<br>MOMENT | TCG               | TRANSVERSE<br>MOMENT  |
|---------------|---------------------------------------------|-------------------|-----|--------------------|-------------------|------------------------|-------------------|-----------------------|
|               | 1 STRUCTURE                                 | 100805.64         |     | 6.97               | 702143.05         | 31.74 A                | 3199243.62        | 0.01 P 961.91         |
|               | 2 PROPULSION SYSTEMS                        | 17068.00          |     | 4.17               | 71137.40          | 42.67 A                | 728367.51         | 0.00 C 0.00           |
|               | 3 ELECTRICAL SYSTEMS                        | 11650.00          |     | 5.96               | 69406.77          | 39.48 A                | 459915.86         | 0.03 P 385.60         |
|               | 4 COMMAND & SURVEILLANCE                    | 3970.93           |     | 16.75              | 66493.94          | 21.91 A                | 87002.28          | 0.00 C 0.00           |
|               | 5 AUXILIARY SYSTEMS                         | 27577.53          |     | 9.43               | 260136.40         | 42.54 A                | 1173149.49        | 0.08 P 2236.66        |
|               | 6 OUTFIT & FURNISHINGS                      | 21571.24          |     | 8.96               | 193194.34         | 30.77 A                | 663664.60         | 0.06 S -1377.27       |
|               | LIGHT SHIP, NO MARGINS                      | 182643.34         |     | 7.46               | 1362511.90        | 34.56 A                | 6311343.35        | 0.01 P 2206.90        |
| M             | MARGINS                                     | 29769.04          |     | 12.09              | 359934.95         | 34.56 A                | 1028685.85        | 0.01 S 359.70         |
|               | <b>LIGHT SHIP CONDITION</b>                 | <b>212412.38</b>  |     | <b>8.11</b>        | <b>1722446.84</b> | <b>34.56 A</b>         | <b>7340029.21</b> | <b>0.01 S 2566.60</b> |
| A             | DEPARTURE LOADS W/ ATON                     | 57482.69          |     | 6.20               | 356504.07         | 32.01 A                | 1839896.11        | 0.04 S -2075.64       |
|               | <b>DESIGN DEPARTURE LOADS W/ ATON, BOSL</b> | <b>269895.07</b>  |     | <b>7.70</b>        | <b>2078950.92</b> | <b>34.01 A</b>         | <b>9179925.32</b> | <b>0.00 P 490.96</b>  |

NOTES: ALL OF THE FOLLOWING QUANTITIES ARE FOR THE DELIVERY CONDITION USING EVEN KEEL HYDROSTATICS

|                                        |                   |                                         |                     |
|----------------------------------------|-------------------|-----------------------------------------|---------------------|
| 1. LENGTH BETWEEN PERPENDICULARS       | 66.53 FEET        | 15. MOMENT TO ALTER TRIM ONE INCH       | 402,865.55 FT - LBS |
| 2. BOTTOM KEEL BELOW BASELINE          | 0 FEET            | 16. LCB ON EVEN KEEL AT ABOVE WATERLINE | 33.29 FEET          |
| 3. C.G. ABOVE BOTTOM OF KEEL           | 7.70 FT ABL       | 17. LONGITUDINAL CENTER OF GRAVITY      | 34.01 FEET          |
| 4. C.G ABOVE BASELINE                  | 7.70 FEET         | 18. TRIMMING LEVER                      | 0.73 FEET AFT LCB   |
| 5. DRAFT AT LCF FOR ABOVE DISPLACEMENT | 6.14 FT ABL       | 19. TRIM                                | 0.49 FEET BY STERN  |
| 6. POUNDS PER INCH IMMERSION           | 6088.83 LB / IN   | 20. L.C.F.                              | 34.97 FEET          |
| 7. TRANSVERSE METACENTER (KEEL)        | 11.78 FEET        | 21. DIFF IN DRAFT, L.C.F. TO MIDSHIP    | 0.01 FEET INCREASE  |
| 8. GM, WITHOUT FREE SURFACE CORR.      | 4.08 FEET         | 22. DRAFTS (+ ABOVE BL):                |                     |
| 9. FREE SURFACE CORR.                  | 0.28 FEET         | FORWARD PERPENDICULAR                   | 5.87 FEET           |
| 10. INCLINING KG MARGIN                | 0.00 FEET         | MEAN                                    | 6.15 FEET           |
| 11. KGv                                | 7.98 FEET         | AFT PERPENDICULAR                       | 6.44 FEET           |
| 12. GM, WITH FREE SURFACE CORR.        | 4.36 FEET         | 23. DESIGNED DRAG                       | 0.00 FEET           |
| 13. MOMENT TO ALTER HEEL ONE DEGREE    | 19228.28 FT - LBS |                                         |                     |
| 14. LIST                               | 0.03 DEG          |                                         |                     |

## HOMELAND SECURITY CUTTER - LIGHT ICEBREAKER

## 1-DIGIT WEIGHT AND MOMENT ESTIMATE

WEIGHT ESTIMATE

11-Mar

IMPERIAL UNITS - POUNDS, LONG TONS, FEET, LB-FT

SHIP: HSC - L

REFERENCED FROM THE FORWARD PERPENDICULAR, CENTERLINE AND BASELINE

| SWBS<br>GROUP | DESCRIPTION                                  | CURRENT<br>WEIGHT | VCG | VERTICAL<br>MOMENT | LCG               | LONGITUDINAL<br>MOMENT | TCG               | TRANSVERSE<br>MOMENT  |
|---------------|----------------------------------------------|-------------------|-----|--------------------|-------------------|------------------------|-------------------|-----------------------|
|               | 1 STRUCTURE                                  | 100805.64         |     | 6.97               | 702143.05         | 31.74 A                | 3199243.62        | 0.01 P 961.91         |
|               | 2 PROPULSION SYSTEMS                         | 17068.00          |     | 4.17               | 71137.40          | 42.67 A                | 728367.51         | 0.00 C 0.00           |
|               | 3 ELECTRICAL SYSTEMS                         | 11650.00          |     | 5.96               | 69406.77          | 39.48 A                | 459915.86         | 0.03 P 385.60         |
|               | 4 COMMAND & SURVEILLANCE                     | 3970.93           |     | 16.75              | 66493.94          | 21.91 A                | 87002.28          | 0.00 C 0.00           |
|               | 5 AUXILIARY SYSTEMS                          | 27577.53          |     | 9.43               | 260136.40         | 42.54 A                | 1173149.49        | 0.08 P 2236.66        |
|               | 6 OUTFIT & FURNISHINGS                       | 21571.24          |     | 8.96               | 193194.34         | 30.77 A                | 663664.60         | 0.06 S -1377.27       |
|               | LIGHT SHIP, NO MARGINS                       | 182643.34         |     | 7.46               | 1362511.90        | 34.56 A                | 6311343.35        | 0.01 P 2206.90        |
| M             | MARGINS                                      | 29769.04          |     | 12.09              | 359934.95         | 34.56 A                | 1028685.85        | 0.01 S 359.70         |
|               | <b>LIGHT SHIP CONDITION</b>                  | <b>212412.38</b>  |     | <b>8.11</b>        | <b>1722446.84</b> | <b>34.56 A</b>         | <b>7340029.21</b> | <b>0.01 S 2566.60</b> |
| B             | DEPARTURE LOADS W/O ATON                     | 33222.69          |     | 3.77               | 125333.27         | 29.89 A                | 993121.81         | 0.03 S -949.24        |
|               | <b>DESIGN DEPARTURE LOADS W/O ATON, BOSL</b> | <b>245635.07</b>  |     | <b>7.52</b>        | <b>1847780.12</b> | <b>33.92 A</b>         | <b>8333151.02</b> | <b>0.01 P 1617.36</b> |
|               |                                              | 109.658516        |     |                    |                   |                        |                   |                       |

NOTES: ALL OF THE FOLLOWING QUANTITIES ARE FOR THE DELIVERY CONDITION USING EVEN KEEL HYDROSTATICS

|                                        |                    |                                         |                      |
|----------------------------------------|--------------------|-----------------------------------------|----------------------|
| 1. LENGTH BETWEEN PERPENDICULARS       | 65.68 FEET         | 15. MOMENT TO ALTER TRIM ONE INCH       | 378,277.16 FT - TONS |
| 2. BOTTOM KEEL BELOW BASELINE          | 0 FEET             | 16. LCB ON EVEN KEEL AT ABOVE WATERLINE | 33.13 FEET           |
| 3. C.G. ABOVE BOTTOM OF KEEL           | 7.52 FT ABL        | 17. LONGITUDINAL CENTER OF GRAVITY      | 33.92 FEET           |
| 4. C.G ABOVE BASELINE                  | 7.52 FEET          | 18. TRIMMING LEVER                      | 0.80 FEET AFT LCB    |
| 5. DRAFT AT LCF FOR ABOVE DISPLACEMENT | 5.80 FT ABL        | 19. TRIM                                | 0.52 FEET BY STERN   |
| 6. TONS PER INCH IMMERSION             | 5940.56 LT / IN    | 20. L.C.F.                              | 34.87 FEET           |
| 7. TRANSVERSE METACENTER (KEEL)        | 12.00 FEET         | 21. DIFF IN DRAFT, L.C.F. TO MIDSHIP    | 0.02 FEET INCREASE   |
| 8. GM, WITHOUT FREE SURFACE CORR.      | 4.48 FEET          | 22. DRAFTS (+ ABOVE BL):                |                      |
| 9. FREE SURFACE CORR.                  | 0.31 FEET          | FORWARD PERPENDICULAR                   | 5.52 FEET            |
| 10. INCLINING KG MARGIN                | 0.00 FEET          | MEAN                                    | 5.82 FEET            |
| 11. KGv                                | 7.83 FEET          | AFT PERPENDICULAR                       | 6.12 FEET            |
| 12. GM, WITH FREE SURFACE CORR.        | 4.79 FEET          | 23. DESIGNED DRAG                       | 0.00 FEET            |
| 13. MOMENT TO ALTER HEEL ONE DEGREE    | 19203.44 FT - TONS |                                         |                      |
| 14. LIST                               | 0.08 DEG           |                                         |                      |

## HOMELAND SECURITY CUTTER - LIGHT ICEBREAKER

## 1-DIGIT WEIGHT AND MOMENT ESTIMATE

WEIGHT ESTIMATE

11-Mar

IMPERIAL UNITS - POUNDS, LONG TONS, FEET, LB-FT

SHIP: HSC - L

REFERENCED FROM THE FORWARD PERPENDICULAR, CENTERLINE AND BASELINE

| SWBS<br>GROUP | DESCRIPTION                            | CURRENT<br>WEIGHT | VCG | VERTICAL<br>MOMENT | LCG               | LONGITUDINAL<br>MOMENT | TCG               | TRANSVERSE<br>MOMENT  |
|---------------|----------------------------------------|-------------------|-----|--------------------|-------------------|------------------------|-------------------|-----------------------|
|               | 1 STRUCTURE                            | 100805.64         |     | 6.97               | 702143.05         | 31.74 A                | 3199243.62        | 0.01 P 961.91         |
|               | 2 PROPULSION SYSTEMS                   | 17068.00          |     | 4.17               | 71137.40          | 42.67 A                | 728367.51         | 0.00 C 0.00           |
|               | 3 ELECTRICAL SYSTEMS                   | 11650.00          |     | 5.96               | 69406.77          | 39.48 A                | 459915.86         | 0.03 P 385.60         |
|               | 4 COMMAND & SURVEILLANCE               | 3970.93           |     | 16.75              | 66493.94          | 21.91 A                | 87002.28          | 0.00 C 0.00           |
|               | 5 AUXILIARY SYSTEMS                    | 27577.53          |     | 9.43               | 260136.40         | 42.54 A                | 1173149.49        | 0.08 P 2236.66        |
|               | 6 OUTFIT & FURNISHINGS                 | 21571.24          |     | 8.96               | 193194.34         | 30.77 A                | 663664.60         | 0.06 S -1377.27       |
|               | LIGHT SHIP, NO MARGINS                 | 182643.34         |     | 7.46               | 1362511.90        | 34.56 A                | 6311343.35        | 0.01 P 2206.90        |
| M             | MARGINS                                | 29769.04          |     | 12.09              | 359934.95         | 34.56 A                | 1028685.85        | 0.01 S 359.70         |
|               | <b>LIGHT SHIP CONDITION</b>            | <b>212412.38</b>  |     | <b>8.11</b>        | <b>1722446.84</b> | <b>34.56 A</b>         | <b>7340029.21</b> | <b>0.01 S 2566.60</b> |
| C             | HALF LOADS W/ ATON                     | 37405.69          |     | 6.43               | 240610.61         | 35.39 A                | 1323729.24        | 0.03 S -1073.11       |
|               | <b>DESIGN HALF LOADS W/ ATON, BOSL</b> | <b>249818.07</b>  |     | <b>7.86</b>        | <b>1963057.46</b> | <b>34.68 A</b>         | <b>8663758.45</b> | <b>0.01 P 1493.49</b> |

NOTES: ALL OF THE FOLLOWING QUANTITIES ARE FOR THE DELIVERY CONDITION USING EVEN KEEL HYDROSTATICS

|                                        |                     |                                         |                      |
|----------------------------------------|---------------------|-----------------------------------------|----------------------|
| 1. LENGTH BETWEEN PERPENDICULARS       | 66.04 FEET          | 15. MOMENT TO ALTER TRIM ONE INCH       | 382,596.38 FT - TONS |
| 2. BOTTOM KEEL BELOW BASELINE          | 0 FEET              | 16. LCB ON EVEN KEEL AT ABOVE WATERLINE | 33.15 FEET           |
| 3. C.G. ABOVE BOTTOM OF KEEL           | 7.86 FT ABL         | 17. LONGITUDINAL CENTER OF GRAVITY      | 34.68 FEET           |
| 4. C.G ABOVE BASELINE                  | 7.86 FEET           | 18. TRIMMING LEVER                      | 1.53 FEET AFT LCB    |
| 5. DRAFT AT LCF FOR ABOVE DISPLACEMENT | 5.86 FT ABL         | 19. TRIM                                | 1.00 FEET BY STERN   |
| 6. TONS PER INCH IMMERSION             | 5967.23 LT / IN     | 20. L.C.F.                              | 34.88 FEET           |
| 7. TRANSVERSE METACENTER (KEEL)        | 11.96 FEET          | 21. DIFF IN DRAFT, L.C.F. TO MIDSHIP    | 0.03 FEET INCREASE   |
| 8. GM, WITHOUT FREE SURFACE CORR.      | 4.10 FEET           | 22. DRAFTS (+ ABOVE BL):                |                      |
| 9. FREE SURFACE CORR.                  | 0.30 FEET           | FORWARD PERPENDICULAR                   | 5.32 FEET            |
| 10. INCLINING KG MARGIN                | 0.00 FEET           | MEAN                                    | 5.89 FEET            |
| 11. KGv                                | 8.16 FEET           | AFT PERPENDICULAR                       | 6.47 FEET            |
| 12. GM, WITH FREE SURFACE CORR.        | 4.41 FEET           | 23. DESIGNED DRAG                       | 0.00 FEET            |
| 13. MOMENT TO ALTER HEEL ONE DEGREE    | 17,889.89 FT - TONS |                                         |                      |
| 14. LIST                               | 0.08 DEG            |                                         |                      |

## HOMELAND SECURITY CUTTER - LIGHT ICEBREAKER

## 1-DIGIT WEIGHT AND MOMENT ESTIMATE

## WEIGHT ESTIMATE

11-Mar

IMPERIAL UNITS - POUNDS, LONG TONS, FEET, LB-FT

SHIP: HSC - L

REFERENCED FROM THE FORWARD PERPENDICULAR, CENTERLINE AND BASELINE

| SWBS<br>GROUP | DESCRIPTION                             | CURRENT<br>WEIGHT | VCG | VERTICAL<br>MOMENT | LCG               | LONGITUDINAL<br>MOMENT | TCG               | TRANSVERSE<br>MOMENT  |
|---------------|-----------------------------------------|-------------------|-----|--------------------|-------------------|------------------------|-------------------|-----------------------|
|               | 1 STRUCTURE                             | 100805.64         |     | 6.97               | 702143.05         | 31.74 A                | 3199243.62        | 0.01 P 961.91         |
|               | 2 PROPULSION SYSTEMS                    | 17068.00          |     | 4.17               | 71137.40          | 42.67 A                | 728367.51         | 0.00 C 0.00           |
|               | 3 ELECTRICAL SYSTEMS                    | 11650.00          |     | 5.96               | 69406.77          | 39.48 A                | 459915.86         | 0.03 P 385.60         |
|               | 4 COMMAND & SURVEILLANCE                | 3970.93           |     | 16.75              | 66493.94          | 21.91 A                | 87002.28          | 0.00 C 0.00           |
|               | 5 AUXILIARY SYSTEMS                     | 27577.53          |     | 9.43               | 260136.40         | 42.54 A                | 1173149.49        | 0.08 P 2236.66        |
|               | 6 OUTFIT & FURNISHINGS                  | 21571.24          |     | 8.96               | 193194.34         | 30.77 A                | 663664.60         | 0.06 S -1377.27       |
|               | LIGHT SHIP, NO MARGINS                  | 182643.34         |     | 7.46               | 1362511.90        | 34.56 A                | 6311343.35        | 0.01 P 2206.90        |
| M             | MARGINS                                 | 29769.04          |     | 12.09              | 359934.95         | 34.56 A                | 1028685.85        | 0.01 S 359.70         |
|               | <b>LIGHT SHIP CONDITION</b>             | <b>212412.38</b>  |     | <b>8.11</b>        | <b>1722446.84</b> | <b>34.56 A</b>         | <b>7340029.21</b> | <b>0.01 S 2566.60</b> |
| D             | HALF LOADS W/O ATON                     | 21065.69          |     | 3.05               | 64272.61          | 27.21 A                | 573109.24         | 0.05 S -1073.11       |
|               | <b>DESIGN HALF LOADS W/O ATON, BOSL</b> | <b>233478.07</b>  |     | <b>7.65</b>        | <b>1786719.46</b> | <b>33.89 A</b>         | <b>7913138.45</b> | <b>0.01 P 1493.49</b> |

NOTES: ALL OF THE FOLLOWING QUANTITIES ARE FOR THE DELIVERY CONDITION USING EVEN KEEL HYDROSTATICS

|                                        |                     |                                         |                      |
|----------------------------------------|---------------------|-----------------------------------------|----------------------|
| 1. LENGTH BETWEEN PERPENDICULARS       | 66.04 FEET          | 15. MOMENT TO ALTER TRIM ONE INCH       | 365,879.70 FT - TONS |
| 2. BOTTOM KEEL BELOW BASELINE          | 0 FEET              | 16. LCB ON EVEN KEEL AT ABOVE WATERLINE | 33.04 FEET           |
| 3. C.G. ABOVE BOTTOM OF KEEL           | 7.65 FT ABL         | 17. LONGITUDINAL CENTER OF GRAVITY      | 33.89 FEET           |
| 4. C.G ABOVE BASELINE                  | 7.65 FEET           | 18. TRIMMING LEVER                      | 0.85 FEET AFT LCB    |
| 5. DRAFT AT LCF FOR ABOVE DISPLACEMENT | 5.63 FT ABL         | 19. TRIM                                | 0.54 FEET BY STERN   |
| 6. TONS PER INCH IMMERSION             | 5862.71 LT / IN     | 20. L.C.F.                              | 34.82 FEET           |
| 7. TRANSVERSE METACENTER (KEEL)        | 12.13 FEET          | 21. DIFF IN DRAFT, L.C.F. TO MIDSHIP    | 0.02 FEET INCREASE   |
| 8. GM, WITHOUT FREE SURFACE CORR.      | 4.48 FEET           | 22. DRAFTS (+ ABOVE BL):                |                      |
| 9. FREE SURFACE CORR.                  | 0.32 FEET           | FORWARD PERPENDICULAR                   | 5.34 FEET            |
| 10. INCLINING KG MARGIN                | 0.00 FEET           | MEAN                                    | 5.65 FEET            |
| 11. KGv                                | 7.98 FEET           | AFT PERPENDICULAR                       | 5.96 FEET            |
| 12. GM, WITH FREE SURFACE CORR.        | 4.80 FEET           | 23. DESIGNED DRAG                       | 0.00 FEET            |
| 13. MOMENT TO ALTER HEEL ONE DEGREE    | 18,246.97 FT - TONS |                                         |                      |
| 14. LIST                               | 0.08 DEG            |                                         |                      |

## HOMELAND SECURITY CUTTER - LIGHT ICEBREAKER

## 1-DIGIT WEIGHT AND MOMENT ESTIMATE

WEIGHT ESTIMATE

11-Mar

IMPERIAL UNITS - POUNDS, LONG TONS, FEET, LB-FT

SHIP: HSC - L

REFERENCED FROM THE FORWARD PERPENDICULAR, CENTERLINE AND BASELINE

| SWBS<br>GROUP | DESCRIPTION                              | CURRENT<br>WEIGHT | VCG | VERTICAL<br>MOMENT | LCG               | LONGITUDINAL<br>MOMENT | TCG               | TRANSVERSE<br>MOMENT  |
|---------------|------------------------------------------|-------------------|-----|--------------------|-------------------|------------------------|-------------------|-----------------------|
|               | 1 STRUCTURE                              | 100805.64         |     | 6.97               | 702143.05         | 31.74 A                | 3199243.62        | 0.01 P 961.91         |
|               | 2 PROPULSION SYSTEMS                     | 17068.00          |     | 4.17               | 71137.40          | 42.67 A                | 728367.51         | 0.00 C 0.00           |
|               | 3 ELECTRICAL SYSTEMS                     | 11650.00          |     | 5.96               | 69406.77          | 39.48 A                | 459915.86         | 0.03 P 385.60         |
|               | 4 COMMAND & SURVEILLANCE                 | 3970.93           |     | 16.75              | 66493.94          | 21.91 A                | 87002.28          | 0.00 C 0.00           |
|               | 5 AUXILIARY SYSTEMS                      | 27577.53          |     | 9.43               | 260136.40         | 42.54 A                | 1173149.49        | 0.08 P 2236.66        |
|               | 6 OUTFIT & FURNISHINGS                   | 21571.24          |     | 8.96               | 193194.34         | 30.77 A                | 663664.60         | 0.06 S -1377.27       |
|               | LIGHT SHIP, NO MARGINS                   | 182643.34         |     | 7.46               | 1362511.90        | 34.56 A                | 6311343.35        | 0.01 P 2206.90        |
| M             | MARGINS                                  | 29769.04          |     | 12.09              | 359934.95         | 34.56 A                | 1028685.85        | 0.01 S 359.70         |
|               | <b>LIGHT SHIP CONDITION</b>              | <b>212412.38</b>  |     | <b>8.11</b>        | <b>1722446.84</b> | <b>34.56 A</b>         | <b>7340029.21</b> | <b>0.01 S 2566.60</b> |
| E             | MIN OP LOADS W/ ATON                     | 31646.69          |     | 6.88               | 217710.74         | 34.17 A                | 1081257.44        | 0.03 S -1100.89       |
|               | <b>DESIGN MIN OP LOADS W/ ATON, BOSL</b> | <b>244059.07</b>  |     | <b>7.95</b>        | <b>1940157.59</b> | <b>34.51 A</b>         | <b>8421286.65</b> | <b>0.01 P 1465.71</b> |

NOTES: ALL OF THE FOLLOWING QUANTITIES ARE FOR THE DELIVERY CONDITION USING EVEN KEEL HYDROSTATICS

|                                        |                     |                                         |                      |
|----------------------------------------|---------------------|-----------------------------------------|----------------------|
| 1. LENGTH BETWEEN PERPENDICULARS       | 65.81 FEET          | 15. MOMENT TO ALTER TRIM ONE INCH       | 376,648.71 FT - TONS |
| 2. BOTTOM KEEL BELOW BASELINE          | 0 FEET              | 16. LCB ON EVEN KEEL AT ABOVE WATERLINE | 33.12 FEET           |
| 3. C.G. ABOVE BOTTOM OF KEEL           | 7.95 FT ABL         | 17. LONGITUDINAL CENTER OF GRAVITY      | 34.51 FEET           |
| 4. C.G ABOVE BASELINE                  | 7.95 FEET           | 18. TRIMMING LEVER                      | 1.39 FEET AFT LCB    |
| 5. DRAFT AT LCF FOR ABOVE DISPLACEMENT | 5.78 FT ABL         | 19. TRIM                                | 0.90 FEET BY STERN   |
| 6. TONS PER INCH IMMERSION             | 5930.44 LT / IN     | 20. L.C.F.                              | 34.86 FEET           |
| 7. TRANSVERSE METACENTER (KEEL)        | 12.02 FEET          | 21. DIFF IN DRAFT, L.C.F. TO MIDSHIP    | 0.03 FEET INCREASE   |
| 8. GM, WITHOUT FREE SURFACE CORR.      | 4.07 FEET           | 22. DRAFTS (+ ABOVE BL):                |                      |
| 9. FREE SURFACE CORR.                  | 0.31 FEET           | FORWARD PERPENDICULAR                   | 5.30 FEET            |
| 10. INCLINING KG MARGIN                | 0.00 FEET           | MEAN                                    | 5.81 FEET            |
| 11. KGv                                | 8.26 FEET           | AFT PERPENDICULAR                       | 6.33 FEET            |
| 12. GM, WITH FREE SURFACE CORR.        | 4.38 FEET           | 23. DESIGNED DRAG                       | 0.00 FEET            |
| 13. MOMENT TO ALTER HEEL ONE DEGREE    | 17,328.11 FT - TONS |                                         |                      |
| 14. LIST                               | 0.08 DEG            |                                         |                      |

## HOMELAND SECURITY CUTTER - LIGHT ICEBREAKER

## 1-DIGIT WEIGHT AND MOMENT ESTIMATE

WEIGHT ESTIMATE

11-Mar

IMPERIAL UNITS - POUNDS, LONG TONS, FEET, LB-FT

SHIP: HSC - L

REFERENCED FROM THE FORWARD PERPENDICULAR, CENTERLINE AND BASELINE

| SWBS<br>GROUP | DESCRIPTION                               | CURRENT<br>WEIGHT | VCG | VERTICAL<br>MOMENT | LCG               | LONGITUDINAL<br>MOMENT | TCG               | TRANSVERSE<br>MOMENT  |
|---------------|-------------------------------------------|-------------------|-----|--------------------|-------------------|------------------------|-------------------|-----------------------|
|               | 1 STRUCTURE                               | 100805.64         |     | 6.97               | 702143.05         | 31.74 A                | 3199243.62        | 0.01 P 961.91         |
|               | 2 PROPULSION SYSTEMS                      | 17068.00          |     | 4.17               | 71137.40          | 42.67 A                | 728367.51         | 0.00 C 0.00           |
|               | 3 ELECTRICAL SYSTEMS                      | 11650.00          |     | 5.96               | 69406.77          | 39.48 A                | 459915.86         | 0.03 P 385.60         |
|               | 4 COMMAND & SURVEILLANCE                  | 3970.93           |     | 16.75              | 66493.94          | 21.91 A                | 87002.28          | 0.00 C 0.00           |
|               | 5 AUXILIARY SYSTEMS                       | 27577.53          |     | 9.43               | 260136.40         | 42.54 A                | 1173149.49        | 0.08 P 2236.66        |
|               | 6 OUTFIT & FURNISHINGS                    | 21571.24          |     | 8.96               | 193194.34         | 30.77 A                | 663664.60         | 0.06 S -1377.27       |
|               | LIGHT SHIP, NO MARGINS                    | 182643.34         |     | 7.46               | 1362511.90        | 34.56 A                | 6311343.35        | 0.01 P 2206.90        |
| M             | MARGINS                                   | 29769.04          |     | 12.09              | 359934.95         | 34.56 A                | 1028685.85        | 0.01 S 359.70         |
|               | <b>LIGHT SHIP CONDITION</b>               | <b>212412.38</b>  |     | <b>8.11</b>        | <b>1722446.84</b> | <b>34.56 A</b>         | <b>7340029.21</b> | <b>0.01 S 2566.60</b> |
| F             | MIN OP LOADS W/O ATON                     | 15306.69          |     | 2.70               | 41372.74          | 21.60 A                | 330637.44         | 0.07 S -1100.89       |
|               | <b>DESIGN MIN OP LOADS W/O ATON, BOSL</b> | <b>227719.07</b>  |     | <b>7.75</b>        | <b>1763819.59</b> | <b>33.68 A</b>         | <b>7670666.65</b> | <b>0.01 P 1465.71</b> |

NOTES: ALL OF THE FOLLOWING QUANTITIES ARE FOR THE DELIVERY CONDITION USING EVEN KEEL HYDROSTATICS

|                                        |                     |                                         |                      |
|----------------------------------------|---------------------|-----------------------------------------|----------------------|
| 1. LENGTH BETWEEN PERPENDICULARS       | 64.92 FEET          | 15. MOMENT TO ALTER TRIM ONE INCH       | 360,011.41 FT - TONS |
| 2. BOTTOM KEEL BELOW BASELINE          | 0 FEET              | 16. LCB ON EVEN KEEL AT ABOVE WATERLINE | 33.00 FEET           |
| 3. C.G. ABOVE BOTTOM OF KEEL           | 7.75 FT ABL         | 17. LONGITUDINAL CENTER OF GRAVITY      | 33.68 FEET           |
| 4. C.G ABOVE BASELINE                  | 7.75 FEET           | 18. TRIMMING LEVER                      | 0.69 FEET AFT LCB    |
| 5. DRAFT AT LCF FOR ABOVE DISPLACEMENT | 5.55 FT ABL         | 19. TRIM                                | 0.43 FEET BY STERN   |
| 6. TONS PER INCH IMMERSION             | 5825.14 LT / IN     | 20. L.C.F.                              | 34.81 FEET           |
| 7. TRANSVERSE METACENTER (KEEL)        | 12.20 FEET          | 21. DIFF IN DRAFT, L.C.F. TO MIDSHIP    | 0.02 FEET INCREASE   |
| 8. GM, WITHOUT FREE SURFACE CORR.      | 4.45 FEET           | 22. DRAFTS (+ ABOVE BL):                |                      |
| 9. FREE SURFACE CORR.                  | 0.33 FEET           | FORWARD PERPENDICULAR                   | 5.32 FEET            |
| 10. INCLINING KG MARGIN                | 0.00 FEET           | MEAN                                    | 5.57 FEET            |
| 11. KGv                                | 8.08 FEET           | AFT PERPENDICULAR                       | 5.81 FEET            |
| 12. GM, WITH FREE SURFACE CORR.        | 4.78 FEET           | 23. DESIGNED DRAG                       | 0.00 FEET            |
| 13. MOMENT TO ALTER HEEL ONE DEGREE    | 17,694.35 FT - TONS |                                         |                      |
| 14. LIST                               | 0.08 DEG            |                                         |                      |

## HOMELAND SECURITY CUTTER - LIGHT ICEBREAKER

## 1-DIGIT WEIGHT AND MOMENT ESTIMATE

WEIGHT ESTIMATE

11-Mar

IMPERIAL UNITS - POUNDS, LONG TONS, FEET, LB-FT

SHIP: HSC - L

REFERENCED FROM THE FORWARD PERPENDICULAR, CENTERLINE AND BASELINE

| SWBS<br>GROUP | DESCRIPTION                                 | CURRENT<br>WEIGHT | VCG | VERTICAL<br>MOMENT | LCG               | LONGITUDINAL<br>MOMENT | TCG               | TRANSVERSE<br>MOMENT |                |
|---------------|---------------------------------------------|-------------------|-----|--------------------|-------------------|------------------------|-------------------|----------------------|----------------|
|               | 1 STRUCTURE                                 | 100805.64         |     | 6.97               | 702143.05         | 31.74 A                | 3199243.62        | 0.01 P               | 961.91         |
|               | 2 PROPULSION SYSTEMS                        | 17068.00          |     | 4.17               | 71137.40          | 42.67 A                | 728367.51         | 0.00 C               | 0.00           |
|               | 3 ELECTRICAL SYSTEMS                        | 11650.00          |     | 5.96               | 69406.77          | 39.48 A                | 459915.86         | 0.03 P               | 385.60         |
|               | 4 COMMAND & SURVEILLANCE                    | 3970.93           |     | 16.75              | 66493.94          | 21.91 A                | 87002.28          | 0.00 C               | 0.00           |
|               | 5 AUXILIARY SYSTEMS                         | 27577.53          |     | 9.43               | 260136.40         | 42.54 A                | 1173149.49        | 0.08 P               | 2236.66        |
|               | 6 OUTFIT & FURNISHINGS                      | 21571.24          |     | 8.96               | 193194.34         | 30.77 A                | 663664.60         | 0.06 S               | -1377.27       |
|               | LIGHT SHIP, NO MARGINS                      | 182643.34         |     | 7.46               | 1362511.90        | 34.56 A                | 6311343.35        | 0.01 P               | 2206.90        |
| M             | MARGINS                                     | 29769.04          |     | 12.09              | 359934.95         | 34.56 A                | 1028685.85        | 0.01 S               | 359.70         |
|               | <b>LIGHT SHIP CONDITION</b>                 | <b>212412.38</b>  |     | <b>8.11</b>        | <b>1722446.84</b> | <b>34.56 A</b>         | <b>7340029.21</b> | <b>0.01 S</b>        | <b>2566.60</b> |
| A             | DEPARTURE LOADS W/ ATON                     | 57482.69          |     | 6.20               | 356504.07         | 32.01 A                | 1839896.11        | 0.04 S               | -2075.64       |
|               | <b>DESIGN DEPARTURE LOADS W/ ATON, BOSL</b> | <b>269895.07</b>  |     | <b>7.70</b>        | <b>2078950.92</b> | <b>34.01 A</b>         | <b>9179925.32</b> | <b>0.00 P</b>        | <b>490.96</b>  |
|               | SERVICE LIFE ALLOWANCE                      | 15930.93          |     | 8.61               | 137148.98         | 34.56 A                | 550502.19         | 0.01 P               | 192.50         |
|               | <b>DESIGN DEPARTURE LOADS W/ ATON, EOSL</b> | <b>285826.00</b>  |     | <b>7.75</b>        | <b>2216099.90</b> | <b>34.04 A</b>         | <b>9730427.51</b> | <b>0.00 P</b>        | <b>683.46</b>  |

NOTES: ALL OF THE FOLLOWING QUANTITIES ARE FOR THE EOSL CONDITION USING EVEN KEEL HYDROSTATICS

|                                        |                    |                                         |                     |
|----------------------------------------|--------------------|-----------------------------------------|---------------------|
| 1. LENGTH BETWEEN PERPENDICULARS       | 67.17 FEET         | 15. MOMENT TO ALTER TRIM ONE INCH       | 417,925.06 FT - LBS |
| 2. BOTTOM KEEL BELOW BASELINE          | 0 FEET             | 16. LCB ON EVEN KEEL AT ABOVE WATERLINE | 33.39 FEET          |
| 3. C.G. ABOVE BOTTOM OF KEEL           | 7.70 FT ABL        | 17. LONGITUDINAL CENTER OF GRAVITY      | 34.01 FEET          |
| 4. C.G ABOVE BASELINE                  | 7.70 FEET          | 18. TRIMMING LEVER                      | 0.63 FEET AFT LCB   |
| 5. DRAFT AT LCF FOR ABOVE DISPLACEMENT | 6.36 FT ABL        | 19. TRIM                                | 0.43 FEET BY STERN  |
| 6. POUNDS PER INCH IMMERSION           | 6175.37 LB / IN    | 20. L.C.F.                              | 35.05 FEET          |
| 7. TRANSVERSE METACENTER (KEEL)        | 11.66 FEET         | 21. DIFF IN DRAFT, L.C.F. TO MIDSHIP    | 0.01 FEET INCREASE  |
| 8. GM, WITHOUT FREE SURFACE CORR.      | 3.96 FEET          | 22. DRAFTS (+ ABOVE BL):                |                     |
| 9. FREE SURFACE CORR.                  | 0.28 FEET          | FORWARD PERPENDICULAR                   | 6.12 FEET           |
| 10. INCLINING KG MARGIN                | 0.00 FEET          | MEAN                                    | 6.37 FEET           |
| 11. KGv                                | 7.98 FEET          | AFT PERPENDICULAR                       | 6.62 FEET           |
| 12. GM, WITH FREE SURFACE CORR.        | 4.24 FEET          | 23. DESIGNED DRAG                       | 0.00 FEET           |
| 13. MOMENT TO ALTER HEEL ONE DEGREE    | 19,742.36 FT - LBS |                                         |                     |
| 14. LIST                               | 0.02 DEG           |                                         |                     |

## HOMELAND SECURITY CUTTER - LIGHT ICEBREAKER

## 1-DIGIT WEIGHT AND MOMENT ESTIMATE

## WEIGHT ESTIMATE

11-Mar

IMPERIAL UNITS - POUNDS, LONG TONS, FEET, LB-FT

SHIP: HSC - L

REFERENCED FROM THE FORWARD PERPENDICULAR, CENTERLINE AND BASELINE

| SWBS<br>GROUP | DESCRIPTION                                  | CURRENT<br>WEIGHT | VCG | VERTICAL<br>MOMENT | LCG               | LONGITUDINAL<br>MOMENT | TCG               | TRANSVERSE<br>MOMENT |                |
|---------------|----------------------------------------------|-------------------|-----|--------------------|-------------------|------------------------|-------------------|----------------------|----------------|
|               | 1 STRUCTURE                                  | 100805.64         |     | 6.97               | 702143.05         | 31.74 A                | 3199243.62        | 0.01 P               | 961.91         |
|               | 2 PROPULSION SYSTEMS                         | 17068.00          |     | 4.17               | 71137.40          | 42.67 A                | 728367.51         | 0.00 C               | 0.00           |
|               | 3 ELECTRICAL SYSTEMS                         | 11650.00          |     | 5.96               | 69406.77          | 39.48 A                | 459915.86         | 0.03 P               | 385.60         |
|               | 4 COMMAND & SURVEILLANCE                     | 3970.93           |     | 16.75              | 66493.94          | 21.91 A                | 87002.28          | 0.00 C               | 0.00           |
|               | 5 AUXILIARY SYSTEMS                          | 27577.53          |     | 9.43               | 260136.40         | 42.54 A                | 1173149.49        | 0.08 P               | 2236.66        |
|               | 6 OUTFIT & FURNISHINGS                       | 21571.24          |     | 8.96               | 193194.34         | 30.77 A                | 663664.60         | 0.06 S               | -1377.27       |
|               | LIGHT SHIP, NO MARGINS                       | 182643.34         |     | 7.46               | 1362511.90        | 34.56 A                | 6311343.35        | 0.01 P               | 2206.90        |
| M             | MARGINS                                      | 29769.04          |     | 12.09              | 359934.95         | 34.56 A                | 1028685.85        | 0.01 S               | 359.70         |
|               | <b>LIGHT SHIP CONDITION</b>                  | <b>212412.38</b>  |     | <b>8.11</b>        | <b>1722446.84</b> | <b>34.56 A</b>         | <b>7340029.21</b> | <b>0.01 S</b>        | <b>2566.60</b> |
| B             | DEPARTURE LOADS W/O ATON                     | 33222.69          |     | 3.77               | 125333.27         | 29.89 A                | 993121.81         | 0.03 S               | -949.24        |
|               | <b>DESIGN DEPARTURE LOADS W/O ATON, BOSL</b> | <b>212412.38</b>  |     | <b>8.11</b>        | <b>1722446.84</b> | <b>34.56 A</b>         | <b>7340029.21</b> | <b>0.01 S</b>        | <b>2566.60</b> |
|               | SERVICE LIFE ALLOWANCE                       | 15930.93          |     | 8.61               | 137148.98         | 34.56 A                | 550502.19         | 0.01 P               | 192.50         |
|               | <b>DESIGN DEPARTURE LOADS W/O ATON, EOSL</b> | <b>261566.00</b>  |     | <b>7.59</b>        | <b>1984929.10</b> | <b>33.96 A</b>         | <b>8883653.21</b> | <b>0.01 P</b>        | <b>1809.86</b> |

NOTES: ALL OF THE FOLLOWING QUANTITIES ARE FOR THE EOSL CONDITION USING EVEN KEEL HYDROSTATICS

|                                        |                     |                                         |                      |
|----------------------------------------|---------------------|-----------------------------------------|----------------------|
| 1. LENGTH BETWEEN PERPENDICULARS       | 66.33 FEET          | 15. MOMENT TO ALTER TRIM ONE INCH       | 394,599.25 FT - TONS |
| 2. BOTTOM KEEL BELOW BASELINE          | 0 FEET              | 16. LCB ON EVEN KEEL AT ABOVE WATERLINE | 33.23 FEET           |
| 3. C.G. ABOVE BOTTOM OF KEEL           | 7.59 FT ABL         | 17. LONGITUDINAL CENTER OF GRAVITY      | 33.96 FEET           |
| 4. C.G ABOVE BASELINE                  | 7.59 FEET           | 18. TRIMMING LEVER                      | 0.73 FEET AFT LCB    |
| 5. DRAFT AT LCF FOR ABOVE DISPLACEMENT | 6.03 FT ABL         | 19. TRIM                                | 0.48 FEET BY STEM    |
| 6. TONS PER INCH IMMERSION             | 6040.00 LT / IN     | 20. L.C.F.                              | 34.94 FEET           |
| 7. TRANSVERSE METACENTER (KEEL)        | 11.85 FEET          | 21. DIFF IN DRAFT, L.C.F. TO MIDSHIP    | 0.01 FEET INCREASE   |
| 8. GM, WITHOUT FREE SURFACE CORR.      | 4.27 FEET           | 22. DRAFTS (+ ABOVE BL):                |                      |
| 9. FREE SURFACE CORR.                  | 0.29 FEET           | FORWARD PERPENDICULAR                   | 5.76 FEET            |
| 10. INCLINING KG MARGIN                | 0.00 FEET           | MEAN                                    | 6.04 FEET            |
| 11. KGv                                | 7.88 FEET           | AFT PERPENDICULAR                       | 6.32 FEET            |
| 12. GM, WITH FREE SURFACE CORR.        | 4.55 FEET           | 23. DESIGNED DRAG                       | 0.00 FEET            |
| 13. MOMENT TO ALTER HEEL ONE DEGREE    | 19,474.72 FT - TONS |                                         |                      |
| 14. LIST                               | 0.09 DEG            |                                         |                      |

## HOMELAND SECURITY CUTTER - LIGHT ICEBREAKER

## 1-DIGIT WEIGHT AND MOMENT ESTIMATE

WEIGHT ESTIMATE

11-Mar

IMPERIAL UNITS - POUNDS, LONG TONS, FEET, LB-FT

SHIP: HSC - L

REFERENCED FROM THE FORWARD PERPENDICULAR, CENTERLINE AND BASELINE

| SWBS<br>GROUP | DESCRIPTION                            | CURRENT<br>WEIGHT | VCG | VERTICAL<br>MOMENT | LCG               | LONGITUDINAL<br>MOMENT | TCG               | TRANSVERSE<br>MOMENT |                |
|---------------|----------------------------------------|-------------------|-----|--------------------|-------------------|------------------------|-------------------|----------------------|----------------|
|               | 1 STRUCTURE                            | 100805.64         |     | 6.97               | 702143.05         | 31.74 A                | 3199243.62        | 0.01 P               | 961.91         |
|               | 2 PROPULSION SYSTEMS                   | 17068.00          |     | 4.17               | 71137.40          | 42.67 A                | 728367.51         | 0.00 C               | 0.00           |
|               | 3 ELECTRICAL SYSTEMS                   | 11650.00          |     | 5.96               | 69406.77          | 39.48 A                | 459915.86         | 0.03 P               | 385.60         |
|               | 4 COMMAND & SURVEILLANCE               | 3970.93           |     | 16.75              | 66493.94          | 21.91 A                | 87002.28          | 0.00 C               | 0.00           |
|               | 5 AUXILIARY SYSTEMS                    | 27577.53          |     | 9.43               | 260136.40         | 42.54 A                | 1173149.49        | 0.08 P               | 2236.66        |
|               | 6 OUTFIT & FURNISHINGS                 | 21571.24          |     | 8.96               | 193194.34         | 30.77 A                | 663664.60         | 0.06 S               | -1377.27       |
|               | LIGHT SHIP, NO MARGINS                 | 182643.34         |     | 7.46               | 1362511.90        | 34.56 A                | 6311343.35        | 0.01 P               | 2206.90        |
| M             | MARGINS                                | 29769.04          |     | 12.09              | 359934.95         | 34.56 A                | 1028685.85        | 0.01 S               | 359.70         |
|               | <b>LIGHT SHIP CONDITION</b>            | <b>212412.38</b>  |     | <b>8.11</b>        | <b>1722446.84</b> | <b>34.56 A</b>         | <b>7340029.21</b> | <b>0.01 S</b>        | <b>2566.60</b> |
| C             | HALF LOADS W/ ATON                     | 37405.69          |     | 6.43               | 240610.61         | 35.39 A                | 1323729.24        | 0.03 S               | -1073.11       |
|               | <b>DESIGN HALF LOADS W/ ATON, BOSL</b> | <b>212412.38</b>  |     | <b>8.11</b>        | <b>1722446.84</b> | <b>34.56 A</b>         | <b>7340029.21</b> | <b>0.01 S</b>        | <b>2566.60</b> |
|               | SERVICE LIFE ALLOWANCE                 | 15930.93          |     | 8.61               | 137148.98         | 34.56 A                | 550502.19         | 0.01 P               | 192.50         |
|               | <b>DESIGN HALF LOADS W/ ATON, EOSL</b> | <b>265749.00</b>  |     | <b>7.90</b>        | <b>2100206.44</b> | <b>34.67 A</b>         | <b>9214260.64</b> | <b>0.01 P</b>        | <b>1685.99</b> |

NOTES: ALL OF THE FOLLOWING QUANTITIES ARE FOR THE EOSL CONDITION USING EVEN KEEL HYDROSTATICS

|                                        |                     |                                         |                      |
|----------------------------------------|---------------------|-----------------------------------------|----------------------|
| 1. LENGTH BETWEEN PERPENDICULARS       | 66.68 FEET          | 15. MOMENT TO ALTER TRIM ONE INCH       | 398,778.72 FT - TONS |
| 2. BOTTOM KEEL BELOW BASELINE          | 0 FEET              | 16. LCB ON EVEN KEEL AT ABOVE WATERLINE | 33.26 FEET           |
| 3. C.G. ABOVE BOTTOM OF KEEL           | 7.90 FT ABL         | 17. LONGITUDINAL CENTER OF GRAVITY      | 34.67 FEET           |
| 4. C.G ABOVE BASELINE                  | 7.90 FEET           | 18. TRIMMING LEVER                      | 1.41 FEET AFT LCB    |
| 5. DRAFT AT LCF FOR ABOVE DISPLACEMENT | 6.08 FT ABL         | 19. TRIM                                | 0.94 FEET BY STEM    |
| 6. TONS PER INCH IMMERSION             | 6064.81 LT / IN     | 20. L.C.F.                              | 34.96 FEET           |
| 7. TRANSVERSE METACENTER (KEEL)        | 11.82 FEET          | 21. DIFF IN DRAFT, L.C.F. TO MIDSHIP    | 0.03 FEET INCREASE   |
| 8. GM, WITHOUT FREE SURFACE CORR.      | 3.92 FEET           | 22. DRAFTS (+ ABOVE BL):                |                      |
| 9. FREE SURFACE CORR.                  | 0.28 FEET           | FORWARD PERPENDICULAR                   | 5.56 FEET            |
| 10. INCLINING KG MARGIN                | 0.00 FEET           | MEAN                                    | 6.11 FEET            |
| 11. KGv                                | 8.19 FEET           | AFT PERPENDICULAR                       | 6.66 FEET            |
| 12. GM, WITH FREE SURFACE CORR.        | 4.20 FEET           | 23. DESIGNED DRAG                       | 0.00 FEET            |
| 13. MOMENT TO ALTER HEEL ONE DEGREE    | 18,163.15 FT - TONS |                                         |                      |
| 14. LIST                               | 0.09 DEG            |                                         |                      |

## HOMELAND SECURITY CUTTER - LIGHT ICEBREAKER

## 1-DIGIT WEIGHT AND MOMENT ESTIMATE

## WEIGHT ESTIMATE

11-Mar

IMPERIAL UNITS - POUNDS, LONG TONS, FEET, LB-FT

SHIP: HSC - L

REFERENCED FROM THE FORWARD PERPENDICULAR, CENTERLINE AND BASELINE

| SWBS<br>GROUP | DESCRIPTION                             | CURRENT<br>WEIGHT | VCG | VERTICAL<br>MOMENT | LCG               | LONGITUDINAL<br>MOMENT | TCG               | TRANSVERSE<br>MOMENT |                |
|---------------|-----------------------------------------|-------------------|-----|--------------------|-------------------|------------------------|-------------------|----------------------|----------------|
|               | 1 STRUCTURE                             | 100805.64         |     | 6.97               | 702143.05         | 31.74 A                | 3199243.62        | 0.01 P               | 961.91         |
|               | 2 PROPULSION SYSTEMS                    | 17068.00          |     | 4.17               | 71137.40          | 42.67 A                | 728367.51         | 0.00 C               | 0.00           |
|               | 3 ELECTRICAL SYSTEMS                    | 11650.00          |     | 5.96               | 69406.77          | 39.48 A                | 459915.86         | 0.03 P               | 385.60         |
|               | 4 COMMAND & SURVEILLANCE                | 3970.93           |     | 16.75              | 66493.94          | 21.91 A                | 87002.28          | 0.00 C               | 0.00           |
|               | 5 AUXILIARY SYSTEMS                     | 27577.53          |     | 9.43               | 260136.40         | 42.54 A                | 1173149.49        | 0.08 P               | 2236.66        |
|               | 6 OUTFIT & FURNISHINGS                  | 21571.24          |     | 8.96               | 193194.34         | 30.77 A                | 663664.60         | 0.06 S               | -1377.27       |
|               | LIGHT SHIP, NO MARGINS                  | 182643.34         |     | 7.46               | 1362511.90        | 34.56 A                | 6311343.35        | 0.01 P               | 2206.90        |
| M             | MARGINS                                 | 29769.04          |     | 12.09              | 359934.95         | 34.56 A                | 1028685.85        | 0.01 S               | 359.70         |
|               | <b>LIGHT SHIP CONDITION</b>             | <b>212412.38</b>  |     | <b>8.11</b>        | <b>1722446.84</b> | <b>34.56 A</b>         | <b>7340029.21</b> | <b>0.01 S</b>        | <b>2566.60</b> |
| D             | HALF LOADS W/O ATON                     | 21065.69          |     | 3.05               | 64272.61          | 27.21 A                | 573109.24         | 0.05 S               | -1073.11       |
|               | <b>DESIGN HALF LOADS W/O ATON, BOSL</b> | <b>212412.38</b>  |     | <b>8.11</b>        | <b>1722446.84</b> | <b>34.56 A</b>         | <b>7340029.21</b> | <b>0.01 S</b>        | <b>2566.60</b> |
|               | SERVICE LIFE ALLOWANCE                  | 15930.93          |     | 8.61               | 137148.98         | 34.56 A                | 550502.19         | 0.01 P               | 192.50         |
|               | <b>DESIGN HALF LOADS W/O ATON, EOSL</b> | <b>249409.00</b>  |     | <b>7.71</b>        | <b>1923868.44</b> | <b>33.93 A</b>         | <b>8463640.64</b> | <b>0.01 P</b>        | <b>1685.99</b> |

NOTES: ALL OF THE FOLLOWING QUANTITIES ARE FOR THE EOSL CONDITION USING EVEN KEEL HYDROSTATICS

|                                        |                     |                                         |                      |
|----------------------------------------|---------------------|-----------------------------------------|----------------------|
| 1. LENGTH BETWEEN PERPENDICULARS       | 65.8 FEET           | 15. MOMENT TO ALTER TRIM ONE INCH       | 382,174.42 FT - TONS |
| 2. BOTTOM KEEL BELOW BASELINE          | 0 FEET              | 16. LCB ON EVEN KEEL AT ABOVE WATERLINE | 33.15 FEET           |
| 3. C.G. ABOVE BOTTOM OF KEEL           | 7.71 FT ABL         | 17. LONGITUDINAL CENTER OF GRAVITY      | 33.93 FEET           |
| 4. C.G ABOVE BASELINE                  | 7.71 FEET           | 18. TRIMMING LEVER                      | 0.78 FEET AFT LCB    |
| 5. DRAFT AT LCF FOR ABOVE DISPLACEMENT | 5.86 FT ABL         | 19. TRIM                                | 0.51 FEET BY STEM    |
| 6. TONS PER INCH IMMERSION             | 5964.63 LT / IN     | 20. L.C.F.                              | 34.88 FEET           |
| 7. TRANSVERSE METACENTER (KEEL)        | 11.96 FEET          | 21. DIFF IN DRAFT, L.C.F. TO MIDSHIP    | 0.02 FEET INCREASE   |
| 8. GM, WITHOUT FREE SURFACE CORR.      | 4.25 FEET           | 22. DRAFTS (+ ABOVE BL):                |                      |
| 9. FREE SURFACE CORR.                  | 0.30 FEET           | FORWARD PERPENDICULAR                   | 5.58 FEET            |
| 10. INCLINING KG MARGIN                | 0.00 FEET           | MEAN                                    | 5.87 FEET            |
| 11. KGv                                | 8.02 FEET           | AFT PERPENDICULAR                       | 6.17 FEET            |
| 12. GM, WITH FREE SURFACE CORR.        | 4.55 FEET           | 23. DESIGNED DRAG                       | 0.00 FEET            |
| 13. MOMENT TO ALTER HEEL ONE DEGREE    | 18,505.57 FT - TONS |                                         |                      |
| 14. LIST                               | 0.09 DEG            |                                         |                      |

## HOMELAND SECURITY CUTTER - LIGHT ICEBREAKER

## 1-DIGIT WEIGHT AND MOMENT ESTIMATE

WEIGHT ESTIMATE

11-Mar

IMPERIAL UNITS - POUNDS, LONG TONS, FEET, LB-FT

SHIP: HSC - L

REFERENCED FROM THE FORWARD PERPENDICULAR, CENTERLINE AND BASELINE

| SWBS<br>GROUP | DESCRIPTION                              | CURRENT<br>WEIGHT | VCG | VERTICAL<br>MOMENT | LCG               | LONGITUDINAL<br>MOMENT | TCG               | TRANSVERSE<br>MOMENT |                |
|---------------|------------------------------------------|-------------------|-----|--------------------|-------------------|------------------------|-------------------|----------------------|----------------|
|               | 1 STRUCTURE                              | 100805.64         |     | 6.97               | 702143.05         | 31.74 A                | 3199243.62        | 0.01 P               | 961.91         |
|               | 2 PROPULSION SYSTEMS                     | 17068.00          |     | 4.17               | 71137.40          | 42.67 A                | 728367.51         | 0.00 C               | 0.00           |
|               | 3 ELECTRICAL SYSTEMS                     | 11650.00          |     | 5.96               | 69406.77          | 39.48 A                | 459915.86         | 0.03 P               | 385.60         |
|               | 4 COMMAND & SURVEILLANCE                 | 3970.93           |     | 16.75              | 66493.94          | 21.91 A                | 87002.28          | 0.00 C               | 0.00           |
|               | 5 AUXILIARY SYSTEMS                      | 27577.53          |     | 9.43               | 260136.40         | 42.54 A                | 1173149.49        | 0.08 P               | 2236.66        |
|               | 6 OUTFIT & FURNISHINGS                   | 21571.24          |     | 8.96               | 193194.34         | 30.77 A                | 663664.60         | 0.06 S               | -1377.27       |
|               | LIGHT SHIP, NO MARGINS                   | 182643.34         |     | 7.46               | 1362511.90        | 34.56 A                | 6311343.35        | 0.01 P               | 2206.90        |
| M             | MARGINS                                  | 29769.04          |     | 12.09              | 359934.95         | 34.56 A                | 1028685.85        | 0.01 S               | 359.70         |
|               | <b>LIGHT SHIP CONDITION</b>              | <b>212412.38</b>  |     | <b>8.11</b>        | <b>1722446.84</b> | <b>34.56 A</b>         | <b>7340029.21</b> | <b>0.01 S</b>        | <b>2566.60</b> |
| E             | MIN OP LOADS W/ ATON                     | 31646.69          |     | 6.88               | 217710.74         | 34.17 A                | 1081257.44        | 0.03 S               | -1100.89       |
|               | <b>DESIGN MIN OP LOADS W/ ATON, BOSL</b> | <b>212412.38</b>  |     | <b>8.11</b>        | <b>1722446.84</b> | <b>34.56 A</b>         | <b>7340029.21</b> | <b>0.01 S</b>        | <b>2566.60</b> |
|               | SERVICE LIFE ALLOWANCE                   | 15930.93          |     | 8.61               | 137148.98         | 34.56 A                | 550502.19         | 0.01 P               | 192.50         |
|               | <b>DESIGN MIN OP LOADS W/ ATON, EOSL</b> | <b>259990.00</b>  |     | <b>7.99</b>        | <b>2077306.57</b> | <b>34.51 A</b>         | <b>8971788.84</b> | <b>0.01 P</b>        | <b>1658.21</b> |

NOTES: ALL OF THE FOLLOWING QUANTITIES ARE FOR THE EOSL CONDITION USING EVEN KEEL HYDROSTATICS

|                                        |                     |                                         |                      |
|----------------------------------------|---------------------|-----------------------------------------|----------------------|
| 1. LENGTH BETWEEN PERPENDICULARS       | 66.46 FEET          | 15. MOMENT TO ALTER TRIM ONE INCH       | 393,011.48 FT - TONS |
| 2. BOTTOM KEEL BELOW BASELINE          | 0 FEET              | 16. LCB ON EVEN KEEL AT ABOVE WATERLINE | 33.22 FEET           |
| 3. C.G. ABOVE BOTTOM OF KEEL           | 7.99 FT ABL         | 17. LONGITUDINAL CENTER OF GRAVITY      | 34.51 FEET           |
| 4. C.G ABOVE BASELINE                  | 7.99 FEET           | 18. TRIMMING LEVER                      | 1.29 FEET AFT LCB    |
| 5. DRAFT AT LCF FOR ABOVE DISPLACEMENT | 6.00 FT ABL         | 19. TRIM                                | 0.85 FEET BY STEM    |
| 6. TONS PER INCH IMMERSION             | 6030.51 LT / IN     | 20. L.C.F.                              | 34.93 FEET           |
| 7. TRANSVERSE METACENTER (KEEL)        | 11.87 FEET          | 21. DIFF IN DRAFT, L.C.F. TO MIDSHIP    | 0.03 FEET INCREASE   |
| 8. GM, WITHOUT FREE SURFACE CORR.      | 3.88 FEET           | 22. DRAFTS (+ ABOVE BL):                |                      |
| 9. FREE SURFACE CORR.                  | 0.29 FEET           | FORWARD PERPENDICULAR                   | 5.54 FEET            |
| 10. INCLINING KG MARGIN                | 0.00 FEET           | MEAN                                    | 6.03 FEET            |
| 11. KGv                                | 8.28 FEET           | AFT PERPENDICULAR                       | 6.52 FEET            |
| 12. GM, WITH FREE SURFACE CORR.        | 4.17 FEET           | 23. DESIGNED DRAG                       | 0.00 FEET            |
| 13. MOMENT TO ALTER HEEL ONE DEGREE    | 17,598.30 FT - TONS |                                         |                      |
| 14. LIST                               | 0.09 DEG            |                                         |                      |

## HOMELAND SECURITY CUTTER - LIGHT ICEBREAKER

## 1-DIGIT WEIGHT AND MOMENT ESTIMATE

WEIGHT ESTIMATE

11-Mar

IMPERIAL UNITS - POUNDS, LONG TONS, FEET, LB-FT

SHIP: HSC - L

REFERENCED FROM THE FORWARD PERPENDICULAR, CENTERLINE AND BASELINE

| SWBS<br>GROUP | DESCRIPTION                               | CURRENT<br>WEIGHT | VCG | VERTICAL<br>MOMENT | LCG               | LONGITUDINAL<br>MOMENT | TCG               | TRANSVERSE<br>MOMENT |                |
|---------------|-------------------------------------------|-------------------|-----|--------------------|-------------------|------------------------|-------------------|----------------------|----------------|
|               | 1 STRUCTURE                               | 100805.64         |     | 6.97               | 702143.05         | 31.74 A                | 3199243.62        | 0.01 P               | 961.91         |
|               | 2 PROPULSION SYSTEMS                      | 17068.00          |     | 4.17               | 71137.40          | 42.67 A                | 728367.51         | 0.00 C               | 0.00           |
|               | 3 ELECTRICAL SYSTEMS                      | 11650.00          |     | 5.96               | 69406.77          | 39.48 A                | 459915.86         | 0.03 P               | 385.60         |
|               | 4 COMMAND & SURVEILLANCE                  | 3970.93           |     | 16.75              | 66493.94          | 21.91 A                | 87002.28          | 0.00 C               | 0.00           |
|               | 5 AUXILIARY SYSTEMS                       | 27577.53          |     | 9.43               | 260136.40         | 42.54 A                | 1173149.49        | 0.08 P               | 2236.66        |
|               | 6 OUTFIT & FURNISHINGS                    | 21571.24          |     | 8.96               | 193194.34         | 30.77 A                | 663664.60         | 0.06 S               | -1377.27       |
|               | LIGHT SHIP, NO MARGINS                    | 182643.34         |     | 7.46               | 1362511.90        | 34.56 A                | 6311343.35        | 0.01 P               | 2206.90        |
| M             | MARGINS                                   | 29769.04          |     | 12.09              | 359934.95         | 34.56 A                | 1028685.85        | 0.01 S               | 359.70         |
|               | <b>LIGHT SHIP CONDITION</b>               | <b>212412.38</b>  |     | <b>8.11</b>        | <b>1722446.84</b> | <b>34.56 A</b>         | <b>7340029.21</b> | <b>0.01 S</b>        | <b>2566.60</b> |
| F             | MIN OP LOADS W/O ATON                     | 15306.69          |     | 2.70               | 41372.74          | 21.60 A                | 330637.44         | 0.07 S               | -1100.89       |
|               | <b>DESIGN MIN OP LOADS W/O ATON, BOSL</b> | <b>212412.38</b>  |     | <b>8.11</b>        | <b>1722446.84</b> | <b>34.56 A</b>         | <b>7340029.21</b> | <b>0.01 S</b>        | <b>2566.60</b> |
|               | SERVICE LIFE ALLOWANCE                    | 15930.93          |     | 8.61               | 137148.98         | 34.56 A                | 550502.19         | 0.01 P               | 192.50         |
|               | <b>DESIGN MIN OP LOADS W/O ATON, EOSL</b> | <b>243650.00</b>  |     | <b>7.80</b>        | <b>1900968.57</b> | <b>33.74 A</b>         | <b>8221168.84</b> | <b>0.01 P</b>        | <b>1658.21</b> |

NOTES: ALL OF THE FOLLOWING QUANTITIES ARE FOR THE EOSL CONDITION USING EVEN KEEL HYDROSTATICS

|                                        |                     |                                         |                      |
|----------------------------------------|---------------------|-----------------------------------------|----------------------|
| 1. LENGTH BETWEEN PERPENDICULARS       | 65.58 FEET          | 15. MOMENT TO ALTER TRIM ONE INCH       | 376,226.07 FT - TONS |
| 2. BOTTOM KEEL BELOW BASELINE          | 0 FEET              | 16. LCB ON EVEN KEEL AT ABOVE WATERLINE | 33.11 FEET           |
| 3. C.G. ABOVE BOTTOM OF KEEL           | 7.80 FT ABL         | 17. LONGITUDINAL CENTER OF GRAVITY      | 33.74 FEET           |
| 4. C.G ABOVE BASELINE                  | 7.80 FEET           | 18. TRIMMING LEVER                      | 0.63 FEET AFT LCB    |
| 5. DRAFT AT LCF FOR ABOVE DISPLACEMENT | 5.78 FT ABL         | 19. TRIM                                | 0.41 FEET BY STEM    |
| 6. TONS PER INCH IMMERSION             | 5927.81 LT / IN     | 20. L.C.F.                              | 34.86 FEET           |
| 7. TRANSVERSE METACENTER (KEEL)        | 12.02 FEET          | 21. DIFF IN DRAFT, L.C.F. TO MIDSHIP    | 0.01 FEET INCREASE   |
| 8. GM, WITHOUT FREE SURFACE CORR.      | 4.22 FEET           | 22. DRAFTS (+ ABOVE BL):                |                      |
| 9. FREE SURFACE CORR.                  | 0.31 FEET           | FORWARD PERPENDICULAR                   | 5.56 FEET            |
| 10. INCLINING KG MARGIN                | 0.00 FEET           | MEAN                                    | 5.79 FEET            |
| 11. KGv                                | 8.11 FEET           | AFT PERPENDICULAR                       | 6.02 FEET            |
| 12. GM, WITH FREE SURFACE CORR.        | 4.53 FEET           | 23. DESIGNED DRAG                       | 0.00 FEET            |
| 13. MOMENT TO ALTER HEEL ONE DEGREE    | 17,943.95 FT - TONS |                                         |                      |
| 14. LIST                               | 0.09 DEG            |                                         |                      |

| SWBS # | SWBS Name                                  | Applied Weight | Vertical |        | Longitudinal |     |         | Transverse |      |          |
|--------|--------------------------------------------|----------------|----------|--------|--------------|-----|---------|------------|------|----------|
|        |                                            |                | VCG ABL  | VMOM   | LCG          | F/A | LMOM    | TCG        | P/C/ | TMOM     |
|        |                                            | lb             | ft       | ft*lb  | ft           |     | ft*lb   | ft         | S    | ft*lb    |
| 1      | Hull Structure                             | 100806         | 6.97     | 702143 | 31.74        | A   | 3199244 | 0.01       | P    | 961.912  |
| 111    | Shell Plating                              | 34528          | 4.81     | 165985 | 29.70        | A   | 1025327 | 0.00       | C    | 0.000    |
| 114    | Shell Appendages                           | 2032           | 0.10     | 3454   | 3.26         | A   | 112694  | 0.00       | C    | 0.000    |
| 115    | Stanchions                                 | 305            | 0.05     | 1733   | 0.43         | A   | 14724   | 0.00       | C    | 0.000    |
| 116    | Longitudinal Framing                       | 2441           | 0.18     | 6285   | 2.44         | A   | 84388   | 0.00       | C    | 0.000    |
| 117    | Transverse Framing                         | 10130          | 1.38     | 47520  | 7.80         | A   | 269402  | 0.00       | S    | 0.000    |
| 121    | Longitudinal Structural Bulkheads          | 516            | 0.01     | 234    | 0.46         | A   | 15732   | 0.00       | C    | 0.000    |
| 122    | Transverse Structural Bulkheads            | 7471           | 1.34     | 46303  | 7.08         | A   | 244366  | 0.00       | S    | -0.027   |
| 123    | Trunks and Enclosures                      | 720            | 0.21     | 7378   | 0.53         | A   | 18174   | 0.00       | C    | 0.000    |
| 131    | Main Deck                                  | 16977          | 4.85     | 167331 | 17.68        | A   | 610505  | 0.00       | C    | 0.000    |
| 149    | Flats                                      | 3207           | 0.32     | 11135  | 2.46         | A   | 85106   | 0.00       | C    | 0.000    |
| 151    | Deckhouse Structure to First Level         | 8130           | 3.84     | 132751 | 6.35         | A   | 219154  | -0.01      | S    | -227.592 |
| 162    | Stacks and Macks (Combined Stack and Mast) | 1022           | 0.53     | 18412  | 1.05         | A   | 36356   | 0.00       | S    | 0.000    |
| 163    | Sea Chests                                 | 100            | 0.01     | 500    | 0.07         | A   | 2417    | 0.00       | C    | 0.000    |
| 167    | Hull Structural Closures                   | 1538           | 0.30     | 10346  | 1.26         | A   | 43362   | 0.01       | P    | 243.288  |
| 168    | Deckhouse Strural Closures                 | 621            | 0.25     | 8749   | 0.51         | A   | 17481   | 0.03       | P    | 913.715  |
| 171    | Masts, Towers, Tetrapods                   | 200            | 0.14     | 5000   | 0.17         | A   | 5833    | 0.00       | C    | 0.000    |
| 182    | Propulsion Plant Foundations               | 931            | 0.05     | 1839   | 1.12         | A   | 38547   | 0.00       | C    | 0.000    |
| 183    | Electrical Plant Foundations               | 1400           | 0.09     | 3045   | 1.64         | A   | 56748   | 0.00       | C    | 0.000    |
| 184    | Command and Surveillance Foundations       | 516            | 0.25     | 8644   | 0.44         | A   | 15056   | 0.00       | C    | 0.000    |
| 185    | Auxiliary Systems Foundations              | 2798           | 0.57     | 19637  | 3.46         | A   | 119419  | 0.00       | C    | 0.000    |
| 186    | Outfit and Furnishing Foundations          | 216            | 0.05     | 1866   | 0.19         | A   | 6579    | 0.00       | C    | 0.000    |
| 196    | Welding (1.5%)                             | 1437           | 0.29     | 10022  | 1.32         | A   | 45621   | 0.00       | P    | 13.941   |
| 197    | Mill Tolerance (2.0)                       | 1916           | 0.39     | 13363  | 1.76         | A   | 60827   | 0.00       | P    | 18.588   |
| 198    | Free Flooding Liquids                      | 0              | 0.00     | 0      | 0.00         |     | 0       | 0.00       | C    | 0.000    |
| NA     | Brackets & Misc. Structure                 | 1653           | 0.31     | 10612  | 1.49         | A   | 51424   | 0.00       | C    | 0.000    |

| SWBS # | SWBS Name                                      | Applied Weight | Vertical |       | Longitudinal |     |        | Transverse |      |       |
|--------|------------------------------------------------|----------------|----------|-------|--------------|-----|--------|------------|------|-------|
|        |                                                |                | VCG ABL  | VMOM  | LCG          | F/A | LMOM   | TCG        | P/C/ | TMOM  |
|        |                                                |                | ft       | ft*lb | ft           |     | ft*lb  | ft         | S    | ft*lb |
| 2      | Propulsion                                     | 17068          | 4.17     | 71137 | 42.67        | A   | 728368 | 0.00       | C    | 0.000 |
| 233    | Propulsion Internal Combustion Systems         | 8598           | 1.01     | 34922 | 9.92         | A   | 342458 | 0.00       | C    | 0.000 |
| 241    | Propulsion Reduction Gears                     | 1516           | 0.12     | 4177  | 1.91         | A   | 65838  | 0.00       | C    | 0.000 |
| 242    | Propulsion Clutches and Couplings              | 140            | 0.02     | 532   | 0.19         | A   | 6720   | 0.00       | C    | 0.000 |
| 243    | Propulsion Shafting                            | 2240           | 0.14     | 4763  | 3.44         | A   | 118853 | 0.00       | C    | 0.000 |
| 244    | Propulsion Shaft Bearings                      | 152            | 0.01     | 339   | 0.22         | A   | 7481   | 0.00       | C    | 0.000 |
| 245    | Propulsors                                     | 614            | 0.03     | 1176  | 1.08         | A   | 37419  | 0.00       | C    | 0.000 |
| 251    | Combustion Air System                          | 300            | 0.07     | 2400  | 0.37         | A   | 12750  | 0.00       | C    | 0.000 |
| 252    | Propulsion Control System                      | 150            | 0.02     | 551   | 0.17         | A   | 6000   | 0.00       | C    | 0.000 |
| 256    | Cooling Sea Water System                       | 0              | 0.00     | 0     | 0.00         |     | 0      | 0.00       | C    | 0.000 |
| 259    | Exhaust System                                 | 1120           | 0.41     | 14195 | 1.20         | A   | 41349  | 0.00       | C    | 0.000 |
| 261    | Fuel Service System                            | 114            | 0.01     | 228   | 0.13         | A   | 4332   | 0.00       | C    | 0.000 |
| 262    | Main Propulsion Lube Oil Sytstem               | 108            | 0.01     | 396   | 0.13         | A   | 4320   | 0.00       | C    | 0.000 |
| 298    | Propulsion Plan Operating Fluids               | 1120           | 0.12     | 4173  | 1.30         | A   | 45008  | 0.00       | C    | 0.000 |
| 299    | Propulsion Plan Repair Parts and Special Tools | 896            | 0.10     | 3285  | 1.04         | A   | 35840  | 0.00       | C    | 0.000 |

| SWBS # | SWBS Name                                     | Applied Weight | Vertical |       | Longitudinal |     |        | Transverse |      |         |
|--------|-----------------------------------------------|----------------|----------|-------|--------------|-----|--------|------------|------|---------|
|        |                                               |                | VCG ABL  | VMOM  | LCG          | F/A | LMOM   | TCG        | P/C/ | TMOM    |
|        |                                               |                | ft       | ft*lb | ft           |     | ft*lb  | ft         | S    | ft*lb   |
| 3      | Electric Plant                                | 11650          | 5.96     | 69407 | 39.48        | A   | 459916 | 0.03       | P    | 385.599 |
| 311    | Ship Service Power Generation                 | 5770           | 0.89     | 30581 | 7.35         | A   | 253612 | 0.01       | P    | 366.723 |
| 313    | Batteries and Service Facilities              | 1120           | 0.11     | 3920  | 1.17         | A   | 40320  | 0.00       | C    | 0.000   |
| 320    | Power Distribution System                     | 2274           | 0.56     | 19329 | 2.31         | A   | 79590  | 0.00       | C    | 0.000   |
| 330    | Lighting System                               | 694            | 0.22     | 7634  | 0.65         | A   | 22486  | 0.00       | C    | 0.000   |
| 342    | Diesel Support System                         | 1120           | 0.12     | 4024  | 1.07         | A   | 37091  | 0.00       | C    | 0.000   |
| 398    | Electric Plan Operating Fluids                | 448            | 0.07     | 2405  | 0.57         | A   | 19560  | 0.00       | P    | 18.876  |
| 399    | Electric Pland Repair Parts and Special Tools | 224            | 0.04     | 1514  | 0.21         | A   | 7258   | 0.00       | C    | 0.000   |
| 4      | Command and Surveillance                      | 3971           | 16.75    | 66494 | 21.91        | A   | 87002  | 0.00       | C    | 0.000   |
| 411    | Data Display Group                            | 448            | 0.19     | 6496  | 0.23         | A   | 8064   | 0.00       | C    | 0.000   |
| 421    | Non-Electrical/Electronic Navigation Aids     | 217            | 0.10     | 3541  | 0.11         | A   | 3906   | 0.00       | C    | 0.000   |
| 422    | Electrical Navigation Aids                    | 448            | 0.21     | 7311  | 0.23         | A   | 8064   | 0.00       | C    | 0.000   |
| 426    | Electrical Navigation Systems                 | 448            | 0.19     | 6496  | 0.23         | A   | 8064   | 0.00       | C    | 0.000   |
| 434    | Entertainment & Training Systems              | 58             | 0.02     | 540   | 0.04         | A   | 1448   | 0.00       | C    | 0.000   |
| 436    | Alarm, Safety, and Warning Systems            | 448            | 0.10     | 3481  | 0.32         | A   | 11200  | 0.00       | C    | 0.000   |
| 437    | Indicating, order, and Metering Systems       | 224            | 0.05     | 1801  | 0.16         | A   | 5600   | 0.00       | C    | 0.000   |
| 438    | Integrated Control System                     | 560            | 0.16     | 5538  | 0.41         | A   | 14000  | 0.00       | C    | 0.000   |
| 441    | Radio Systems                                 | 448            | 0.13     | 4596  | 0.32         | A   | 11200  | 0.00       | C    | 0.000   |
| 443    | Visual and Audible Communication Systems      | 224            | 0.12     | 4283  | 0.16         | A   | 5600   | 0.00       | C    | 0.000   |
| 446    | Security Equipment                            | 0              | 0.00     | 0     | 0.00         |     | 0      | 0.00       | C    | 0.000   |
| 451    | Surface Search Radar                          | 448            | 0.65     | 22409 | 0.29         | A   | 9856   | 0.00       | C    | 0.000   |

| SWBS # | SWBS Name                                | Applied Weight | Vertical |        | Longitudinal |     |         | Transverse |           |          |
|--------|------------------------------------------|----------------|----------|--------|--------------|-----|---------|------------|-----------|----------|
|        |                                          |                | VCG ABL  | VMOM   | LCG          | F/A | LMOM    | TCG        | P/C/<br>S | TMOM     |
|        |                                          | lb             | ft       | ft*lb  | ft           |     | ft*lb   | ft         |           | ft*lb    |
| 5      | Auxiliary Systems                        | 27578          | 9.43     | 260136 | 42.54        | A   | 1173149 | 0.08       | P         | 2236.661 |
| 511    | Compartment Heating System               | 184            | 0.04     | 1450   | 0.17         | A   | 5953    | 0.00       | C         | 0.000    |
| 512    | Ventilation System                       | 407            | 0.14     | 4885   | 0.31         | A   | 10584   | 0.00       | C         | 0.000    |
| 513    | Machinery Space Ventilation System       | 694            | 0.42     | 14582  | 0.80         | A   | 27776   | 0.00       | C         | 0.000    |
| 514    | Air Conditioning System                  | 283            | 0.10     | 3392   | 0.21         | A   | 7350    | 0.00       | C         | 0.000    |
| 516    | Refrigeration System                     | 605            | 0.11     | 3629   | 0.26         | A   | 9072    | 0.00       | C         | 0.000    |
| 523    | Washdown System                          | 192            | 0.06     | 1921   | 0.23         | A   | 8069    | 0.00       | C         | 0.000    |
| 526    | Scuppers and Deck Drains                 | 641            | 0.19     | 6606   | 0.60         | A   | 20759   | 0.00       | C         | 0.000    |
| 529    | Drainage and Ballasting System           | 603            | 0.07     | 2297   | 0.57         | A   | 19536   | 0.00       | C         | 0.000    |
| 532    | Cooling Water                            | 2192           | 0.08     | 2689   | 2.73         | A   | 94239   | 0.00       | C         | 0.000    |
| 533    | Potable Water                            | 1352           | 0.16     | 5407   | 0.90         | A   | 31090   | 0.00       | C         | 0.000    |
| 555    | Fire Extinguishing Systems               | 850            | 0.15     | 5101   | 0.94         | A   | 32305   | 0.00       | C         | 0.000    |
| 556    | Hydraulic System                         | 1965           | 0.34     | 11790  | 2.96         | A   | 102180  | 0.00       | C         | 0.000    |
| 561    | Steering Control System                  | 2240           | 0.39     | 13440  | 4.02         | A   | 138880  | 0.00       | C         | 0.000    |
| 562    | Rudder                                   | 672            | 0.06     | 2016   | 1.24         | A   | 42672   | 0.00       | C         | 0.000    |
| 581    | Anchor Handling and Stowage              | 811            | 0.24     | 8337   | 0.09         | A   | 3206    | 0.06       | P         | 2004.886 |
| 582    | Mooring and Towing Systems               | 763            | 0.29     | 10181  | 0.33         | A   | 11412   | -0.01      | S         | -307.914 |
| 583    | Boat Handling and Stowage Systems        | 168            | 0.09     | 3110   | 0.12         | A   | 4137    | 0.02       | P         | 539.690  |
| 589    | Crane                                    | 10940          | 4.34     | 150000 | 15.12        | A   | 522114  | 0.00       | C         | 0.000    |
| 598    | Auxiliary Systems Operating Fluids       | 0              | 0.00     | 0      | 0.00         |     | 0       | 0.00       | C         | 0.000    |
| 593    | Environmental Pollution Control Systems  | 1545           | 0.20     | 6952   | 1.88         | A   | 64883   | 0.00       | C         | 0.000    |
| 599    | Auxiliary Systems Repair Parts and Tools | 470            | 0.07     | 2352   | 0.49         | A   | 16934   | 0.00       | C         | 0.000    |

| SWBS #                        | SWBS Name                                     | Applied Weight | Vertical    |                | Longitudinal |          |                | Transverse  |          |             |
|-------------------------------|-----------------------------------------------|----------------|-------------|----------------|--------------|----------|----------------|-------------|----------|-------------|
|                               |                                               |                | VCG ABL     | VMOM           | LCG          | F/A      | LMOM           | TCG         | P/C/     | TMOM        |
|                               |                                               | lb             | ft          | ft*lb          | ft           |          | ft*lb          | ft          | S        | ft*lb       |
| 6                             | Outfit and Furnishings                        | 21571          | 8.96        | 193194         | 30.77        | A        | 663665         | 0.06        | S        | -1377.274   |
| 611                           | Hull Fittings                                 | 1131           | 0.35        | 11947          | 1.05         | A        | 36422          | -0.01       | S        | -198.794    |
| 612                           | Rails, Stanchions, & Lifelines                | 854            | 0.31        | 10865          | 0.90         | A        | 30994          | 0.00        | C        | 0.000       |
| 621                           | Non-Structural Bulkheads                      | 915            | 0.19        | 6529           | 0.70         | A        | 24301          | -0.04       | S        | -1220.313   |
| 622                           | Floors & Gratings                             | 2584           | 0.26        | 8915           | 3.17         | A        | 109615         | 0.00        | S        | -5.738      |
| 623                           | Ladders                                       | 937            | 0.24        | 8407           | 0.85         | A        | 29179          | 0.02        | P        | 805.346     |
| 624                           | Non-Structural Closures                       | 365            | 0.07        | 2552           | 0.29         | A        | 9943           | -0.05       | S        | -1864.196   |
| 625                           | Airports, Fixed Portlights, & Windows         | 1096           | 1.08        | 37423          | 1.49         | A        | 51516          | 0.00        | P        | 88.975      |
| 631                           | Painting                                      | 1528           | 0.29        | 10026          | 1.43         | A        | 49220          | 0.00        | C        | 0.000       |
| 633                           | Cathodic Protection                           | 300            | 0.03        | 900            | 0.28         | A        | 9720           | 0.00        | C        | 0.000       |
| 634                           | Deck Coverings                                | 715            | 0.15        | 5031           | 0.49         | A        | 16927          | -0.02       | S        | -542.642    |
| 635                           | Hull Insulation                               | 4242           | 1.03        | 35392          | 3.92         | A        | 135241         | 0.00        | P        | 91.080      |
| 637                           | Sheathing                                     | 696            | 0.19        | 6535           | 0.67         | A        | 23226          | 0.00        | S        | -41.760     |
| 641                           | Officer Berthing                              | 848            | 0.17        | 5985           | 0.54         | A        | 18726          | -0.01       | S        | -202.973    |
| 642                           | Enlisted Berthing                             | 948            | 0.19        | 6714           | 0.23         | A        | 8037           | 0.00        | C        | 0.000       |
| 643                           | Mess Spaces                                   | 692            | 0.11        | 3728           | 0.33         | A        | 11252          | 0.11        | P        | 3868.696    |
| 644                           | Sanitary Spaces & Fixtures                    | 146            | 0.02        | 719            | 0.13         | A        | 4440           | -0.02       | S        | -726.701    |
| 651                           | Commissary Spaces                             | 1137           | 0.23        | 7901           | 0.53         | A        | 18438          | -0.11       | S        | -3814.488   |
| 652                           | Medical Spaces                                | 22             | 0.01        | 209            | 0.02         | A        | 726            | 0.00        | C        | 0.000       |
| 661                           | Offices Furniture                             | 778            | 0.32        | 11025          | 0.61         | A        | 21050          | 0.03        | P        | 1136.388    |
| 665                           | Workshop                                      | 400            | 0.08        | 2709           | 0.36         | A        | 12480          | 0.08        | P        | 2933.333    |
| 671                           | Lockers & Special Stowage                     | 452            | 0.12        | 4211           | 0.49         | A        | 16810          | -0.05       | S        | -1683.487   |
| 699                           | Outfit and Furnishing Parts and Special Tools | 784            | 0.16        | 5472           | 0.74         | A        | 25402          | 0.00        | C        | 0.000       |
|                               |                                               |                |             |                |              |          |                |             |          |             |
| <b>Light Ship, No Margins</b> |                                               | <b>182643</b>  | <b>7.46</b> | <b>1362512</b> | <b>34.56</b> | <b>A</b> | <b>6311343</b> | <b>0.01</b> | <b>P</b> | <b>2207</b> |
|                               |                                               |                |             |                |              |          |                |             |          |             |

| SWBS #               | SWBS Name                                    | Applied Weight | Vertical |         | Longitudinal |     |         | Transverse |      |             |
|----------------------|----------------------------------------------|----------------|----------|---------|--------------|-----|---------|------------|------|-------------|
|                      |                                              |                | VCG ABL  | VMOM    | LCG          | F/A | LMOM    | TCG        | P/C/ | TMOM        |
|                      |                                              | lb             | ft       | ft*lb   | ft           |     | ft*lb   | ft         | S    | ft*lb       |
| M                    | Margins                                      | 29769          | 12.09    | 359935  | 34.56        | A   | 1028686 | 0.01       | S    | 359.7024585 |
| M11                  | Design Margin Weight (10.3%)                 | 18810          | 7.46     | 140325  | 34.56        | A   | 650005  | 0.01       | S    | 227.288522  |
| M11                  | Design Margin KG (5.0%)                      | 0              | 0.37     | 0       | 0.00         |     | 0       |            |      | 0           |
| M21                  | Construction Margin Weight (5.0%)            | 9132           | 7.46     | 68126   | 34.56        | A   | 315567  | 0.01       | S    | 110.3449471 |
| M21                  | Construction Margin KG (3.0%)                | 0              | 0.22     | 0       | 0.00         |     | 0       |            |      | 0           |
| M31                  | Contract Modification Weight (0.7%)          | 1279           | 7.46     | 9538    | 34.56        | A   | 44179   | 0.01       | S    | 15.44829259 |
| M31                  | Contract Modification KG (0.5%)              | 0              | 0.04     | 0       | 0.00         |     | 0       |            |      | 0           |
| M41                  | Government Furnished Equipment Weight (0.3%) | 548            | 7.46     | 4088    | 34.56        | A   | 18934   | 0.01       | S    | 6.620696826 |
| M41                  | Government Furnished Equipment KG (0.2%)     | 0              | 0.01     | 0       | 0.00         |     | 0       |            |      | 0           |
|                      |                                              |                |          |         |              |     |         |            |      |             |
| Light Ship Condition |                                              | 212412         | 8.11     | 1722447 | 34.56        | A   | 7340029 | 0.01       | S    | 2566.6014   |

| SWBS #                                      | SWBS Name                 | Applied Weight | Vertical    |                | Longitudinal |          |                | Transverse  |          |                   |
|---------------------------------------------|---------------------------|----------------|-------------|----------------|--------------|----------|----------------|-------------|----------|-------------------|
|                                             |                           |                | VCG ABL     | VMOM           | LCG          | F/A      | LMOM           | TCG         | P/C/     | TMOM              |
|                                             |                           | lb             | ft          | ft*lb          | ft           |          | ft*lb          | ft          | S        | ft*lb             |
| A                                           | Departure Loads W/ AtoN   | 57483          | 6.20        | 356504         | 32.01        | A        | 1839896        | 0.04        | S        | -2075.638         |
| A11                                         | Ship's Officers           | 0              | 0.00        | 0              | 0.00         |          | 0              | 0.00        | C        | 0                 |
| A13                                         | Ship's Enlisted Men       | 7920           | 1.59        | 54833          | 2.78         | A        | 96154          | -0.03       | S        | -1126.4           |
| A29                                         | Special mission Sys.      | 0              | 0.00        | 0              | 0.00         |          | 0              | 0.00        | C        | 0                 |
| A31                                         | Provisions & Pers. Stores | 217            | 0.05        | 1630           | 0.09         | A        | 3179           | -0.03       | S        | -1116.158153      |
| A32                                         | General Stores            | 1750           | 0.34        | 11830          | 0.43         | A        | 14875          | 0.00        | C        | 0                 |
| A41                                         | Diesel Fuel               | 21207          | 1.31        | 45084          | 18.71        | A        | 645965         | 0.00        | P        | 166.92            |
| A43                                         | Gasoline                  | 0              | 0.00        | 0              | 0.00         |          | 0              | 0.00        | C        | 0                 |
| A46                                         | Lubricating Oil           | 0              | 0.00        | 0              | 0.00         |          | 0              | 0.00        | C        | 0                 |
| A51                                         | Sea Water                 | 0              | 0.00        | 0              | 0.00         |          | 0              | 0.00        | C        | 0                 |
| A52                                         | Fresh Water               | 8160           | 1.76        | 60874          | 7.19         | A        | 248390         | 0.00        | C        | 0                 |
| A54                                         | Hydraulic Fluid           | 1400           | 0.16        | 5532           | 2.06         | A        | 71296          | 0.00        | C        | 0                 |
| A55                                         | Sanitary Tank Liquid      | 489            | 0.01        | 384            | 0.27         | A        | 9416           | 0.00        | C        | 0                 |
| A62                                         | Cargo Stores              | 0              | 0.00        | 0              | 0.00         |          | 0              | 0.00        | C        | 0                 |
| A69                                         | Miscellaneous Cargo       | 16340          | 5.11        | 176338         | 21.74        | A        | 750620         | 0.00        | C        | 0                 |
| <b>Design Departure Loads W/ AtoN, BOSL</b> |                           | <b>269895</b>  | <b>7.70</b> | <b>2078951</b> | <b>34.01</b> | <b>A</b> | <b>9179925</b> | <b>0.00</b> | <b>P</b> | <b>491</b>        |
| Service Life Allowance (7.5%)               |                           | <b>15931</b>   | <b>8.61</b> | <b>137149</b>  | <b>34.56</b> | <b>A</b> | <b>550502</b>  | <b>0.01</b> | <b>P</b> | <b>192.495105</b> |
| <b>Design Departure Loads W/ AtoN, EOSL</b> |                           | <b>285826</b>  | <b>7.75</b> | <b>2216100</b> | <b>34.04</b> | <b>A</b> | <b>9730428</b> | <b>0.00</b> | <b>P</b> | <b>683</b>        |

| SWBS #                                       | SWBS Name                 | Applied Weight | Vertical    |                | Longitudinal |          |                | Transverse  |          |                   |
|----------------------------------------------|---------------------------|----------------|-------------|----------------|--------------|----------|----------------|-------------|----------|-------------------|
|                                              |                           |                | VCG ABL     | VMOM           | LCG          | F/A      | LMOM           | TCG         | P/C/     | TMOM              |
|                                              |                           | lb             | ft          | ft*lb          | ft           |          | ft*lb          | ft          | S        | ft*lb             |
| B                                            | Departure Loads W/O AtoN  | 33223          | 3.77        | 125333         | 29.89        | A        | 993122         | 0.03        | S        | -949.238          |
| B11                                          | Ship's Officers           | 0              | 0.00        | 0              | 0.00         |          | 0              | 0.00        | C        | 0                 |
| B13                                          | Ship's Enlisted Men       | 0              | 0.00        | 0              | 0.00         |          | 0              | 0.00        | C        | 0                 |
| B29                                          | Special mission Sys.      | 0              | 0.00        | 0              | 0.00         |          | 0              | 0.00        | C        | 0                 |
| B31                                          | Provisions & Pers. Stores | 217            | 0.05        | 1630           | 0.09         | A        | 3179           | -0.03       | S        | -1116.158153      |
| B32                                          | General Stores            | 1750           | 0.34        | 11830          | 0.43         | A        | 14875          | 0.00        | C        | 0                 |
| B41                                          | Diesel Fuel               | 21207          | 1.31        | 45084          | 18.71        | A        | 645965         | 0.00        | P        | 166.92            |
| B43                                          | Gasoline                  | 0              | 0.00        | 0              | 0.00         |          | 0              | 0.00        | C        | 0                 |
| B46                                          | Lubricating Oil           | 0              | 0.00        | 0              | 0.00         |          | 0              | 0.00        | C        | 0                 |
| B51                                          | Sea Water                 | 0              | 0.00        | 0              | 0.00         |          | 0              | 0.00        | C        | 0                 |
| B52                                          | Fresh Water               | 8160           | 1.76        | 60874          | 7.19         | A        | 248390         | 0.00        | C        | 0                 |
| B54                                          | Hydraulic Fluid           | 1400           | 0.16        | 5532           | 2.06         | A        | 71296          | 0.00        | C        | 0                 |
| B55                                          | Sanitary Tank Liquid      | 489            | 0.01        | 384            | 0.27         | A        | 9416           | 0.00        | C        | 0                 |
| B62                                          | Cargo Stores              | 0              | 0.00        | 0              | 0.00         |          | 0              | 0.00        | C        | 0                 |
| B69                                          | Miscellaneous Cargo       | 0              | 0.00        | 0              | 0.00         |          | 0              | 0.00        | C        | 0                 |
| <b>Design Departure Loads W/O AtoN, BOSL</b> |                           | <b>245635</b>  | <b>7.52</b> | <b>1847780</b> | <b>33.92</b> | <b>A</b> | <b>8333151</b> | <b>0.01</b> | <b>P</b> | <b>1617</b>       |
| Service Life Allowance (7.5%)                |                           | <b>15931</b>   | <b>8.61</b> | <b>137149</b>  | <b>34.56</b> | <b>A</b> | <b>550502</b>  | <b>0.01</b> | <b>P</b> | <b>192.495105</b> |
| <b>Design Departure Loads W/O AtoN, EOSL</b> |                           | <b>261566</b>  | <b>7.59</b> | <b>1984929</b> | <b>33.96</b> | <b>A</b> | <b>8883653</b> | <b>0.01</b> | <b>P</b> | <b>1810</b>       |

| SWBS #                                 | SWBS Name                 | Applied Weight | Vertical |         | Longitudinal |     |         | Transverse |      |              |
|----------------------------------------|---------------------------|----------------|----------|---------|--------------|-----|---------|------------|------|--------------|
|                                        |                           |                | VCG ABL  | VMOM    | LCG          | F/A | LMOM    | TCG        | P/C/ | TMOM         |
|                                        |                           | lb             | ft       | ft*lb   | ft           |     | ft*lb   | ft         | S    | ft*lb        |
|                                        |                           |                |          |         |              |     |         |            |      |              |
| C                                      | Half Loads W/ AtoN        | 37406          | 6.43     | 240611  | 35.39        | A   | 1323729 | 0.03       | S    | -1073.108153 |
|                                        |                           |                |          |         |              |     |         |            |      |              |
| C11                                    | Ship's Officers           | 0              | 0.00     | 0       | 0.00         |     | 0       | 0.00       | C    | 0            |
| C13                                    | Ship's Enlisted Men       | 0              | 0.00     | 0       | 0.00         |     | 0       | 0.00       | C    | 0            |
| C29                                    | Special mission Sys.      | 0              | 0.00     | 0       | 0.00         |     | 0       | 0.00       | C    | 0            |
| C31                                    | Provisions & Pers. Stores | 217            | 0.05     | 1630    | 0.09         | A   | 3179    | -0.03      | S    | -1116.158153 |
| C32                                    | General Stores            | 1750           | 0.34     | 11830   | 0.43         | A   | 14875   | 0.00       | C    | 0            |
| C41                                    | Diesel Fuel               | 8522           | 0.35     | 11956   | 7.51         | A   | 259436  | 0.00       | P    | 43.05        |
| C43                                    | Gasoline                  | 0              | 0.00     | 0       | 0.00         |     | 0       | 0.00       | C    | 0            |
| C46                                    | Lubricating Oil           | 0              | 0.00     | 0       | 0.00         |     | 0       | 0.00       | C    | 0            |
| C51                                    | Sea Water                 | 0              | 0.00     | 0       | 0.00         |     | 0       | 0.00       | C    | 0            |
| C52                                    | Fresh Water               | 4294           | 0.75     | 25936   | 3.78         | A   | 130581  | 0.00       | C    | 0            |
| C54                                    | Hydraulic Fluid           | 1400           | 0.16     | 5532    | 2.06         | A   | 71296   | 0.00       | C    | 0            |
| C55                                    | Sanitary Tank Liquid      | 4883           | 0.21     | 7389    | 2.71         | A   | 93742   | 0.00       | C    | 0            |
| C62                                    | Cargo Stores              | 0              | 0.00     | 0       | 0.00         |     | 0       | 0.00       | C    | 0            |
| C69                                    | Miscellaneous Cargo       | 16340          | 5.11     | 176338  | 21.74        | A   | 750620  | 0.00       | C    | 0            |
|                                        |                           |                |          |         |              |     |         |            |      |              |
| <b>Design Half Loads W/ AtoN, BOSL</b> |                           | 249818         | 7.86     | 1963057 | 34.68        | A   | 8663758 | 0.01       | P    | 1493.493248  |
| Service Life Allowance (7.5%)          |                           | 15931          | 8.61     | 137149  | 34.56        | A   | 550502  | 0.01       | P    | 192.495105   |
| <b>Design Half Loads W/ AtoN, EOSL</b> |                           | 265749         | 7.90     | 2100206 | 34.67        | A   | 9214261 | 0.01       | P    | 1685.988353  |

| SWBS #                                  | SWBS Name                 | Applied Weight | Vertical    |                | Longitudinal |          |                | Transverse  |          |                    |
|-----------------------------------------|---------------------------|----------------|-------------|----------------|--------------|----------|----------------|-------------|----------|--------------------|
|                                         |                           |                | VCG ABL     | VMOM           | LCG          | F/A      | LMOM           | TCG         | P/C/     | TMOM               |
|                                         |                           | lb             | ft          | ft*lb          | ft           |          | ft*lb          | ft          | S        | ft*lb              |
| D                                       | Half Loads W/O AtoN       | 21066          | 3.05        | 64273          | 27.21        | A        | 573109         | 0.05        | S        | -1073.108153       |
| D11                                     | Ship's Officers           | 0              | 0.00        | 0              | 0.00         |          | 0              | 0.00        | C        | 0                  |
| D13                                     | Ship's Enlisted Men       | 0              | 0.00        | 0              | 0.00         |          | 0              | 0.00        | C        | 0                  |
| D29                                     | Special mission Sys.      | 0              | 0.00        | 0              | 0.00         |          | 0              | 0.00        | C        | 0                  |
| D31                                     | Provisions & Pers. Stores | 217            | 0.05        | 1630           | 0.09         | A        | 3179           | -0.03       | S        | -1116.158153       |
| D32                                     | General Stores            | 1750           | 0.34        | 11830          | 0.43         | A        | 14875          | 0.00        | C        | 0                  |
| D41                                     | Diesel Fuel               | 8522           | 0.35        | 11956          | 7.51         | A        | 259436         | 0.00        | P        | 43.05              |
| D43                                     | Gasoline                  | 0              | 0.00        | 0              | 0.00         |          | 0              | 0.00        | C        | 0                  |
| D46                                     | Lubricating Oil           | 0              | 0.00        | 0              | 0.00         |          | 0              | 0.00        | C        | 0                  |
| D51                                     | Sea Water                 | 0              | 0.00        | 0              | 0.00         |          | 0              | 0.00        | C        | 0                  |
| D52                                     | Fresh Water               | 4294           | 0.75        | 25936          | 3.78         | A        | 130581         | 0.00        | C        | 0                  |
| D54                                     | Hydraulic Fluid           | 1400           | 0.16        | 5532           | 2.06         | A        | 71296          | 0.00        | C        | 0                  |
| D55                                     | Sanitary Tank Liquid      | 4883           | 0.21        | 7389           | 2.71         | A        | 93742          | 0.00        | C        | 0                  |
| D62                                     | Cargo Stores              | 0              | 0.00        | 0              | 0.00         |          | 0              | 0.00        | C        | 0                  |
| D69                                     | Miscellaneous Cargo       | 0              | 0.00        | 0              | 0.00         |          | 0              | 0.00        | C        | 0                  |
| <b>Design Half Loads W/O AtoN, BOSL</b> |                           | <b>233478</b>  | <b>7.65</b> | <b>1786719</b> | <b>33.89</b> | <b>A</b> | <b>7913138</b> | <b>0.01</b> | <b>P</b> | <b>1493.493248</b> |
| Service Life Allowance (7.5%)           |                           | <b>15931</b>   | <b>8.61</b> | <b>137149</b>  | <b>34.56</b> | <b>A</b> | <b>550502</b>  | <b>0.01</b> | <b>P</b> | <b>192.495105</b>  |
| <b>Design Half Loads W/O AtoN, EOSL</b> |                           | <b>249409</b>  | <b>7.71</b> | <b>1923868</b> | <b>33.93</b> | <b>A</b> | <b>8463641</b> | <b>0.01</b> | <b>P</b> | <b>1685.988353</b> |

| SWBS #                                   | SWBS Name                 | Applied Weight | Vertical    |                | Longitudinal |          |                | Transverse  |          |                    |
|------------------------------------------|---------------------------|----------------|-------------|----------------|--------------|----------|----------------|-------------|----------|--------------------|
|                                          |                           |                | VCG ABL     | VMOM           | LCG          | F/A      | LMOM           | TCG         | P/C/     | TMOM               |
|                                          |                           | lb             | ft          | ft*lb          | ft           |          | ft*lb          | ft          | S        | ft*lb              |
| E                                        | Min Op Loads W/ AtoN      | 31647          | 6.88        | 217711         | 34.17        | A        | 1081257        | 0.03        | S        | -1100.888153       |
| E11                                      | Ship's Officers           | 0              | 0.00        | 0              | 0.00         |          | 0              | 0.00        | C        | 0                  |
| E13                                      | Ship's Enlisted Men       | 0              | 0.00        | 0              | 0.00         |          | 0              | 0.00        | C        | 0                  |
| E29                                      | Special mission Sys.      | 0              | 0.00        | 0              | 0.00         |          | 0              | 0.00        | C        | 0                  |
| E31                                      | Provisions & Pers. Stores | 217            | 0.05        | 1630           | 0.09         | A        | 3179           | -0.03       | S        | -1116.158153       |
| E32                                      | General Stores            | 1750           | 0.34        | 11830          | 0.43         | A        | 14875          | 0.00        | C        | 0                  |
| E41                                      | Diesel Fuel               | 2234           | 0.06        | 1972           | 1.50         | A        | 51674          | 0.00        | P        | 15.27              |
| E43                                      | Gasoline                  | 0              | 0.00        | 0              | 0.00         |          | 0              | 0.00        | C        | 0                  |
| E46                                      | Lubricating Oil           | 0              | 0.00        | 0              | 0.00         |          | 0              | 0.00        | C        | 0                  |
| E51                                      | Sea Water                 | 0              | 0.00        | 0              | 0.00         |          | 0              | 0.00        | C        | 0                  |
| E52                                      | Fresh Water               | 430            | 0.05        | 1862           | 0.38         | A        | 12965          | 0.00        | C        | 0                  |
| E54                                      | Hydraulic Fluid           | 1400           | 0.16        | 5532           | 2.06         | A        | 71296          | 0.00        | C        | 0                  |
| E55                                      | Sanitary Tank Liquid      | 9276           | 0.54        | 18547          | 5.12         | A        | 176649         | 0.00        | C        | 0                  |
| E62                                      | Cargo Stores              | 0              | 0.00        | 0              | 0.00         |          | 0              | 0.00        | C        | 0                  |
| E69                                      | Miscellaneous Cargo       | 16340          | 5.11        | 176338         | 21.74        | A        | 750620         | 0.00        | C        | 0                  |
| <b>Design Min Op Loads W/ AtoN, BOSL</b> |                           | <b>244059</b>  | <b>7.95</b> | <b>1940158</b> | <b>34.51</b> | <b>A</b> | <b>8421287</b> | <b>0.01</b> | <b>P</b> | <b>1465.713248</b> |
| Service Life Allowance (7.5%)            |                           | <b>15931</b>   | <b>8.61</b> | <b>137149</b>  | <b>34.56</b> | <b>A</b> | <b>550502</b>  | <b>0.01</b> | <b>P</b> | <b>192.495105</b>  |
| <b>Design Min Op Loads W/ AtoN, EOSL</b> |                           | <b>259990</b>  | <b>7.99</b> | <b>2077307</b> | <b>34.51</b> | <b>A</b> | <b>8971789</b> | <b>0.01</b> | <b>P</b> | <b>1658.208353</b> |

| SWBS #                                    | SWBS Name                 | Applied Weight | Vertical    |                | Longitudinal |          |                | Transverse  |          |                    |
|-------------------------------------------|---------------------------|----------------|-------------|----------------|--------------|----------|----------------|-------------|----------|--------------------|
|                                           |                           |                | VCG ABL     | VMOM           | LCG          | F/A      | LMOM           | TCG         | P/C/     | TMOM               |
|                                           |                           | lb             | ft          | ft*lb          | ft           |          | ft*lb          | ft          | S        | ft*lb              |
| F                                         | Min Op Loads W/O AtoN     | 15307          | 2.70        | 41373          | 21.60        | A        | 330637         | 0.07        | S        | -1100.888153       |
| F11                                       | Ship's Officers           | 0              | 0.00        | 0              | 0.00         |          | 0              | 0.00        | C        | 0                  |
| F13                                       | Ship's Enlisted Men       | 0              | 0.00        | 0              | 0.00         |          | 0              | 0.00        | C        | 0                  |
| F29                                       | Special mission Sys.      | 0              | 0.00        | 0              | 0.00         |          | 0              | 0.00        | C        | 0                  |
| F31                                       | Provisions & Pers. Stores | 217            | 0.05        | 1630           | 0.09         | A        | 3179           | -0.03       | S        | -1116.158153       |
| F32                                       | General Stores            | 1750           | 0.34        | 11830          | 0.43         | A        | 14875          | 0.00        | C        | 0                  |
| F41                                       | Diesel Fuel               | 2234           | 0.06        | 1972           | 1.50         | A        | 51674          | 0.00        | P        | 15.27              |
| F43                                       | Gasoline                  | 0              | 0.00        | 0              | 0.00         |          | 0              | 0.00        | C        | 0                  |
| F46                                       | Lubricating Oil           | 0              | 0.00        | 0              | 0.00         |          | 0              | 0.00        | C        | 0                  |
| F51                                       | Sea Water                 | 0              | 0.00        | 0              | 0.00         |          | 0              | 0.00        | C        | 0                  |
| F52                                       | Fresh Water               | 430            | 0.05        | 1862           | 0.38         | A        | 12965          | 0.00        | C        | 0                  |
| F54                                       | Hydraulic Fluid           | 1400           | 0.16        | 5532           | 2.06         | A        | 71296          | 0.00        | C        | 0                  |
| F55                                       | Sanitary Tank Liquid      | 9276           | 0.54        | 18547          | 5.12         | A        | 176649         | 0.00        | C        | 0                  |
| F62                                       | Cargo Stores              | 0              | 0.00        | 0              | 0.00         |          | 0              | 0.00        | C        | 0                  |
| F69                                       | Miscellaneous Cargo       | 0              | 0.00        | 0              | 0.00         |          | 0              | 0.00        | C        | 0                  |
| <b>Design Min Op Loads W/O AtoN, BOSL</b> |                           | <b>227719</b>  | <b>7.75</b> | <b>1763820</b> | <b>33.68</b> | <b>A</b> | <b>7670667</b> | <b>0.01</b> | <b>P</b> | <b>1465.713248</b> |
| Service Life Allowance (7.5%)             |                           | 15931          | 8.61        | 137149         | 34.56        | A        | 550502         | 0.01        | P        | 192.495105         |
| <b>Design Min Op Loads W/O AtoN, EOSL</b> |                           | <b>243650</b>  | <b>7.80</b> | <b>1900969</b> | <b>33.74</b> | <b>A</b> | <b>8221169</b> | <b>0.01</b> | <b>P</b> | <b>1658.208353</b> |

| SWBS # | ITEM | DESCRIPTION             | UNIT WEIGHT | NO. UNITS | CURRENT WEIGHT | E C A | VERTICAL |        | LONGITUDINAL |   |        | TRANSVERSE |   |        |
|--------|------|-------------------------|-------------|-----------|----------------|-------|----------|--------|--------------|---|--------|------------|---|--------|
|        |      |                         |             |           |                |       | VCG      | MOMENT | LCG          | S | MOMENT | TCG        | S | MOMENT |
|        |      |                         | lb          | #         | lb             |       | ft       | ft*lb  | ft           |   | ft*lb  | ft         |   | ft*lb  |
| 11101  | 1    | PLATE, 7/16 IN, STL     | 17.85       | 255.11    | 4553.71        | C     | 9.17     | 41758  | 11.75        | A | 53506  | 0.00       | C | 0      |
| 11101  | 2    | PLATE, 7/16 IN, STL     | 17.85       | 350.01    | 6247.68        | C     | 4.92     | 30739  | 12.78        | A | 79845  | 0.00       | C | 0      |
| 11101  | 3    | PLATE, 7/16 IN, STL     | 17.85       | 301.67    | 5384.81        | C     | 1.36     | 7323   | 19.61        | A | 105596 | 0.00       | C | 0      |
| 11101  | 4    | PLATE, 7/16 IN, STL     | 17.85       | 52.34     | 934.27         | C     | 9.16     | 8558   | 31.48        | A | 29411  | 0.00       | C | 0      |
| 11101  | 5    | PLATE, 7/16 IN, STL     | 17.85       | 66.41     | 1185.42        | C     | 4.98     | 5903   | 31.48        | A | 37317  | 0.00       | C | 0      |
| 11101  | 6    | PLATE, 7/16 IN, STL     | 17.85       | 143.58    | 2562.90        | C     | 1.07     | 2742   | 31.48        | A | 80680  | 0.00       | C | 0      |
| 11102  | 7    | PLATE, 5/16 IN, STL     | 12.75       | 133.05    | 1696.39        | C     | 8.16     | 13843  | 53.87        | A | 91384  | 0.00       | C | 0      |
| 11102  | 8    | PLATE, 5/16 IN, STL     | 12.75       | 502.54    | 6407.39        | C     | 4.95     | 31717  | 54.12        | A | 346768 | 0.00       | C | 0      |
| 11102  | 9    | PLATE, 5/16 IN, STL     | 12.75       | 318.17    | 4056.67        | C     | 1.35     | 5477   | 43.33        | A | 175775 | 0.00       | C | 0      |
| 11103  | 10   | PLATE, 1/4 IN, STL      | 10.20       | 95.91     | 978.28         | C     | 11.82    | 11565  | 17.05        | A | 16676  | 0.00       | C | 0      |
| 11103  | 11   | PLATE, 1/4 IN, STL      | 195.55      | 1.00      | 195.55         | C     | 11.79    | 2306   | 16.34        | A | 3196   | 0.00       | C | 0      |
| 11104  | 12   | PIPE, 2 IN, SCH 40      | 3.65        | 89.15     | 325.40         | C     | 12.46    | 4054   | 15.90        | A | 5173   | 0.00       | C | 0      |
| 11401  | 1    | SKEG                    | 1015.81     | 2.00      | 2031.62        | E     | 1.70     | 3454   | 55.47        | A | 112694 | 0.00       | C | 0      |
| 11501  | 1    | STANCHION, 5 IN, STL    | 14.60       | 6.23      | 90.96          | C     | 5.34     | 485    | 44.05        | A | 4007   | 5.63       | S | -512   |
| 11501  | 2    | STANCHION, 5 IN, STL    | 14.60       | 6.23      | 90.96          | C     | 5.34     | 485    | 44.05        | A | 4007   | 5.63       | P | 512    |
| 11501  | 3    | STANCHION, 3.5 IN, STL  | 9.12        | 5.04      | 45.96          | C     | 5.95     | 273    | 54.50        | A | 2505   | 0.00       | C | 0      |
| 11501  | 4    | STANCHION, 3.5 IN, STL  | 9.12        | 4.23      | 38.58          | C     | 6.34     | 244    | 54.50        | A | 2102   | 6.00       | S | -231   |
| 11501  | 5    | STANCHION, 3.5 IN, STL  | 9.12        | 4.23      | 38.58          | C     | 6.34     | 244    | 54.50        | A | 2102   | 6.00       | P | 231    |
| 11601  | 1    | 8 X 4 X 10 I-T, STL     | 7.19        | 10.38     | 74.62          | C     | 5.69     | 425    | 1.42         | A | 106    | 0.00       | C | 0      |
| 11602  | 2    | 8 X 5 1/2 X 21 I-T, STL | 13.87       | 9.67      | 134.12         | C     | 2.29     | 307    | 8.49         | A | 1139   | 0.00       | C | 0      |
| 11602  | 3    | 8 X 5 1/2 X 21 I-T, STL | 13.87       | 9.77      | 135.45         | C     | 3.45     | 467    | 55.26        | A | 7485   | 0.00       | C | 0      |
| 11603  | 4    | 8 X 5 1/2 X 13 T, STL   | 12.83       | 13.00     | 166.79         | C     | 0.52     | 87     | 19.49        | A | 3251   | 0.00       | C | 0      |
| 11603  | 5    | 8 X 5 1/2 X 13 T, STL   | 12.83       | 9.00      | 115.47         | C     | 0.48     | 55     | 30.50        | A | 3522   | 0.00       | C | 0      |
| 11603  | 6    | 8 X 5 1/2 X 13 T, STL   | 12.83       | 15.04     | 192.96         | C     | 0.56     | 108    | 42.51        | A | 8203   | 0.00       | C | 0      |
| 11603  | 7    | 8 X 5 1/2 X 13 T, STL   | 12.83       | 30.10     | 386.18         | C     | 1.84     | 711    | 42.51        | A | 16417  | 0.00       | C | 0      |
| 11604  | 8    | 8 X 4 X 10 I-T, STL     | 7.19        | 10.12     | 72.74          | C     | 5.29     | 385    | 63.01        | A | 4583   | 0.00       | C | 0      |
| 11605  | 9    | 7 X 5 X 11 T, STL       | 10.81       | 18.98     | 205.17         | C     | 4.21     | 864    | 8.50         | A | 1744   | 0.00       | C | 0      |
| 11605  | 10   | 7 X 5 X 11 T, STL       | 10.81       | 26.18     | 283.01         | C     | 2.15     | 608    | 19.49        | A | 5516   | 0.00       | C | 0      |
| 11605  | 11   | 7 X 5 X 11 T, STL       | 10.81       | 18.68     | 201.89         | C     | 4.13     | 834    | 55.26        | A | 11157  | 0.00       | C | 0      |

| SWBS # | ITEM | DESCRIPTION        | UNIT WEIGHT | NO. UNITS | CURRENT WEIGHT | E | VERTICAL |        | LONGITUDINAL |   |        | TRANSVERSE |   |        |
|--------|------|--------------------|-------------|-----------|----------------|---|----------|--------|--------------|---|--------|------------|---|--------|
|        |      |                    |             |           |                |   | VCG      | MOMENT | LCG          | S | MOMENT | TCG        | S | MOMENT |
|        |      |                    | lb          | #         | lb             | A | ft       | ft*lb  | ft           |   | ft*lb  | ft         |   | ft*lb  |
| 11605  |      |                    |             |           |                |   |          |        |              |   |        |            |   |        |
| 11606  | 12   | 6 X 4 X 9.5 T, STL | 9.34        | 30.10     | 281.13         | C | 0.98     | 274    | 42.52        | A | 11955  | 0.00       | C | 0      |
| 11606  |      |                    |             |           |                | C |          |        |              |   |        |            |   |        |
| 11607  | 13   | 6 X 4 X 7 T, STL   | 6.94        | 21.53     | 149.44         | C | 5.23     | 782    | 62.42        | A | 9328   | 0.00       | C | 0      |
| 11602  |      |                    |             |           |                |   |          |        |              |   |        |            |   |        |
| 11603  | 14   | 2 X 1/4 FB, STL    | 1.70        | 3.15      | 5.35           | C | 9.78     | 52     | 0.54         | F | -3     | 1.49       | P | 8      |
| 11603  | 15   | 2 X 1/4 FB, STL    | 1.70        | 3.06      | 5.20           | C | 9.78     | 51     | 0.45         | F | -2     | 1.46       | P | 8      |
| 11603  | 16   | 2 X 1/4 FB, STL    | 1.70        | 3.12      | 5.30           | C | 8.22     | 44     | 0.33         | F | -2     | 1.38       | P | 7      |
| 11603  | 17   | 2 X 1/4 FB, STL    | 1.70        | 3.03      | 5.15           | C | 8.22     | 42     | 0.24         | F | -1     | 1.36       | P | 7      |
| 11603  | 18   | 2 X 1/4 FB, STL    | 1.70        | 3.15      | 5.35           | C | 9.78     | 52     | 0.54         | F | -3     | 1.49       | S | -8     |
| 11603  | 19   | 2 X 1/4 FB, STL    | 1.70        | 3.06      | 5.20           | C | 9.78     | 51     | 0.45         | F | -2     | 1.46       | S | -8     |
| 11603  | 20   | 2 X 1/4 FB, STL    | 1.70        | 3.12      | 5.30           | C | 8.22     | 44     | 0.33         | F | -2     | 1.38       | S | -7     |
| 11603  | 21   | 2 X 1/4 FB, STL    | 1.70        | 3.03      | 5.15           | C | 8.22     | 42     | 0.24         | F | -1     | 1.36       | S | -7     |
|        |      |                    |             |           |                |   |          |        |              |   |        |            |   |        |
| 11701  | 1    | 5 X 4 X 6 T, STL   | 5.88        | 4.81      | 28.30          | C | 8.99     | 254    | 0.48         | F | -14    | 0.96       | P | 27     |
| 11701  | 2    | 5 X 4 X 6 T, STL   | 5.88        | 1.77      | 10.42          | C | 6.26     | 65     | 0.32         | A | 3      | 0.46       | P | 5      |
| 11701  | 3    | 5 X 4 X 6 T, STL   | 5.88        | 5.02      | 29.54          | C | 8.88     | 262    | 0.01         | A | 0      | 1.84       | P | 54     |
| 11701  | 4    | 5 X 4 X 6 T, STL   | 5.88        | 2.71      | 15.96          | C | 5.88     | 94     | 1.22         | A | 20     | 0.84       | P | 13     |
| 11701  | 5    | 5 X 4 X 6 T, STL   | 5.88        | 5.28      | 31.07          | C | 8.78     | 273    | 0.59         | A | 18     | 2.63       | P | 82     |
| 11701  | 6    | 5 X 4 X 6 T, STL   | 5.88        | 3.36      | 19.77          | C | 5.56     | 110    | 2.03         | A | 40     | 1.13       | P | 22     |
| 11701  | 7    | 5 X 4 X 6 T, STL   | 5.88        | 5.54      | 32.56          | C | 8.67     | 282    | 1.23         | A | 40     | 3.33       | P | 108    |
| 11701  | 8    | 5 X 4 X 6 T, STL   | 5.88        | 3.96      | 23.28          | C | 5.26     | 122    | 2.83         | A | 66     | 1.38       | P | 32     |
| 11701  | 9    | 5 X 4 X 6 T, STL   | 5.88        | 5.79      | 34.03          | C | 8.50     | 289    | 1.96         | A | 67     | 3.95       | P | 134    |
| 11701  | 10   | 5 X 4 X 6 T, STL   | 5.88        | 3.41      | 20.02          | C | 5.25     | 105    | 3.37         | A | 67     | 2.04       | P | 41     |
| 11701  | 11   | 5 X 4 X 6 T, STL   | 5.88        | 6.01      | 35.36          | C | 8.39     | 297    | 2.72         | A | 96     | 4.53       | P | 160    |
| 11701  | 12   | 5 X 4 X 6 T, STL   | 5.88        | 1.69      | 9.93           | C | 5.48     | 54     | 3.74         | A | 37     | 3.07       | P | 30     |
| 11701  | 13   | 5 X 4 X 6 T, STL   | 5.88        | 5.73      | 33.71          | C | 8.79     | 296    | 3.39         | A | 114    | 5.27       | P | 178    |
| 11701  | 14   | 5 X 4 X 6 T, STL   | 5.88        | 4.81      | 28.30          | C | 8.99     | 254    | 0.48         | F | -14    | 0.96       | S | -27    |
| 11701  | 15   | 5 X 4 X 6 T, STL   | 5.88        | 1.77      | 10.42          | C | 6.26     | 65     | 0.32         | A | 3      | 0.46       | S | -5     |
| 11701  | 16   | 5 X 4 X 6 T, STL   | 5.88        | 5.02      | 29.54          | C | 8.88     | 262    | 0.01         | A | 0      | 1.84       | S | -54    |
| 11701  | 17   | 5 X 4 X 6 T, STL   | 5.88        | 2.71      | 15.96          | C | 5.88     | 94     | 1.22         | A | 20     | 0.84       | S | -13    |
| 11701  | 18   | 5 X 4 X 6 T, STL   | 5.88        | 5.28      | 31.07          | C | 8.78     | 273    | 0.59         | A | 18     | 2.63       | S | -82    |
| 11701  | 19   | 5 X 4 X 6 T, STL   | 5.88        | 3.36      | 19.77          | C | 5.56     | 110    | 2.03         | A | 40     | 1.13       | S | -22    |
| 11701  | 20   | 5 X 4 X 6 T, STL   | 5.88        | 5.54      | 32.56          | C | 8.67     | 282    | 1.23         | A | 40     | 3.33       | S | -108   |
| 11701  | 21   | 5 X 4 X 6 T, STL   | 5.88        | 3.96      | 23.28          | C | 5.26     | 122    | 2.83         | A | 66     | 1.38       | S | -32    |
| 11701  | 22   | 5 X 4 X 6 T, STL   | 5.88        | 5.79      | 34.03          | C | 8.50     | 289    | 1.96         | A | 67     | 3.95       | S | -134   |
| 11701  | 23   | 5 X 4 X 6 T, STL   | 5.88        | 3.41      | 20.02          | C | 5.25     | 105    | 3.37         | A | 67     | 2.04       | S | -41    |
| 11701  | 24   | 5 X 4 X 6 T, STL   | 5.88        | 6.01      | 35.36          | C | 8.39     | 297    | 2.72         | A | 96     | 4.53       | S | -160   |
| 11701  | 25   | 5 X 4 X 6 T, STL   | 5.88        | 1.69      | 9.93           | C | 5.48     | 54     | 3.74         | A | 37     | 3.07       | S | -30    |
| 11701  | 26   | 5 X 4 X 6 T, STL   | 5.88        | 5.73      | 33.71          | C | 8.79     | 296    | 3.39         | A | 114    | 5.27       | S | -178   |
| 11701  |      |                    |             |           |                |   |          |        |              |   |        |            |   |        |

| SWBS # | ITEM | DESCRIPTION        | UNIT WEIGHT | NO. UNITS | CURRENT WEIGHT | E | VERTICAL |        | LONGITUDINAL |   |        | TRANSVERSE |   |        |
|--------|------|--------------------|-------------|-----------|----------------|---|----------|--------|--------------|---|--------|------------|---|--------|
|        |      |                    |             |           |                |   | VCG      | MOMENT | LCG          | S | MOMENT | TCG        | S | MOMENT |
|        |      |                    | lb          | #         | lb             | A | ft       | ft*lb  | ft           |   | ft*lb  | ft         |   | ft*lb  |
| 11702  | 33   | 5 X 4 X 7.5 T, STL | 7.50        | 23.10     | 173.22         | C | 6.64     | 1150   | 5.00         | A | 866    | 0.00       | C | 0      |
| 11702  | 34   | 5 X 4 X 7.5 T, STL | 7.50        | 24.50     | 183.77         | C | 6.35     | 1167   | 6.00         | A | 1103   | 0.00       | C | 0      |
| 11702  | 35   | 5 X 4 X 7.5 T, STL | 7.50        | 25.81     | 193.58         | C | 6.08     | 1177   | 7.00         | A | 1355   | 0.00       | C | 0      |
| 11702  | 36   | 5 X 4 X 7.5 T, STL | 7.50        | 27.03     | 202.75         | C | 5.83     | 1182   | 8.00         | A | 1622   | 0.00       | C | 0      |
| 11702  | 37   | 5 X 4 X 7.5 T, STL | 7.50        | 28.18     | 211.33         | C | 5.59     | 1182   | 9.00         | A | 1902   | 0.00       | C | 0      |
| 11702  | 38   | 5 X 4 X 7.5 T, STL | 7.50        | 29.24     | 219.31         | C | 5.37     | 1178   | 10.00        | A | 2193   | 0.00       | C | 0      |
| 11702  | 39   | 5 X 4 X 7.5 T, STL | 7.50        | 24.18     | 181.37         | C | 5.91     | 1072   | 11.00        | A | 1995   | 0.00       | C | 0      |
| 11702  |      |                    |             |           |                |   |          |        |              |   |        |            |   |        |
| 11703  | 40   | 5 X 4 X 8.5 T, STL | 8.50        | 31.09     | 264.29         | C | 4.98     | 1317   | 12.00        | A | 3171   | 0.00       | C | 0      |
| 11703  | 41   | 5 X 4 X 8.5 T, STL | 8.50        | 32.55     | 276.65         | C | 4.69     | 1297   | 14.00        | A | 3873   | 0.00       | C | 0      |
| 11703  | 42   | 5 X 4 X 8.5 T, STL | 8.50        | 33.14     | 281.66         | C | 4.58     | 1290   | 15.00        | A | 4225   | 0.00       | C | 0      |
| 11703  | 43   | 5 X 4 X 8.5 T, STL | 8.50        | 33.66     | 286.09         | C | 4.49     | 1284   | 16.00        | A | 4577   | 0.00       | C | 0      |
| 11703  | 44   | 5 X 4 X 8.5 T, STL | 8.50        | 34.13     | 290.15         | C | 4.41     | 1279   | 17.00        | A | 4932   | 0.00       | C | 0      |
| 11703  | 45   | 5 X 4 X 8.5 T, STL | 8.50        | 34.58     | 293.94         | C | 4.34     | 1274   | 18.00        | A | 5291   | 0.00       | C | 0      |
| 11703  | 46   | 5 X 4 X 8.5 T, STL | 8.50        | 34.99     | 297.44         | C | 4.27     | 1269   | 19.00        | A | 5651   | 0.00       | C | 0      |
| 11703  | 47   | 5 X 4 X 8.5 T, STL | 8.50        | 35.37     | 300.65         | C | 4.20     | 1264   | 20.00        | A | 6013   | 0.00       | C | 0      |
| 11703  | 48   | 5 X 4 X 8.5 T, STL | 8.50        | 35.72     | 303.59         | C | 4.14     | 1257   | 21.00        | A | 6375   | 0.00       | C | 0      |
| 11703  | 49   | 5 X 4 X 8.5 T, STL | 8.50        | 36.03     | 306.26         | C | 4.09     | 1253   | 22.00        | A | 6738   | 0.00       | C | 0      |
| 11703  | 50   | 5 X 4 X 8.5 T, STL | 8.50        | 36.32     | 308.68         | C | 4.04     | 1248   | 23.00        | A | 7100   | 0.00       | C | 0      |
| 11703  | 51   | 5 X 4 X 8.5 T, STL | 8.50        | 36.57     | 310.81         | C | 4.00     | 1244   | 24.00        | A | 7459   | 0.00       | C | 0      |
| 11703  | 52   | 5 X 4 X 8.5 T, STL | 8.50        | 36.78     | 312.63         | C | 3.96     | 1240   | 25.00        | A | 7816   | 0.00       | C | 0      |
| 11703  | 53   | 5 X 4 X 8.5 T, STL | 8.50        | 37.10     | 315.33         | C | 3.91     | 1233   | 27.00        | A | 8514   | 0.00       | C | 0      |
| 11703  | 54   | 5 X 4 X 8.5 T, STL | 8.50        | 37.19     | 316.10         | C | 3.89     | 1230   | 28.00        | A | 8851   | 0.00       | C | 0      |
| 11703  |      |                    |             |           |                |   |          |        |              |   |        |            |   |        |
| 11704  | 55   | 4 X 4 X 5 T, STL   | 4.91        | 37.25     | 182.91         | C | 3.85     | 704    | 29.00        | A | 5304   | 0.00       | C | 0      |
| 11704  | 56   | 4 X 4 X 5 T, STL   | 4.91        | 37.30     | 183.12         | C | 3.83     | 702    | 30.50        | A | 5585   | 0.00       | C | 0      |
| 11704  | 57   | 4 X 4 X 5 T, STL   | 4.91        | 37.30     | 183.14         | C | 3.83     | 702    | 32.00        | A | 5860   | 0.00       | C | 0      |
| 11704  | 58   | 4 X 4 X 5 T, STL   | 4.91        | 37.25     | 182.89         | C | 3.84     | 702    | 33.50        | A | 6127   | 0.00       | C | 0      |
| 11704  | 59   | 4 X 4 X 5 T, STL   | 4.91        | 32.97     | 161.88         | C | 3.10     | 502    | 36.50        | A | 5909   | 0.00       | C | 0      |
| 11704  | 60   | 4 X 4 X 5 T, STL   | 4.91        | 32.79     | 161.00         | C | 3.13     | 504    | 38.00        | A | 6118   | 0.00       | C | 0      |
| 11704  | 61   | 4 X 4 X 5 T, STL   | 4.91        | 32.58     | 159.95         | C | 3.17     | 506    | 39.50        | A | 6318   | 0.00       | C | 0      |
| 11704  | 62   | 4 X 4 X 5 T, STL   | 4.91        | 32.33     | 158.74         | C | 3.21     | 509    | 41.00        | A | 6508   | 0.00       | C | 0      |
| 11704  | 63   | 4 X 4 X 5 T, STL   | 4.91        | 13.77     | 67.61          | C | 5.87     | 397    | 42.50        | A | 2874   | 0.00       | C | 0      |
| 11704  | 64   | 4 X 4 X 5 T, STL   | 4.91        | 31.75     | 155.87         | C | 3.31     | 515    | 44.00        | A | 6858   | 0.00       | C | 0      |
| 11704  | 65   | 4 X 4 X 5 T, STL   | 4.91        | 31.40     | 154.16         | C | 3.37     | 520    | 45.50        | A | 7014   | 0.00       | C | 0      |
| 11704  | 66   | 4 X 4 X 5 T, STL   | 4.91        | 30.98     | 152.10         | C | 3.47     | 528    | 47.00        | A | 7148   | 0.00       | C | 0      |
| 11704  | 67   | 4 X 4 X 5 T, STL   | 4.91        | 30.45     | 149.52         | C | 3.62     | 542    | 48.50        | A | 7252   | 0.00       | C | 0      |
| 11704  | 68   | 4 X 4 X 5 T, STL   | 4.91        | 29.03     | 142.56         | C | 4.11     | 586    | 51.50        | A | 7342   | 0.00       | C | 0      |
| 11704  | 69   | 4 X 4 X 5 T, STL   | 4.91        | 28.19     | 138.39         | C | 4.46     | 617    | 53.00        | A | 7335   | 0.00       | C | 0      |
| 11704  | 70   | 4 X 4 X 5 T, STL   | 4.91        | 27.37     | 134.39         | C | 4.80     | 645    | 54.50        | A | 7324   | 0.00       | C | 0      |
| 11704  | 71   | 4 X 4 X 5 T, STL   | 4.91        | 26.58     | 130.51         | C | 5.14     | 671    | 56.00        | A | 7308   | 0.00       | C | 0      |

| SWBS # | ITEM | DESCRIPTION           | UNIT WEIGHT | NO. UNITS | CURRENT WEIGHT | E C A | VERTICAL |        | LONGITUDINAL |   |        | TRANSVERSE |   |        |
|--------|------|-----------------------|-------------|-----------|----------------|-------|----------|--------|--------------|---|--------|------------|---|--------|
|        |      |                       |             |           |                |       | VCG      | MOMENT | LCG          | S | MOMENT | TCG        | S | MOMENT |
|        |      |                       | lb          | #         | lb             |       | ft       | ft*lb  | ft           |   | ft*lb  | ft         |   | ft*lb  |
| 11704  | 72   | 4 X 4 X 5 T, STL      | 4.91        | 25.81     | 126.74         | C     | 5.44     | 690    | 57.50        | A | 7288   | 0.00       | C | 0      |
| 11704  | 73   | 4 X 4 X 5 T, STL      | 4.91        | 24.20     | 118.81         | C     | 5.50     | 654    | 60.50        | A | 7188   | 0.00       | C | 0      |
| 11704  | 74   | 4 X 4 X 5 T, STL      | 4.91        | 23.23     | 114.05         | C     | 5.84     | 666    | 62.00        | A | 7071   | 0.00       | C | 0      |
| 11704  | 75   | 4 X 4 X 5 T, STL      | 4.91        | 21.69     | 106.50         | C     | 5.90     | 629    | 63.50        | A | 6763   | 0.00       | C | 0      |
| 11704  | 76   | 4 X 4 X 5 T, STL      | 4.91        | 19.36     | 95.05          | C     | 6.08     | 578    | 65.00        | A | 6178   | 0.00       | C | 0      |
| 11704  | 77   | 4 X 4 X 5 T, STL      | 4.91        | 15.96     | 78.35          | C     | 6.68     | 523    | 66.50        | A | 5211   | 0.00       | C | 0      |
| 11704  |      |                       |             |           |                |       |          |        |              |   |        |            |   |        |
| 11705  | 78   | 8 X 5 1/2 X 13 T, STL | 12.83       | 18.28     | 234.57         | C     | 1.44     | 337    | 42.50        | A | 9969   | 0.00       | C | 0      |
| 11707  |      |                       |             |           |                |       |          |        |              |   |        |            |   |        |
| 11708  | 79   | 12 X 6.5 X 26 MT, STL | 19.35       | 2.58      | 49.82          | C     | 8.24     | 411    | 47.00        | A | 2341   | 0.00       | C | 0      |
| 11708  |      |                       |             |           |                |       |          |        |              |   |        |            |   | 0      |
| 11709  | 80   | PLATE, 1/4 IN, STL    | 10.20       | 0.12      | 1.24           | C     | 8.50     | 11     | 49.50        | A | 61     | 0.00       | C | 0      |
| 11709  | 81   | PLATE, 1/4 IN, STL    | 10.20       | 0.12      | 1.24           | C     | 8.50     | 11     | 49.50        | A | 61     | 0.00       | C | 0      |
| 11709  |      |                       |             |           |                |       |          |        |              |   |        |            |   | 0      |
| 11710  | 82   | PLATE, 1/2 IN, STL    | 20.40       | 6.22      | 126.95         | C     | 8.38     | 1064   | 50.02        | A | 6350   | 0.00       | C | 0      |
|        |      |                       |             |           |                |       |          |        |              |   |        |            |   |        |
| 12101  | 1    | PLATE, 1/4 IN, STL    | 10.20       | 22.08     | 225.17         | C     | 0.19     | 43     | 30.53        | A | 6874   | 5.01       | S | -1128  |
| 12101  | 2    | PLATE, 1/4 IN, STL    | 10.20       | 22.08     | 225.17         | C     | 0.19     | 43     | 30.53        | A | 6874   | 5.01       | P | 1128   |
| 12101  |      |                       |             |           |                |       |          |        |              |   |        |            |   |        |
| 12102  | 3    | 3 X 2.2 MT, STL       | 2.20        | 2.46      | 5.40           | C     | 2.28     | 12     | 27.00        | A | 146    | 4.76       | P | 26     |
| 12102  | 4    | 3 X 2.2 MT, STL       | 2.20        | 2.46      | 5.40           | C     | 2.28     | 12     | 27.00        | A | 146    | 4.76       | S | -26    |
| 12102  | 5    | 3 X 2.2 MT, STL       | 2.20        | 2.48      | 5.46           | C     | 2.27     | 12     | 28.00        | A | 153    | 4.76       | P | 26     |
| 12102  | 6    | 3 X 2.2 MT, STL       | 2.20        | 2.48      | 5.46           | C     | 2.27     | 12     | 28.00        | A | 153    | 4.76       | S | -26    |
| 12102  | 7    | 3 X 2.2 MT, STL       | 2.20        | 2.50      | 5.51           | C     | 2.25     | 12     | 29.00        | A | 160    | 4.76       | P | 26     |
| 12102  | 8    | 3 X 2.2 MT, STL       | 2.20        | 2.50      | 5.51           | C     | 2.25     | 12     | 29.00        | A | 160    | 4.76       | S | -26    |
| 12102  | 9    | 3 X 2.2 MT, STL       | 2.20        | 2.52      | 5.55           | C     | 2.24     | 12     | 30.50        | A | 169    | 4.76       | P | 26     |
| 12102  | 10   | 3 X 2.2 MT, STL       | 2.20        | 2.52      | 5.55           | C     | 2.24     | 12     | 30.50        | A | 169    | 4.76       | S | -26    |
| 12102  | 11   | 3 X 2.2 MT, STL       | 2.20        | 2.53      | 5.57           | C     | 2.24     | 12     | 32.00        | A | 178    | 4.76       | P | 27     |
| 12102  | 12   | 3 X 2.2 MT, STL       | 2.20        | 2.53      | 5.57           | C     | 2.24     | 12     | 32.00        | A | 178    | 4.76       | S | -27    |
| 12102  | 13   | 3 X 2.2 MT, STL       | 2.20        | 2.53      | 5.56           | C     | 2.24     | 12     | 33.50        | A | 186    | 4.76       | P | 26     |
| 12102  | 14   | 3 X 2.2 MT, STL       | 2.20        | 2.53      | 5.56           | C     | 2.24     | 12     | 33.50        | A | 186    | 4.76       | S | -26    |
|        |      |                       |             |           |                |       |          |        |              |   |        |            |   |        |
| 12201  | 1    | PLATE, 3/16 IN, STL   | 7.65        | 136.30    | 1042.67        | C     | 7.51     | 7828   | 13.00        | A | 13555  | 0.00       | C | 0      |
| 12201  | 2    | PLATE, 3/16 IN, STL   | 7.65        | 21.41     | 163.81         | C     | 2.44     | 399    | 13.00        | A | 2130   | 0.00       | C | 0      |
| 12201  | 3    | PLATE, 3/16 IN, STL   | 7.65        | 156.55    | 1197.60        | C     | 7.35     | 8807   | 26.00        | A | 31138  | 0.00       | C | 0      |
| 12201  | 4    | PLATE, 3/16 IN, STL   | 7.65        | 155.93    | 1192.87        | C     | 7.36     | 8781   | 35.00        | A | 41750  | 0.00       | C | 0      |
| 12201  | 5    | PLATE, 3/16 IN, STL   | 7.65        | 132.13    | 1010.79        | C     | 5.61     | 5668   | 50.00        | A | 50540  | 0.00       | C | 0      |
| 12201  | 6    | PLATE, 3/16 IN, STL   | 7.65        | 78.84     | 603.13         | C     | 6.92     | 4174   | 59.00        | A | 35585  | 0.00       | C | 0      |
| 12201  |      |                       |             |           |                |       |          |        |              |   |        |            |   |        |
| 12202  | 7    | PLATE, 1/4 IN, STL    | 10.20       | 45.50     | 464.14         | C     | 2.21     | 1028   | 26.00        | A | 12068  | 0.00       | C | 0      |
| 12202  | 8    | PLATE, 1/4 IN, STL    | 10.20       | 47.06     | 479.97         | C     | 2.18     | 1048   | 35.00        | A | 16799  | 0.00       | C | 0      |

| SWBS # | ITEM | DESCRIPTION     | UNIT WEIGHT | NO. UNITS | CURRENT WEIGHT | E C A | VERTICAL |        | LONGITUDINAL |   |        | TRANSVERSE |   |        |
|--------|------|-----------------|-------------|-----------|----------------|-------|----------|--------|--------------|---|--------|------------|---|--------|
|        |      |                 |             |           |                |       | VCG      | MOMENT | LCG          | S | MOMENT | TCG        | S | MOMENT |
|        |      |                 | lb          | #         | lb             |       | ft       | ft*lb  | ft           |   | ft*lb  | ft         |   | ft*lb  |
| 12202  |      |                 |             |           |                |       |          |        |              |   |        |            |   |        |
| 12203  | 9    | 3 X 2.2 MT, STL | 2.20        | 2.54      | 5.59           | C     | 9.91     | 55     | 3.82         | A | 21     | 6.00       | P | 34     |
| 12203  | 10   | 3 X 2.2 MT, STL | 2.20        | 5.00      | 10.99          | C     | 8.71     | 96     | 3.82         | A | 42     | 4.50       | P | 50     |
| 12203  | 11   | 3 X 2.2 MT, STL | 2.20        | 6.11      | 13.43          | C     | 8.12     | 109    | 3.82         | A | 51     | 3.00       | P | 40     |
| 12203  | 12   | 3 X 2.2 MT, STL | 2.20        | 6.75      | 14.85          | C     | 7.83     | 116    | 3.82         | A | 57     | 1.50       | P | 22     |
| 12203  | 13   | 3 X 2.2 MT, STL | 2.20        | 7.38      | 16.24          | C     | 7.54     | 122    | 3.82         | A | 62     | 0.00       | C | 0      |
| 12203  | 14   | 3 X 2.2 MT, STL | 2.20        | 6.75      | 14.85          | C     | 7.83     | 116    | 3.82         | A | 57     | 1.50       | S | -22    |
| 12203  | 15   | 3 X 2.2 MT, STL | 2.20        | 6.11      | 13.44          | C     | 8.12     | 109    | 3.82         | A | 51     | 3.00       | S | -40    |
| 12203  | 16   | 3 X 2.2 MT, STL | 2.20        | 5.00      | 10.99          | C     | 8.71     | 96     | 3.82         | A | 42     | 4.50       | S | -50    |
| 12203  | 17   | 3 X 2.2 MT, STL | 2.20        | 2.54      | 5.59           | C     | 9.91     | 55     | 3.82         | A | 21     | 6.00       | S | -34    |
| 12203  | 18   | 3 X 2.2 MT, STL | 2.20        | 4.32      | 9.51           | C     | 8.91     | 85     | 12.82        | A | 122    | 9.00       | P | 86     |
| 12203  | 19   | 3 X 2.2 MT, STL | 2.20        | 6.97      | 15.33          | C     | 7.62     | 117    | 12.82        | A | 197    | 7.50       | P | 115    |
| 12203  | 20   | 3 X 2.2 MT, STL | 2.20        | 7.91      | 17.41          | C     | 7.66     | 133    | 12.82        | A | 223    | 6.00       | P | 104    |
| 12203  | 21   | 3 X 2.2 MT, STL | 2.20        | 8.65      | 19.03          | C     | 7.68     | 146    | 12.82        | A | 244    | 4.50       | P | 86     |
| 12203  | 22   | 3 X 2.2 MT, STL | 2.20        | 9.39      | 20.66          | C     | 7.69     | 159    | 12.82        | A | 265    | 3.00       | P | 62     |
| 12203  | 23   | 3 X 2.2 MT, STL | 2.20        | 10.13     | 22.28          | C     | 7.71     | 172    | 12.82        | A | 286    | 1.50       | P | 33     |
| 12203  | 24   | 3 X 2.2 MT, STL | 2.20        | 10.87     | 23.91          | C     | 7.72     | 185    | 12.82        | A | 307    | 0.00       | C | 0      |
| 12203  | 25   | 3 X 2.2 MT, STL | 2.20        | 10.13     | 22.28          | C     | 7.71     | 172    | 12.82        | A | 286    | 1.50       | S | -33    |
| 12203  | 26   | 3 X 2.2 MT, STL | 2.20        | 9.39      | 20.66          | C     | 7.69     | 159    | 12.82        | A | 265    | 3.00       | S | -62    |
| 12203  | 27   | 3 X 2.2 MT, STL | 2.20        | 8.65      | 19.03          | C     | 7.68     | 146    | 12.82        | A | 244    | 4.50       | S | -86    |
| 12203  | 28   | 3 X 2.2 MT, STL | 2.20        | 7.91      | 17.41          | C     | 7.66     | 133    | 12.82        | A | 223    | 6.00       | S | -104   |
| 12203  | 29   | 3 X 2.2 MT, STL | 2.20        | 6.97      | 15.33          | C     | 7.62     | 117    | 12.82        | A | 197    | 7.50       | S | -115   |
| 12203  | 30   | 3 X 2.2 MT, STL | 2.20        | 4.32      | 9.51           | C     | 8.91     | 85     | 12.82        | A | 122    | 9.00       | S | -86    |
| 12203  | 31   | 3 X 2.2 MT, STL | 2.20        | 0.34      | 0.75           | C     | 3.34     | 2      | 12.82        | A | 10     | 6.00       | P | 4      |
| 12203  | 32   | 3 X 2.2 MT, STL | 2.20        | 1.05      | 2.30           | C     | 2.95     | 7      | 12.82        | A | 30     | 4.50       | P | 10     |
| 12203  | 33   | 3 X 2.2 MT, STL | 2.20        | 1.76      | 3.87           | C     | 2.55     | 10     | 12.82        | A | 50     | 3.00       | P | 12     |
| 12203  | 34   | 3 X 2.2 MT, STL | 2.20        | 2.47      | 5.44           | C     | 2.16     | 12     | 12.82        | A | 70     | 1.50       | P | 8      |
| 12203  | 35   | 3 X 2.2 MT, STL | 2.20        | 3.19      | 7.01           | C     | 1.77     | 12     | 12.82        | A | 90     | 0.00       | C | 0      |
| 12203  | 36   | 3 X 2.2 MT, STL | 2.20        | 2.47      | 5.44           | C     | 2.16     | 12     | 12.82        | A | 70     | 1.50       | S | -8     |
| 12203  | 37   | 3 X 2.2 MT, STL | 2.20        | 1.76      | 3.87           | C     | 2.55     | 10     | 12.82        | A | 50     | 3.00       | S | -12    |
| 12203  | 38   | 3 X 2.2 MT, STL | 2.20        | 1.05      | 2.30           | C     | 2.95     | 7      | 12.82        | A | 30     | 4.50       | S | -10    |
| 12203  | 39   | 3 X 2.2 MT, STL | 2.20        | 0.34      | 0.75           | C     | 3.34     | 2      | 12.82        | A | 10     | 6.00       | S | -4     |
| 12203  | 40   | 3 X 2.2 MT, STL | 2.20        | 7.53      | 16.56          | C     | 7.64     | 127    | 25.82        | A | 428    | 9.00       | P | 149    |
| 12203  | 41   | 3 X 2.2 MT, STL | 2.20        | 7.55      | 16.62          | C     | 7.65     | 127    | 25.82        | A | 429    | 7.50       | P | 125    |
| 12203  | 42   | 3 X 2.2 MT, STL | 2.20        | 7.58      | 16.68          | C     | 7.67     | 128    | 25.82        | A | 431    | 6.00       | P | 100    |
| 12203  | 43   | 3 X 2.2 MT, STL | 2.20        | 7.61      | 16.73          | C     | 7.68     | 128    | 25.82        | A | 432    | 4.50       | P | 75     |
| 12203  | 44   | 3 X 2.2 MT, STL | 2.20        | 7.63      | 16.79          | C     | 7.69     | 129    | 25.82        | A | 433    | 3.00       | P | 50     |
| 12203  | 45   | 3 X 2.2 MT, STL | 2.20        | 7.66      | 16.85          | C     | 7.71     | 130    | 25.82        | A | 435    | 1.50       | P | 25     |
| 12203  | 46   | 3 X 2.2 MT, STL | 2.20        | 7.68      | 16.90          | C     | 7.72     | 130    | 25.82        | A | 436    | 0.00       | C | 0      |
| 12203  | 47   | 3 X 2.2 MT, STL | 2.20        | 7.66      | 16.85          | C     | 7.71     | 130    | 25.82        | A | 435    | 1.50       | S | -25    |
| 12203  | 48   | 3 X 2.2 MT, STL | 2.20        | 7.63      | 16.79          | C     | 7.69     | 129    | 25.82        | A | 433    | 3.00       | S | -50    |

| SWBS # | ITEM | DESCRIPTION     | UNIT WEIGHT | NO. UNITS | CURRENT WEIGHT | E | VERTICAL |        | LONGITUDINAL |   |        | TRANSVERSE |   |        |
|--------|------|-----------------|-------------|-----------|----------------|---|----------|--------|--------------|---|--------|------------|---|--------|
|        |      |                 |             |           |                |   | VCG      | MOMENT | LCG          | S | MOMENT | TCG        | S | MOMENT |
|        |      |                 | lb          | #         | lb             | A | ft       | ft*lb  | ft           |   | ft*lb  | ft         |   | ft*lb  |
| 12203  | 49   | 3 X 2.2 MT, STL | 2.20        | 7.61      | 16.73          | C | 7.68     | 128    | 25.82        | A | 432    | 4.50       | S | -75    |
| 12203  | 50   | 3 X 2.2 MT, STL | 2.20        | 7.58      | 16.68          | C | 7.67     | 128    | 25.82        | A | 431    | 6.00       | S | -100   |
| 12203  | 51   | 3 X 2.2 MT, STL | 2.20        | 7.55      | 16.62          | C | 7.65     | 127    | 25.82        | A | 429    | 7.50       | S | -125   |
| 12203  | 52   | 3 X 2.2 MT, STL | 2.20        | 7.53      | 16.56          | C | 7.64     | 127    | 25.82        | A | 428    | 9.00       | S | -149   |
| 12203  | 53   | 3 X 2.2 MT, STL | 2.20        | 1.43      | 3.14           | C | 2.72     | 9      | 25.82        | A | 81     | 9.00       | P | 28     |
| 12203  | 54   | 3 X 2.2 MT, STL | 2.20        | 1.77      | 3.90           | C | 2.53     | 10     | 25.82        | A | 101    | 7.50       | P | 29     |
| 12203  | 55   | 3 X 2.2 MT, STL | 2.20        | 2.12      | 4.66           | C | 2.34     | 11     | 25.82        | A | 120    | 6.00       | P | 28     |
| 12203  | 56   | 3 X 2.2 MT, STL | 2.20        | 2.46      | 5.42           | C | 2.15     | 12     | 25.82        | A | 140    | 4.50       | P | 24     |
| 12203  | 57   | 3 X 2.2 MT, STL | 2.20        | 2.81      | 6.18           | C | 1.96     | 12     | 25.82        | A | 160    | 3.00       | P | 19     |
| 12203  | 58   | 3 X 2.2 MT, STL | 2.20        | 3.15      | 6.94           | C | 1.77     | 12     | 25.82        | A | 179    | 1.50       | P | 10     |
| 12203  | 59   | 3 X 2.2 MT, STL | 2.20        | 3.50      | 7.70           | C | 1.58     | 12     | 25.82        | A | 199    | 0.00       | C | 0      |
| 12203  | 60   | 3 X 2.2 MT, STL | 2.20        | 3.15      | 6.94           | C | 1.77     | 12     | 25.82        | A | 179    | 1.50       | S | -10    |
| 12203  | 61   | 3 X 2.2 MT, STL | 2.20        | 2.81      | 6.18           | C | 1.96     | 12     | 25.82        | A | 160    | 3.00       | S | -19    |
| 12203  | 62   | 3 X 2.2 MT, STL | 2.20        | 2.46      | 5.42           | C | 2.15     | 12     | 25.82        | A | 140    | 4.50       | S | -24    |
| 12203  | 63   | 3 X 2.2 MT, STL | 2.20        | 2.12      | 4.66           | C | 2.34     | 11     | 25.82        | A | 120    | 6.00       | S | -28    |
| 12203  | 64   | 3 X 2.2 MT, STL | 2.20        | 1.77      | 3.90           | C | 2.53     | 10     | 25.82        | A | 101    | 7.50       | S | -29    |
| 12203  | 65   | 3 X 2.2 MT, STL | 2.20        | 1.43      | 3.14           | C | 2.72     | 9      | 25.82        | A | 81     | 9.00       | S | -28    |
| 12203  | 66   | 3 X 2.2 MT, STL | 2.20        | 7.53      | 16.57          | C | 7.64     | 127    | 34.82        | A | 577    | 9.00       | P | 149    |
| 12203  | 67   | 3 X 2.2 MT, STL | 2.20        | 7.56      | 16.63          | C | 7.65     | 127    | 34.82        | A | 579    | 7.50       | P | 125    |
| 12203  | 68   | 3 X 2.2 MT, STL | 2.20        | 7.58      | 16.68          | C | 7.67     | 128    | 34.82        | A | 581    | 6.00       | P | 100    |
| 12203  | 69   | 3 X 2.2 MT, STL | 2.20        | 7.61      | 16.74          | C | 7.68     | 129    | 34.82        | A | 583    | 4.50       | P | 75     |
| 12203  | 70   | 3 X 2.2 MT, STL | 2.20        | 7.63      | 16.79          | C | 7.69     | 129    | 34.82        | A | 585    | 3.00       | P | 50     |
| 12203  | 71   | 3 X 2.2 MT, STL | 2.20        | 7.66      | 16.85          | C | 7.71     | 130    | 34.82        | A | 587    | 1.50       | P | 25     |
| 12203  | 72   | 3 X 2.2 MT, STL | 2.20        | 7.68      | 16.90          | C | 7.72     | 130    | 34.82        | A | 589    | 0.00       | C | 0      |
| 12203  | 73   | 3 X 2.2 MT, STL | 2.20        | 7.66      | 16.85          | C | 7.71     | 130    | 34.82        | A | 587    | 1.50       | S | -25    |
| 12203  | 74   | 3 X 2.2 MT, STL | 2.20        | 7.63      | 16.79          | C | 7.69     | 129    | 34.82        | A | 585    | 3.00       | S | -50    |
| 12203  | 75   | 3 X 2.2 MT, STL | 2.20        | 7.61      | 16.74          | C | 7.68     | 129    | 34.82        | A | 583    | 4.50       | S | -75    |
| 12203  | 76   | 3 X 2.2 MT, STL | 2.20        | 7.58      | 16.68          | C | 7.67     | 128    | 34.82        | A | 581    | 6.00       | S | -100   |
| 12203  | 77   | 3 X 2.2 MT, STL | 2.20        | 7.56      | 16.63          | C | 7.65     | 127    | 34.82        | A | 579    | 7.50       | S | -125   |
| 12203  | 78   | 3 X 2.2 MT, STL | 2.20        | 7.53      | 16.57          | C | 7.64     | 127    | 34.82        | A | 577    | 9.00       | S | -149   |
| 12203  | 79   | 3 X 2.2 MT, STL | 2.20        | 1.60      | 3.52           | C | 2.62     | 9      | 34.82        | A | 123    | 9.00       | P | 32     |
| 12203  | 80   | 3 X 2.2 MT, STL | 2.20        | 1.92      | 4.22           | C | 2.44     | 10     | 34.82        | A | 147    | 7.50       | P | 32     |
| 12203  | 81   | 3 X 2.2 MT, STL | 2.20        | 2.23      | 4.92           | C | 2.27     | 11     | 34.82        | A | 171    | 6.00       | P | 29     |
| 12203  | 82   | 3 X 2.2 MT, STL | 2.20        | 2.55      | 5.61           | C | 2.10     | 12     | 34.82        | A | 195    | 4.50       | P | 25     |
| 12203  | 83   | 3 X 2.2 MT, STL | 2.20        | 2.87      | 6.31           | C | 1.92     | 12     | 34.82        | A | 220    | 3.00       | P | 19     |
| 12203  | 84   | 3 X 2.2 MT, STL | 2.20        | 3.18      | 7.00           | C | 1.75     | 12     | 34.82        | A | 244    | 1.50       | P | 11     |
| 12203  | 85   | 3 X 2.2 MT, STL | 2.20        | 3.50      | 7.70           | C | 1.58     | 12     | 34.82        | A | 268    | 0.00       | C | 0      |
| 12203  | 86   | 3 X 2.2 MT, STL | 2.20        | 3.18      | 7.00           | C | 1.75     | 12     | 34.82        | A | 244    | 1.50       | S | -11    |
| 12203  | 87   | 3 X 2.2 MT, STL | 2.20        | 2.87      | 6.31           | C | 1.92     | 12     | 34.82        | A | 220    | 3.00       | S | -19    |
| 12203  | 88   | 3 X 2.2 MT, STL | 2.20        | 2.55      | 5.61           | C | 2.10     | 12     | 34.82        | A | 195    | 4.50       | S | -25    |
| 12203  | 89   | 3 X 2.2 MT, STL | 2.20        | 2.23      | 4.92           | C | 2.27     | 11     | 34.82        | A | 171    | 6.00       | S | -29    |

| SWBS # | ITEM | DESCRIPTION           | UNIT WEIGHT | NO. UNITS | CURRENT WEIGHT | E<br>C<br>A | VERTICAL |        | LONGITUDINAL |   |        | TRANSVERSE |   |        |
|--------|------|-----------------------|-------------|-----------|----------------|-------------|----------|--------|--------------|---|--------|------------|---|--------|
|        |      |                       |             |           |                |             | VCG      | MOMENT | LCG          | S | MOMENT | TCG        | S | MOMENT |
|        |      |                       | lb          | #         | lb             |             | ft       | ft*lb  | ft           |   | ft*lb  | ft         |   | ft*lb  |
| 12203  | 90   | 3 X 2.2 MT, STL       | 2.20        | 1.92      | 4.22           | C           | 2.44     | 10     | 34.82        | A | 147    | 7.50       | S | -32    |
| 12203  | 91   | 3 X 2.2 MT, STL       | 2.20        | 1.60      | 3.52           | C           | 2.62     | 9      | 34.82        | A | 123    | 9.00       | S | -32    |
| 12203  | 92   | 3 X 2.2 MT, STL       | 2.20        | 4.57      | 10.05          | C           | 6.70     | 67     | 49.82        | A | 501    | 9.00       | P | 90     |
| 12203  | 93   | 3 X 2.2 MT, STL       | 2.20        | 6.02      | 13.25          | C           | 5.97     | 79     | 49.82        | A | 660    | 7.50       | P | 99     |
| 12203  | 94   | 3 X 2.2 MT, STL       | 2.20        | 6.48      | 14.25          | C           | 5.74     | 82     | 49.82        | A | 710    | 6.00       | P | 85     |
| 12203  | 95   | 3 X 2.2 MT, STL       | 2.20        | 6.93      | 15.25          | C           | 5.52     | 84     | 49.82        | A | 760    | 4.50       | P | 69     |
| 12203  | 96   | 3 X 2.2 MT, STL       | 2.20        | 7.39      | 16.25          | C           | 5.29     | 86     | 49.82        | A | 810    | 3.00       | P | 49     |
| 12203  | 97   | 4 X 4 X 5 MT, STL     | 4.91        | 7.84      | 38.50          | C           | 5.11     | 197    | 49.76        | A | 1916   | 1.78       | P | 69     |
| 12203  | 98   | 4 X 4 X 5 MT, STL     | 4.91        | 8.29      | 40.73          | C           | 4.68     | 190    | 49.76        | A | 2026   | 0.00       | C | 0      |
| 12203  | 99   | 4 X 4 X 5 MT, STL     | 4.91        | 7.84      | 38.50          | C           | 5.11     | 197    | 49.82        | A | 1918   | 1.78       | S | -69    |
| 12203  | 100  | 3 X 2.2 MT, STL       | 2.20        | 7.39      | 16.25          | C           | 5.29     | 86     | 49.82        | A | 810    | 3.00       | S | -49    |
| 12203  | 101  | 3 X 2.2 MT, STL       | 2.20        | 6.93      | 15.25          | C           | 5.52     | 84     | 49.82        | A | 760    | 4.50       | S | -69    |
| 12203  | 102  | 3 X 2.2 MT, STL       | 2.20        | 6.48      | 14.25          | C           | 5.74     | 82     | 49.82        | A | 710    | 6.00       | S | -85    |
| 12203  | 103  | 3 X 2.2 MT, STL       | 2.20        | 6.02      | 13.25          | C           | 5.97     | 79     | 49.82        | A | 660    | 7.50       | S | -99    |
| 12203  | 104  | 3 X 2.2 MT, STL       | 2.20        | 4.57      | 10.05          | C           | 6.70     | 67     | 49.82        | A | 501    | 9.00       | S | -90    |
| 12203  | 105  | 3 X 2.2 MT, STL       | 2.20        | 2.81      | 6.18           | C           | 7.58     | 47     | 58.82        | A | 364    | 9.00       | P | 56     |
| 12203  | 106  | 3 X 2.2 MT, STL       | 2.20        | 3.99      | 8.77           | C           | 6.99     | 61     | 58.82        | A | 516    | 7.50       | P | 66     |
| 12203  | 107  | 3 X 2.2 MT, STL       | 2.20        | 4.29      | 9.45           | C           | 6.84     | 65     | 58.82        | A | 556    | 6.00       | P | 57     |
| 12203  | 108  | 3 X 2.2 MT, STL       | 2.20        | 4.34      | 9.56           | C           | 6.82     | 65     | 58.82        | A | 562    | 4.50       | P | 43     |
| 12203  | 109  | 3 X 2.2 MT, STL       | 2.20        | 4.39      | 9.67           | C           | 6.79     | 66     | 58.82        | A | 569    | 3.00       | P | 29     |
| 12203  | 110  | 3 X 2.2 MT, STL       | 2.20        | 4.44      | 9.78           | C           | 6.76     | 66     | 58.82        | A | 575    | 1.50       | P | 15     |
| 12203  | 111  | 3 X 2.2 MT, STL       | 2.20        | 4.49      | 9.88           | C           | 6.73     | 67     | 58.82        | A | 581    | 0.00       | C | 0      |
| 12203  | 112  | 3 X 2.2 MT, STL       | 2.20        | 4.44      | 9.78           | C           | 6.76     | 66     | 58.82        | A | 575    | 1.50       | S | -15    |
| 12203  | 113  | 3 X 2.2 MT, STL       | 2.20        | 4.39      | 9.67           | C           | 6.79     | 66     | 58.82        | A | 569    | 3.00       | S | -29    |
| 12203  | 114  | 3 X 2.2 MT, STL       | 2.20        | 4.34      | 9.56           | C           | 6.82     | 65     | 58.82        | A | 562    | 4.50       | S | -43    |
| 12203  | 115  | 3 X 2.2 MT, STL       | 2.20        | 4.29      | 9.45           | C           | 6.84     | 65     | 58.82        | A | 556    | 6.00       | S | -57    |
| 12203  | 116  | 3 X 2.2 MT, STL       | 2.20        | 3.99      | 8.77           | C           | 6.99     | 61     | 58.82        | A | 516    | 7.50       | S | -66    |
| 12203  | 117  | 3 X 2.2 MT, STL       | 2.20        | 2.81      | 6.18           | C           | 7.58     | 47     | 58.82        | A | 364    | 9.00       | S | -56    |
| 12301  | 1    | TRUNKS AND ENCLOSURES | 719.76      | 1.00      | 719.76         | E           | 10.25    | 7378   | 25.25        | A | 18174  | 0.00       | C | 0      |
| 13101  | 1    | PLATE, 3/16 IN, STL   | 7.65        | 482.32    | 3689.76        | C           | 11.09    | 40905  | 14.23        | A | 52508  | 0.00       | C | 0      |
| 13101  |      |                       |             |           |                |             |          |        |              |   |        |            |   |        |
| 13102  | 2    | PLATE, 1/4 IN, STL    | 10.20       | 193.51    | 1973.81        | C           | 11.08    | 21865  | 30.49        | A | 60182  | 0.00       | C | 0      |
| 13102  | 3    | PLATE, 1/4 IN, STL    | 10.20       | 640.34    | 6531.46        | C           | 9.00     | 58790  | 49.31        | A | 322049 | 0.00       | C | 0      |
| 13101  |      |                       |             |           |                |             |          |        |              |   |        |            |   |        |
| 13103  | 4    | 6 X 4 X 7 T, STL      | 6.94        | 9.00      | 62.46          | C           | 10.84    | 677    | 1.19         | A | 74     | 0.00       | C | 0      |
| 13103  | 5    | 6 X 4 X 7 T, STL      | 6.94        | 9.00      | 62.46          | C           | 10.84    | 677    | 8.94         | A | 558    | 0.00       | C | 0      |
| 13103  |      |                       |             |           |                |             |          |        |              |   |        |            |   |        |
| 13104  | 6    | 7 X 5 X 11 T, STL     | 10.81       | 13.00     | 140.53         | C           | 10.76    | 1512   | 19.50        | A | 2740   | 0.00       | C | 0      |
| 13104  | 7    | 7 X 5 X 11 T, STL     | 10.81       | 26.15     | 282.71         | C           | 10.64    | 3009   | 19.48        | A | 5507   | 0.00       | C | 0      |

| SWBS # | ITEM | DESCRIPTION           | UNIT WEIGHT | NO. UNITS | CURRENT WEIGHT | E C A | VERTICAL |        | LONGITUDINAL |   |        | TRANSVERSE |   |        |
|--------|------|-----------------------|-------------|-----------|----------------|-------|----------|--------|--------------|---|--------|------------|---|--------|
|        |      |                       |             |           |                |       | VCG      | MOMENT | LCG          | S | MOMENT | TCG        | S | MOMENT |
|        |      |                       | lb          | #         | lb             |       | ft       | ft*lb  | ft           |   | ft*lb  | ft         |   | ft*lb  |
| 13104  | 8    | 7 X 5 X 11 T, STL     | 10.81       | 7.50      | 81.08          | C     | 8.54     | 692    | 38.75        | A | 3142   | 0.00       | C | 0      |
| 13104  | 9    | 7 X 5 X 11 T, STL     | 10.81       | 30.22     | 326.71         | C     | 8.56     | 2796   | 42.61        | A | 13919  | 0.00       | C | 0      |
| 13104  | 10   | 7 X 5 X 11 T, STL     | 10.81       | 18.37     | 198.58         | C     | 8.54     | 1695   | 54.50        | A | 10823  | 0.00       | C | 0      |
| 13104  | 11   | 7 X 5 X 11 T, STL     | 10.81       | 36.74     | 397.16         | C     | 8.56     | 3399   | 54.58        | A | 21678  | 0.00       | C | 0      |
| 13104  | 12   | 7 X 5 X 11 T, STL     | 10.81       | 9.37      | 101.29         | C     | 8.54     | 865    | 63.69        | A | 6451   | 0.00       | C | 0      |
| 13104  | 13   | 7 X 5 X 11 T, STL     | 10.81       | 16.31     | 176.31         | C     | 8.55     | 1508   | 61.97        | A | 10927  | 0.00       | C | 0      |
| 13104  |      |                       |             |           |                |       |          |        |              |   |        |            |   |        |
| 13105  | 14   | 6 X 4 X 7 T, STL      | 6.94        | 9.00      | 62.46          | C     | 10.83    | 677    | 30.50        | A | 1905   | 0.00       | C | 0      |
| 13105  | 15   | 6 X 4 X 7 T, STL      | 6.94        | 2.64      | 18.33          | C     | 9.74     | 179    | 38.69        | A | 709    | 0.00       | C | 0      |
| 13105  |      |                       |             |           |                |       |          |        |              |   |        |            |   |        |
| 13106  | 16   | 5 X 4 X 6 T, STL      | 5.88        | 18.00     | 105.84         | C     | 10.80    | 1143   | 8.50         | A | 900    | 0.00       | C | 0      |
| 13106  | 17   | 5 X 4 X 6 T, STL      | 5.88        | 18.00     | 105.84         | C     | 10.77    | 1140   | 30.50        | A | 3228   | 0.00       | C | 0      |
| 13106  | 18   | 5 X 4 X 6 T, STL      | 5.88        | 5.28      | 31.07          | C     | 9.74     | 303    | 38.69        | A | 1202   | 0.00       | C | 0      |
| 13106  |      |                       |             |           |                |       |          |        |              |   |        |            |   |        |
| 13107  | 19   | 12 X 6.5 X 26 MT, STL | 19.35       | 4.07      | 78.83          | C     | 8.33     | 657    | 47.00        | A | 3705   | 0.00       | C | 0      |
| 13107  |      |                       |             |           |                |       |          |        |              |   |        |            |   |        |
| 13108  | 20   | PLATE, 1/4 IN, STL    | 10.20       | 1.83      | 18.67          | C     | 8.60     | 160    | 47.00        | A | 877    | 0.00       | C | 0      |
| 13108  | 21   | PLATE, 1/4 IN, STL    | 10.20       | 1.01      | 10.33          | C     | 8.60     | 89     | 43.41        | A | 448    | 0.00       | C | 0      |
| 13108  |      |                       |             |           |                |       |          |        |              |   |        |            |   |        |
| 13109  | 22   | PLATE, 3/8 IN, STL    | 15.30       | 1.09      | 16.65          | C     | 8.22     | 137    | 47.00        | A | 783    | 0.00       | C | 0      |
| 13109  | 23   | PLATE, 3/8 IN, STL    | 15.30       | 0.65      | 9.98           | C     | 8.21     | 82     | 43.38        | A | 433    | 0.00       | C | 0      |
| 13109  |      |                       |             |           |                |       |          |        |              |   |        |            |   |        |
| 13110  | 24   | 3 X 2.2 MT, STL       | 2.20        | 1.72      | 3.78           | C     | 10.82    | 41     | 1.00         | F | -4     | 0.00       | C | 0      |
| 13110  | 25   | 3 X 2.2 MT, STL       | 2.20        | 6.71      | 14.76          | C     | 10.82    | 160    | 0.00         | A | 0      | 0.00       | C | 0      |
| 13110  | 26   | 3 X 2.2 MT, STL       | 2.20        | 9.21      | 20.26          | C     | 10.82    | 219    | 1.00         | A | 20     | 0.00       | C | 0      |
| 13110  | 27   | 3 X 2.2 MT, STL       | 2.20        | 11.00     | 24.20          | C     | 10.82    | 262    | 2.00         | A | 48     | 0.00       | C | 0      |
| 13110  | 28   | 3 X 2.2 MT, STL       | 2.20        | 12.42     | 27.32          | C     | 10.82    | 296    | 3.00         | A | 82     | 0.00       | C | 0      |
| 13110  | 29   | 3 X 2.2 MT, STL       | 2.20        | 14.62     | 32.16          | C     | 10.82    | 348    | 5.00         | A | 161    | 0.00       | C | 0      |
| 13110  | 30   | 3 X 2.2 MT, STL       | 2.20        | 15.54     | 34.19          | C     | 10.82    | 370    | 6.00         | A | 205    | 0.00       | C | 0      |
| 13110  | 31   | 3 X 2.2 MT, STL       | 2.20        | 16.40     | 36.08          | C     | 10.82    | 390    | 7.00         | A | 253    | 0.00       | C | 0      |
| 13110  | 32   | 3 X 2.2 MT, STL       | 2.20        | 17.21     | 37.86          | C     | 10.82    | 410    | 8.00         | A | 303    | 0.00       | C | 0      |
| 13110  | 33   | 3 X 2.2 MT, STL       | 2.20        | 17.95     | 39.49          | C     | 10.82    | 427    | 9.00         | A | 355    | 0.00       | C | 0      |
| 13110  | 34   | 3 X 2.2 MT, STL       | 2.20        | 18.62     | 40.96          | C     | 10.82    | 443    | 10.00        | A | 410    | 0.00       | C | 0      |
| 13110  | 35   | 3 X 2.2 MT, STL       | 2.20        | 19.19     | 42.22          | C     | 10.82    | 457    | 11.00        | A | 464    | 0.00       | C | 0      |
| 13110  | 36   | 3 X 2.2 MT, STL       | 2.20        | 19.66     | 43.25          | C     | 10.82    | 468    | 12.00        | A | 519    | 0.00       | C | 0      |
| 13110  | 37   | 3 X 2.2 MT, STL       | 2.20        | 20.37     | 44.81          | C     | 10.82    | 485    | 14.00        | A | 627    | 0.00       | C | 0      |
| 13110  | 38   | 3 X 2.2 MT, STL       | 2.20        | 20.62     | 45.36          | C     | 10.82    | 491    | 15.00        | A | 680    | 0.00       | C | 0      |
| 13110  | 39   | 3 X 2.2 MT, STL       | 2.20        | 20.83     | 45.83          | C     | 10.82    | 496    | 16.00        | A | 733    | 0.00       | C | 0      |
| 13110  | 40   | 3 X 2.2 MT, STL       | 2.20        | 21.00     | 46.20          | C     | 10.82    | 500    | 17.00        | A | 785    | 0.00       | C | 0      |
| 13110  | 41   | 3 X 2.2 MT, STL       | 2.20        | 21.15     | 46.53          | C     | 10.82    | 503    | 18.00        | A | 838    | 0.00       | C | 0      |
| 13110  | 42   | 3 X 2.2 MT, STL       | 2.20        | 21.25     | 46.75          | C     | 10.82    | 506    | 19.00        | A | 888    | 0.00       | C | 0      |

| SWBS # | ITEM | DESCRIPTION         | UNIT WEIGHT | NO. UNITS | CURRENT WEIGHT | E | VERTICAL |        | LONGITUDINAL |   |        | TRANSVERSE |   |        |
|--------|------|---------------------|-------------|-----------|----------------|---|----------|--------|--------------|---|--------|------------|---|--------|
|        |      |                     |             |           |                |   | VCG      | MOMENT | LCG          | S | MOMENT | TCG        | S | MOMENT |
|        |      |                     | lb          | #         | lb             | A | ft       | ft*lb  | ft           |   | ft*lb  | ft         |   | ft*lb  |
| 13110  | 43   | 3 X 2.2 MT, STL     | 2.20        | 21.28     | 46.82          | C | 10.82    | 507    | 20.00        | A | 936    | 0.00       | C | 0      |
| 13110  | 44   | 3 X 2.2 MT, STL     | 2.20        | 21.28     | 46.82          | C | 10.82    | 507    | 21.00        | A | 983    | 0.00       | C | 0      |
| 13110  | 45   | 3 X 2.2 MT, STL     | 2.20        | 21.29     | 46.84          | C | 10.82    | 507    | 22.00        | A | 1030   | 0.00       | C | 0      |
| 13110  | 46   | 3 X 2.2 MT, STL     | 2.20        | 21.29     | 46.84          | C | 10.82    | 507    | 23.00        | A | 1077   | 0.00       | C | 0      |
| 13110  | 47   | 3 X 2.2 MT, STL     | 2.20        | 21.30     | 46.86          | C | 10.82    | 507    | 24.00        | A | 1125   | 0.00       | C | 0      |
| 13110  | 48   | 3 X 2.2 MT, STL     | 2.20        | 21.30     | 46.86          | C | 10.82    | 507    | 25.00        | A | 1172   | 0.00       | C | 0      |
| 13110  | 49   | 3 X 2.2 MT, STL     | 2.20        | 21.30     | 46.86          | C | 10.82    | 507    | 27.00        | A | 1265   | 0.00       | C | 0      |
| 13110  | 50   | 3 X 2.2 MT, STL     | 2.20        | 21.30     | 46.86          | C | 10.82    | 507    | 28.00        | A | 1312   | 0.00       | C | 0      |
| 13110  | 51   | 3 X 2.2 MT, STL     | 2.20        | 21.30     | 46.86          | C | 10.82    | 507    | 29.00        | A | 1359   | 0.00       | C | 0      |
| 13110  | 52   | 3 X 2.2 MT, STL     | 2.20        | 21.30     | 46.86          | C | 10.82    | 507    | 30.50        | A | 1429   | 0.00       | C | 0      |
| 13110  | 53   | 3 X 2.2 MT, STL     | 2.20        | 21.31     | 46.88          | C | 10.82    | 507    | 32.00        | A | 1500   | 0.00       | C | 0      |
| 13110  | 54   | 3 X 2.2 MT, STL     | 2.20        | 21.31     | 46.88          | C | 10.82    | 507    | 33.50        | A | 1571   | 0.00       | C | 0      |
| 13110  |      |                     |             |           |                |   |          |        |              |   |        |            |   |        |
| 13111  | 55   | 4 X 3.25 MT, STL    | 3.25        | 20.98     | 68.19          | C | 8.77     | 598    | 36.50        | A | 2489   | 0.00       | C | 0      |
| 13111  | 56   | 4 X 3.25 MT, STL    | 3.25        | 20.96     | 68.12          | C | 8.77     | 597    | 38.00        | A | 2589   | 0.00       | C | 0      |
| 13111  | 57   | 4 X 3.25 MT, STL    | 3.25        | 20.94     | 68.06          | C | 8.77     | 597    | 39.50        | A | 2688   | 0.00       | C | 0      |
| 13111  | 58   | 4 X 3.25 MT, STL    | 3.25        | 20.92     | 67.99          | C | 8.77     | 596    | 41.00        | A | 2788   | 0.00       | C | 0      |
| 13111  | 59   | 4 X 3.25 MT, STL    | 3.25        | 20.90     | 67.93          | C | 8.77     | 596    | 42.50        | A | 2887   | 0.00       | C | 0      |
| 13111  | 60   | 4 X 3.25 MT, STL    | 3.25        | 20.88     | 67.86          | C | 8.77     | 595    | 44.00        | A | 2986   | 0.00       | C | 0      |
| 13111  | 61   | 4 X 3.25 MT, STL    | 3.25        | 16.36     | 53.17          | C | 8.77     | 466    | 45.50        | A | 2419   | 0.00       | C | 0      |
| 13111  | 62   | 4 X 3.25 MT, STL    | 3.25        | 13.34     | 43.36          | C | 8.77     | 380    | 47.00        | A | 2038   | 0.00       | C | 0      |
| 13111  | 63   | 4 X 3.25 MT, STL    | 3.25        | 16.31     | 53.01          | C | 8.77     | 465    | 48.50        | A | 2571   | 0.00       | C | 0      |
| 13111  | 64   | 4 X 3.25 MT, STL    | 3.25        | 20.75     | 67.44          | C | 8.77     | 591    | 51.50        | A | 3473   | 0.00       | C | 0      |
| 13111  | 65   | 4 X 3.25 MT, STL    | 3.25        | 20.72     | 67.34          | C | 8.77     | 591    | 53.00        | A | 3569   | 0.00       | C | 0      |
| 13111  | 66   | 4 X 3.25 MT, STL    | 3.25        | 20.65     | 67.11          | C | 8.77     | 589    | 54.50        | A | 3658   | 0.00       | C | 0      |
| 13111  | 67   | 4 X 3.25 MT, STL    | 3.25        | 20.48     | 66.56          | C | 8.77     | 584    | 56.00        | A | 3727   | 0.00       | C | 0      |
| 13111  | 68   | 4 X 3.25 MT, STL    | 3.25        | 20.17     | 65.55          | C | 8.77     | 575    | 57.50        | A | 3769   | 0.00       | C | 0      |
| 13111  | 69   | 4 X 3.25 MT, STL    | 3.25        | 18.94     | 61.56          | C | 8.77     | 540    | 60.50        | A | 3724   | 0.00       | C | 0      |
| 13111  | 70   | 4 X 3.25 MT, STL    | 3.25        | 17.92     | 58.24          | C | 8.77     | 511    | 62.00        | A | 3611   | 0.00       | C | 0      |
| 13111  | 71   | 4 X 3.25 MT, STL    | 3.25        | 16.54     | 53.76          | C | 8.77     | 471    | 63.50        | A | 3413   | 0.00       | C | 0      |
| 13111  | 72   | 4 X 3.25 MT, STL    | 3.25        | 14.65     | 47.61          | C | 8.77     | 418    | 65.00        | A | 3095   | 0.00       | C | 0      |
| 13111  | 73   | 4 X 3.25 MT, STL    | 3.25        | 11.88     | 38.61          | C | 8.77     | 339    | 66.50        | A | 2568   | 0.00       | C | 0      |
| 13111  | 74   | 4 X 3.25 MT, STL    | 3.25        | 6.45      | 20.96          | C | 8.77     | 184    | 68.00        | A | 1425   | 0.00       | C | 0      |
| 13111  |      |                     |             |           |                |   |          |        |              |   |        |            |   |        |
| 13112  | 75   | PLATE, 1/4 IN, STL  | 10.20       | 1.62      | 16.50          | C | 8.60     | 142    | 47.00        | A | 776    | 0.00       | C | 0      |
| 13112  | 76   | PLATE, 1/4 IN, STL  | 10.20       | 2.35      | 24.01          | C | 8.60     | 206    | 47.00        | A | 1128   | 0.00       | C | 0      |
| 13113  |      |                     |             |           |                |   |          |        |              |   |        |            |   |        |
| 13113  | 133  | PLATE, 3/8 IN, STL  | 15.30       | 1.62      | 24.75          | C | 8.22     | 204    | 47.00        | A | 1163   | 0.00       | C | 0      |
| 13113  | 134  | PLATE, 3/8 IN, STL  | 15.30       | 1.49      | 22.87          | C | 8.21     | 188    | 47.00        | A | 1075   | 0.00       | C | 0      |
|        |      |                     |             |           |                |   |          |        |              |   |        |            |   |        |
| 14901  | 1    | PLATE, 3/16 IN, STL | 7.65        | 63.11     | 482.81         | C | 3.50     | 1690   | 10.00        | A | 4827   | 0.00       | C | 0      |

| SWBS # | ITEM | DESCRIPTION         | UNIT WEIGHT | NO. UNITS | CURRENT WEIGHT | E C A | VERTICAL |        | LONGITUDINAL |   |        | TRANSVERSE |   |        |
|--------|------|---------------------|-------------|-----------|----------------|-------|----------|--------|--------------|---|--------|------------|---|--------|
|        |      |                     |             |           |                |       | VCG      | MOMENT | LCG          | S | MOMENT | TCG        | S | MOMENT |
|        |      |                     | lb          | #         | lb             |       | ft       | ft*lb  | ft           |   | ft*lb  | ft         |   | ft*lb  |
| 14901  |      |                     |             |           |                |       |          |        |              |   |        |            |   |        |
| 14902  | 2    | PLATE, 5/16 IN, STL | 12.75       | 174.21    | 2221.22        | C     | 3.50     | 7774   | 30.49        | A | 67734  | 0.00       | C | 0      |
| 14902  |      |                     |             |           |                |       |          |        |              |   |        |            |   |        |
| 14903  | 3    | 3 X 2.2# MT, STL    | 2.20        | 0.99      | 2.18           | C     | 3.43     | 7      | 5.00         | A | 11     | 0.00       | C | 0      |
| 14903  | 4    | 3 X 2.2# MT, STL    | 2.20        | 3.11      | 6.85           | C     | 3.33     | 23     | 6.00         | A | 41     | 0.00       | C | 0      |
| 14903  | 5    | 3 X 2.2# MT, STL    | 2.20        | 4.99      | 10.98          | C     | 3.33     | 37     | 7.00         | A | 77     | 0.00       | C | 0      |
| 14903  | 6    | 3 X 2.2# MT, STL    | 2.20        | 6.68      | 14.70          | C     | 3.33     | 49     | 8.00         | A | 118    | 0.00       | C | 0      |
| 14903  | 7    | 3 X 2.2# MT, STL    | 2.20        | 8.22      | 18.09          | C     | 3.33     | 60     | 9.00         | A | 163    | 0.00       | C | 0      |
| 14903  | 8    | 3 X 2.2# MT, STL    | 2.20        | 9.65      | 21.23          | C     | 3.33     | 71     | 10.00        | A | 212    | 0.00       | C | 0      |
| 14903  | 9    | 3 X 2.2# MT, STL    | 2.20        | 10.98     | 24.16          | C     | 3.33     | 80     | 11.00        | A | 266    | 0.00       | C | 0      |
| 14903  | 10   | 3 X 2.2# MT, STL    | 2.20        | 12.24     | 26.92          | C     | 3.33     | 90     | 12.00        | A | 323    | 0.00       | C | 0      |
| 14903  |      |                     |             |           |                |       |          |        |              |   |        |            |   |        |
| 14904  | 11   | 4 X 3.25# MT, STL   | 3.25        | 19.38     | 62.97          | C     | 3.32     | 209    | 27.00        | A | 1700   | 0.00       | C | 0      |
| 14904  | 12   | 4 X 3.25# MT, STL   | 3.25        | 19.41     | 63.09          | C     | 3.32     | 209    | 28.00        | A | 1766   | 0.00       | C | 0      |
| 14904  | 13   | 4 X 3.25# MT, STL   | 3.25        | 19.42     | 63.13          | C     | 3.32     | 210    | 29.00        | A | 1831   | 0.00       | C | 0      |
| 14904  | 14   | 4 X 3.25# MT, STL   | 3.25        | 19.41     | 63.08          | C     | 3.32     | 209    | 30.50        | A | 1924   | 0.00       | C | 0      |
| 14904  | 15   | 4 X 3.25# MT, STL   | 3.25        | 19.36     | 62.91          | C     | 3.32     | 209    | 32.00        | A | 2013   | 0.00       | C | 0      |
| 14904  | 16   | 4 X 3.25# MT, STL   | 3.25        | 19.28     | 62.67          | C     | 3.32     | 208    | 33.50        | A | 2100   | 0.00       | C | 0      |
| 14904  |      |                     |             |           |                |       |          |        |              |   |        |            | C |        |
|        |      |                     |             |           |                |       |          |        |              |   |        |            |   |        |
| 15101  | 1    | PLATE, 3/16 IN, STL | 7.65        | 275.34    | 2106.33        | C     | 19.33    | 40722  | 25.25        | A | 53180  | 0.00       | C | 0      |
| 15101  | 2    | PLATE, 3/16 IN, STL | 7.65        | 109.75    | 839.56         | C     | 14.93    | 12533  | 28.77        | A | 24157  | 6.83       | P | 5737   |
| 15101  | 3    | PLATE, 3/16 IN, STL | 7.65        | 109.75    | 839.56         | C     | 14.93    | 12533  | 28.77        | A | 24157  | 6.83       | S | -5737  |
| 15101  | 4    | PLATE, 3/16 IN, STL | 7.65        | 9.45      | 72.28          | C     | 12.24    | 885    | 29.70        | A | 2147   | 5.83       | P | 422    |
| 15101  | 5    | PLATE, 3/16 IN, STL | 7.65        | 2.15      | 16.46          | C     | 12.20    | 201    | 27.50        | A | 453    | 6.33       | P | 104    |
| 15101  | 6    | PLATE, 3/16 IN, STL | 7.65        | 4.31      | 32.95          | C     | 13.29    | 438    | 29.65        | A | 977    | 6.33       | P | 209    |
| 15101  | 7    | PLATE, 3/16 IN, STL | 7.65        | 9.45      | 72.28          | C     | 12.24    | 885    | 29.70        | A | 2147   | 5.83       | S | -422   |
| 15101  | 8    | PLATE, 3/16 IN, STL | 7.65        | 2.15      | 16.46          | C     | 12.20    | 201    | 27.50        | A | 453    | 6.33       | S | -104   |
| 15101  | 9    | PLATE, 3/16 IN, STL | 7.65        | 4.31      | 32.95          | C     | 13.29    | 438    | 29.65        | A | 977    | 6.33       | S | -209   |
| 15101  | 10   | PLATE, 3/16 IN, STL | 7.65        | 12.78     | 97.80          | C     | 11.58    | 1133   | 36.30        | A | 3550   | 4.50       | P | 440    |
| 15101  | 11   | PLATE, 3/16 IN, STL | 7.65        | 12.78     | 97.80          | C     | 11.58    | 1133   | 36.30        | A | 3550   | 4.50       | S | -440   |
| 15101  | 12   | PLATE, 3/16 IN, STL | 7.65        | 33.18     | 253.86         | C     | 13.27    | 3369   | 35.00        | A | 8885   | 0.80       | S | -204   |
| 15101  | 13   | PLATE, 3/16 IN, STL | 7.65        | 32.88     | 251.52         | C     | 17.55    | 4414   | 35.81        | A | 9006   | 0.06       | P | 14     |
| 15101  | 14   | PLATE, 3/16 IN, STL | 7.65        | 10.71     | 81.96          | C     | 13.60    | 1114   | 36.50        | A | 2992   | 5.67       | P | 464    |
| 15101  | 15   | PLATE, 3/16 IN, STL | 7.65        | 5.89      | 45.04          | C     | 10.60    | 477    | 38.00        | A | 1712   | 5.67       | P | 255    |
| 15101  | 16   | PLATE, 3/16 IN, STL | 7.65        | 10.71     | 81.96          | C     | 13.60    | 1114   | 36.50        | A | 2992   | 5.67       | S | -464   |
| 15101  | 17   | PLATE, 3/16 IN, STL | 7.65        | 5.89      | 45.04          | C     | 10.60    | 477    | 38.00        | A | 1712   | 5.67       | S | -255   |
| 15101  |      |                     |             |           |                |       |          |        |              |   |        |            |   |        |
| 15102  | 18   | PLATE, 1/8 IN, STL  | 5.10        | 13.80     | 70.36          | C     | 13.40    | 943    | 33.40        | A | 2350   | 4.50       | P | 317    |
| 15102  | 19   | PLATE, 1/8 IN, STL  | 5.10        | 4.06      | 20.71          | C     | 16.01    | 332    | 33.37        | A | 691    | 4.29       | P | 89     |
| 15102  | 20   | PLATE, 1/8 IN, STL  | 5.10        | 13.21     | 67.37          | C     | 17.98    | 1212   | 33.62        | A | 2265   | 4.08       | P | 275    |

| SWBS # | ITEM | DESCRIPTION        | UNIT WEIGHT | NO. UNITS | CURRENT WEIGHT | E | VERTICAL |        | LONGITUDINAL |   |        | TRANSVERSE |   |        |
|--------|------|--------------------|-------------|-----------|----------------|---|----------|--------|--------------|---|--------|------------|---|--------|
|        |      |                    |             |           |                |   | VCG      | MOMENT | LCG          | S | MOMENT | TCG        | S | MOMENT |
|        |      |                    | lb          | #         | lb             | A | ft       | ft*lb  | ft           |   | ft*lb  | ft         |   | ft*lb  |
| 15102  | 21   | PLATE, 1/8 IN, STL | 5.10        | 13.80     | 70.36          | C | 13.40    | 943    | 33.40        | A | 2350   | 4.50       | S | -317   |
| 15102  | 22   | PLATE, 1/8 IN, STL | 5.10        | 4.06      | 20.71          | C | 16.01    | 332    | 33.37        | A | 691    | 4.29       | S | -89    |
| 15102  | 23   | PLATE, 1/8 IN, STL | 5.10        | 13.21     | 67.37          | C | 17.98    | 1212   | 33.62        | A | 2265   | 4.08       | S | -275   |
| 15102  | 24   | PLATE, 1/8 IN, STL | 5.10        | 10.28     | 52.43          | C | 13.35    | 700    | 28.65        | A | 1502   | 5.67       | P | 297    |
| 15102  | 25   | PLATE, 1/8 IN, STL | 5.10        | 2.54      | 12.96          | C | 16.00    | 207    | 28.36        | A | 368    | 5.56       | P | 72     |
| 15102  | 26   | PLATE, 1/8 IN, STL | 5.10        | 7.83      | 39.95          | C | 17.92    | 716    | 28.18        | A | 1126   | 5.46       | P | 218    |
| 15102  | 27   | PLATE, 1/8 IN, STL | 5.10        | 10.28     | 52.43          | C | 13.35    | 700    | 28.65        | A | 1502   | 5.67       | S | -297   |
| 15102  | 28   | PLATE, 1/8 IN, STL | 5.10        | 2.54      | 12.96          | C | 16.00    | 207    | 28.36        | A | 368    | 5.56       | S | -72    |
| 15102  | 29   | PLATE, 1/8 IN, STL | 5.10        | 7.83      | 39.95          | C | 17.92    | 716    | 28.18        | A | 1126   | 5.46       | S | -218   |
| 15102  | 30   | PLATE, 1/8 IN, STL | 5.10        | 47.04     | 239.88         | C | 19.19    | 4604   | 26.07        | A | 6253   | 0.00       | C | 0      |
| 15102  | 31   | PLATE, 1/8 IN, STL | 5.10        | 0.08      | 0.40           | C | 19.17    | 8      | 14.30        | A | 6      | 0.00       | C | 0      |
| 15102  | 32   | PLATE, 1/8 IN, STL | 5.10        | 0.08      | 0.40           | C | 19.17    | 8      | 36.63        | A | 15     | 0.00       | C | 0      |
| 15102  | 33   | PLATE, 1/8 IN, STL | 5.10        | 0.08      | 0.40           | C | 19.17    | 8      | 36.63        | A | 15     | 4.08       | P | 2      |
| 15102  | 34   | PLATE, 1/8 IN, STL | 5.10        | 0.08      | 0.40           | C | 19.17    | 8      | 36.63        | A | 15     | 4.08       | S | -2     |
| 15102  | 35   | PLATE, 1/8 IN, STL | 5.10        | 0.07      | 0.34           | C | 19.17    | 7      | 20.00        | A | 7      | 7.01       | P | 2      |
| 15102  | 36   | PLATE, 1/8 IN, STL | 5.10        | 0.07      | 0.34           | C | 19.17    | 7      | 26.00        | A | 9      | 7.01       | P | 2      |
| 15102  | 37   | PLATE, 1/8 IN, STL | 5.10        | 0.07      | 0.34           | C | 19.17    | 7      | 31.17        | A | 11     | 7.01       | P | 2      |
| 15102  | 38   | PLATE, 1/8 IN, STL | 5.10        | 0.07      | 0.34           | C | 19.17    | 7      | 20.00        | A | 7      | 7.01       | S | -2     |
| 15102  | 39   | PLATE, 1/8 IN, STL | 5.10        | 0.07      | 0.34           | C | 19.17    | 7      | 26.00        | A | 9      | 7.01       | S | -2     |
| 15102  | 40   | PLATE, 1/8 IN, STL | 5.10        | 0.07      | 0.34           | C | 19.17    | 7      | 31.17        | A | 11     | 7.01       | S | -2     |
| 15102  |      |                    |             |           |                |   |          |        |              |   |        |            |   |        |
| 15103  | 41   | PLATE, 1/4 IN, STL | 10.20       | 27.01     | 275.54         | C | 13.31    | 3667   | 16.00        | A | 4409   | 0.00       | C | 0      |
| 15103  | 42   | PLATE, 1/4 IN, STL | 10.20       | 15.63     | 159.38         | C | 17.99    | 2867   | 15.03        | A | 2396   | 0.00       | C | 0      |
| 15103  | 43   | PLATE, 1/4 IN, STL | 10.20       | 19.17     | 195.55         | C | 13.25    | 2592   | 17.50        | A | 3422   | 5.08       | P | 994    |
| 15103  | 44   | PLATE, 1/4 IN, STL | 10.20       | 8.46      | 86.27          | C | 17.90    | 1545   | 16.56        | A | 1428   | 5.27       | P | 455    |
| 15103  | 45   | PLATE, 1/4 IN, STL | 10.20       | 19.17     | 195.55         | C | 13.25    | 2592   | 17.50        | A | 3422   | 5.08       | S | -994   |
| 15103  | 46   | PLATE, 1/4 IN, STL | 10.20       | 8.46      | 86.27          | C | 17.90    | 1545   | 16.56        | A | 1428   | 5.27       | S | -455   |
| 15103  | 47   | PLATE, 1/4 IN, STL | 10.20       | 0.27      | 2.80           | C | 18.61    | 52     | 25.68        | A | 72     | 0.00       | C | 0      |
| 15103  | 48   | PLATE, 1/4 IN, STL | 10.20       | 0.27      | 2.80           | C | 18.61    | 52     | 26.26        | A | 73     | 0.00       | C | 0      |
| 15103  | 49   | PLATE, 1/4 IN, STL | 10.20       | 0.03      | 0.26           | C | 18.93    | 5      | 26.00        | A | 7      | 0.13       | P | 0      |
| 15103  | 50   | PLATE, 1/4 IN, STL | 10.20       | 0.03      | 0.26           | C | 18.93    | 5      | 26.00        | A | 7      | 0.13       | S | 0      |
| 15103  |      |                    |             |           |                |   |          |        |              |   |        |            |   |        |
| 15104  | 51   | 6 X 7 WT, STL      | 7.07        | 21.73     | 153.63         | C | 18.98    | 2916   | 25.50        | A | 3917   | 0.00       | C | 0      |
| 15104  |      |                    |             |           |                |   |          |        |              |   |        |            |   |        |
| 15105  | 52   | 5 X 6 WT, STL      | 6.02        | 6.67      | 40.12          | C | 15.08    | 605    | 16.30        | A | 654    | 0.00       | C | 0      |
| 15105  |      |                    |             |           |                |   |          |        |              |   |        |            |   |        |
| 15106  | 53   | 5 X 3.75 MT, STL   | 3.75        | 7.43      | 27.84          | C | 18.22    | 507    | 15.47        | A | 431    | 0.00       | C | 0      |
| 15106  |      |                    |             |           |                |   |          |        |              |   |        |            |   |        |
| 15107  | 54   | 4 X 3.1 MT, STL    | 3.10        | 13.67     | 42.33          | C | 19.10    | 808    | 20.00        | A | 847    | 0.00       | C | 0      |
| 15107  | 55   | 4 X 3.1 MT, STL    | 3.10        | 13.67     | 42.33          | C | 19.10    | 808    | 26.00        | A | 1100   | 0.00       | C | 0      |
| 15107  | 56   | 4 X 3.1 MT, STL    | 3.10        | 8.17      | 25.29          | C | 19.10    | 483    | 31.17        | A | 788    | 0.00       | C | 0      |

| SWBS # | ITEM | DESCRIPTION        | UNIT WEIGHT | NO. UNITS | CURRENT WEIGHT | E | VERTICAL |        | LONGITUDINAL |   |        | TRANSVERSE |   |        |
|--------|------|--------------------|-------------|-----------|----------------|---|----------|--------|--------------|---|--------|------------|---|--------|
|        |      |                    |             |           |                |   | VCG      | MOMENT | LCG          | S | MOMENT | TCG        | S | MOMENT |
|        |      |                    | lb          | #         | lb             | A | ft       | ft*lb  | ft           |   | ft*lb  | ft         |   | ft*lb  |
| 15107  | 57   | 4 X 3.1 MT, STL    | 3.10        | 4.61      | 14.28          | C | 15.08    | 215    | 17.71        | A | 253    | 4.99       | P | 71     |
| 15107  | 58   | 4 X 3.1 MT, STL    | 3.10        | 4.61      | 14.28          | C | 15.08    | 215    | 17.71        | A | 253    | 4.99       | S | -71    |
| 15107  | 59   | 4 X 3.1 MT, STL    | 3.10        | 4.11      | 12.72          | C | 18.22    | 232    | 16.52        | A | 210    | 5.28       | P | 67     |
| 15107  | 60   | 4 X 3.1 MT, STL    | 3.10        | 4.11      | 12.72          | C | 18.22    | 232    | 16.52        | A | 210    | 5.28       | S | -67    |
| 15107  | 61   | 4 X 3.1 MT, STL    | 3.10        | 7.88      | 24.40          | C | 15.06    | 368    | 20.00        | A | 488    | 6.60       | P | 161    |
| 15107  | 62   | 4 X 3.1 MT, STL    | 3.10        | 7.88      | 24.40          | C | 15.06    | 368    | 23.00        | A | 561    | 6.60       | P | 161    |
| 15107  | 63   | 4 X 3.1 MT, STL    | 3.10        | 7.88      | 24.40          | C | 15.06    | 368    | 26.00        | A | 635    | 6.60       | P | 161    |
| 15107  | 64   | 4 X 3.1 MT, STL    | 3.10        | 5.71      | 17.69          | C | 16.14    | 286    | 29.00        | A | 513    | 6.60       | P | 117    |
| 15107  | 65   | 4 X 3.1 MT, STL    | 3.10        | 7.88      | 24.40          | C | 15.06    | 368    | 20.00        | A | 488    | 6.60       | S | -161   |
| 15107  | 66   | 4 X 3.1 MT, STL    | 3.10        | 7.88      | 24.40          | C | 15.06    | 368    | 23.00        | A | 561    | 6.60       | S | -161   |
| 15107  | 67   | 4 X 3.1 MT, STL    | 3.10        | 7.88      | 24.40          | C | 15.06    | 368    | 26.00        | A | 635    | 6.60       | S | -161   |
| 15107  | 68   | 4 X 3.1 MT, STL    | 3.10        | 5.71      | 17.69          | C | 16.14    | 286    | 29.00        | A | 513    | 6.60       | S | -117   |
| 15107  | 69   | 4 X 3.1 MT, STL    | 3.10        | 4.05      | 12.55          | C | 13.31    | 167    | 34.77        | A | 436    | 1.67       | P | 21     |
| 15107  | 70   | 4 X 3.1 MT, STL    | 3.10        | 4.26      | 13.19          | C | 17.33    | 229    | 35.78        | A | 472    | 1.67       | P | 22     |
| 15107  | 71   | 4 X 3.1 MT, STL    | 3.10        | 6.17      | 19.10          | C | 15.08    | 288    | 34.77        | A | 664    | 1.42       | S | -27    |
| 15107  | 72   | 4 X 3.1 MT, STL    | 3.10        | 5.75      | 17.81          | C | 18.15    | 323    | 35.78        | A | 637    | 1.21       | S | -22    |
| 15107  |      |                    |             |           |                |   |          |        |              |   |        |            |   |        |
| 15108  | 73   | 3 X 1.85 MT, STL   | 1.85        | 3.75      | 6.95           | C | 13.21    | 92     | 15.18        | A | 105    | 0.00       | C | 0      |
| 15108  | 74   | 3 X 1.85 MT, STL   | 1.85        | 3.98      | 7.37           | C | 17.08    | 126    | 15.54        | A | 114    | 0.00       | C | 0      |
| 15108  | 75   | 3 X 1.85 MT, STL   | 1.85        | 3.80      | 7.05           | C | 13.18    | 93     | 15.18        | A | 107    | 1.67       | P | 12     |
| 15108  | 76   | 3 X 1.85 MT, STL   | 1.85        | 3.89      | 7.21           | C | 13.14    | 95     | 17.16        | A | 124    | 4.47       | P | 32     |
| 15108  | 77   | 3 X 1.85 MT, STL   | 1.85        | 3.93      | 7.28           | C | 13.12    | 96     | 18.24        | A | 133    | 5.74       | P | 42     |
| 15108  | 78   | 3 X 1.85 MT, STL   | 1.85        | 3.80      | 7.05           | C | 13.18    | 93     | 15.18        | A | 107    | 1.67       | S | -12    |
| 15108  | 79   | 3 X 1.85 MT, STL   | 1.85        | 3.89      | 7.21           | C | 13.14    | 95     | 17.16        | A | 124    | 4.47       | S | -32    |
| 15108  | 80   | 3 X 1.85 MT, STL   | 1.85        | 3.93      | 7.28           | C | 13.12    | 96     | 18.24        | A | 133    | 5.74       | S | -42    |
| 15108  |      |                    |             |           |                |   |          |        |              |   |        |            |   |        |
| 15109  | 81   | 2.5 X 3/16 FB, STL | 1.59        | 21.98     | 35.03          | C | 19.23    | 674    | 25.47        | A | 892    | 1.38       | P | 48     |
| 15109  | 82   | 2.5 X 3/16 FB, STL | 1.59        | 21.98     | 35.03          | C | 19.23    | 674    | 25.47        | A | 892    | 2.75       | P | 96     |
| 15109  | 83   | 2.5 X 3/16 FB, STL | 1.59        | 21.82     | 34.77          | C | 19.23    | 669    | 25.55        | A | 888    | 4.08       | P | 142    |
| 15109  | 84   | 2.5 X 3/16 FB, STL | 1.59        | 15.35     | 24.46          | C | 19.23    | 470    | 23.49        | A | 575    | 6.83       | P | 167    |
| 15109  | 85   | 2.5 X 3/16 FB, STL | 1.59        | 21.98     | 35.03          | C | 19.23    | 674    | 25.47        | A | 892    | 1.38       | S | -48    |
| 15109  | 86   | 2.5 X 3/16 FB, STL | 1.59        | 21.98     | 35.03          | C | 19.23    | 674    | 25.47        | A | 892    | 2.75       | S | -96    |
| 15109  | 87   | 2.5 X 3/16 FB, STL | 1.59        | 21.82     | 34.77          | C | 19.23    | 669    | 25.55        | A | 888    | 4.08       | S | -142   |
| 15109  | 88   | 2.5 X 3/16 FB, STL | 1.59        | 15.35     | 24.46          | C | 19.23    | 470    | 23.49        | A | 575    | 6.83       | S | -167   |
| 15109  | 89   | 2.5 X 3/16 FB, STL | 1.59        | 2.75      | 4.38           | C | 19.23    | 84     | 35.17        | A | 154    | 5.46       | P | 24     |
| 15109  | 90   | 2.5 X 3/16 FB, STL | 1.59        | 2.75      | 4.38           | C | 19.23    | 84     | 35.17        | A | 154    | 5.46       | S | -24    |
| 15109  |      |                    |             |           |                |   |          |        |              |   |        |            |   |        |
| 15110  | 91   | 2.25 X 1/4 FB, STL | 1.91        | 2.91      | 5.57           | C | 11.15    | 62     | 36.50        | A | 203    | 6.74       | P | 38     |
| 15110  | 92   | 2.25 X 1/4 FB, STL | 1.91        | 5.33      | 10.20          | C | 12.79    | 130    | 34.53        | A | 352    | 6.74       | P | 69     |
| 15110  | 93   | 2.25 X 1/4 FB, STL | 1.91        | 4.09      | 7.82           | C | 14.20    | 111    | 33.72        | A | 264    | 6.74       | P | 53     |
| 15110  | 94   | 2.25 X 1/4 FB, STL | 1.91        | 3.79      | 7.26           | C | 16.12    | 117    | 33.39        | A | 242    | 6.74       | P | 49     |

| SWBS<br># | ITEM | DESCRIPTION        | UNIT<br>WEIGHT | NO.<br>UNITS | CURRENT<br>WEIGHT | E<br>C<br>A | VERTICAL  |                 | LONGITUDINAL |   |                 | TRANSVERSE |   |                 |
|-----------|------|--------------------|----------------|--------------|-------------------|-------------|-----------|-----------------|--------------|---|-----------------|------------|---|-----------------|
|           |      |                    |                |              |                   |             | VCG<br>ft | MOMENT<br>ft*lb | LCG<br>ft    | S | MOMENT<br>ft*lb | TCG<br>ft  | S | MOMENT<br>ft*lb |
| 15110     | 95   | 2.25 X 1/4 FB, STL | 1.91           | 4.61         | 8.81              | C           | 17.79     | 157             | 33.59        | A | 296             | 6.74       | P | 59              |
| 15110     | 96   | 2.25 X 1/4 FB, STL | 1.91           | 2.91         | 5.57              | C           | 11.19     | 62              | 36.50        | A | 203             | 4.59       | P | 26              |
| 15110     | 97   | 2.25 X 1/4 FB, STL | 1.91           | 5.33         | 10.20             | C           | 12.79     | 130             | 34.53        | A | 352             | 4.59       | P | 47              |
| 15110     | 98   | 2.25 X 1/4 FB, STL | 1.91           | 4.09         | 7.82              | C           | 14.20     | 111             | 33.72        | A | 264             | 4.59       | P | 36              |
| 15110     | 99   | 2.25 X 1/4 FB, STL | 1.91           | 3.79         | 7.26              | C           | 16.12     | 117             | 33.39        | A | 242             | 4.34       | P | 31              |
| 15110     | 100  | 2.25 X 1/4 FB, STL | 1.91           | 4.61         | 8.81              | C           | 17.79     | 157             | 33.59        | A | 296             | 4.18       | P | 37              |
| 15110     | 101  | 2.25 X 1/4 FB, STL | 1.91           | 2.91         | 5.57              | C           | 11.15     | 62              | 36.50        | A | 203             | 6.74       | S | -38             |
| 15110     | 102  | 2.25 X 1/4 FB, STL | 1.91           | 5.33         | 10.20             | C           | 12.79     | 130             | 34.53        | A | 352             | 6.74       | S | -69             |
| 15110     | 103  | 2.25 X 1/4 FB, STL | 1.91           | 4.09         | 7.82              | C           | 14.20     | 111             | 33.72        | A | 264             | 6.74       | S | -53             |
| 15110     | 104  | 2.25 X 1/4 FB, STL | 1.91           | 3.79         | 7.26              | C           | 16.12     | 117             | 33.39        | A | 242             | 6.74       | S | -49             |
| 15110     | 105  | 2.25 X 1/4 FB, STL | 1.91           | 4.61         | 8.81              | C           | 17.79     | 157             | 33.59        | A | 296             | 6.74       | S | -59             |
| 15110     | 106  | 2.25 X 1/4 FB, STL | 1.91           | 2.91         | 5.57              | C           | 11.19     | 62              | 36.50        | A | 203             | 4.59       | S | -26             |
| 15110     | 107  | 2.25 X 1/4 FB, STL | 1.91           | 5.33         | 10.20             | C           | 12.79     | 130             | 34.53        | A | 352             | 4.59       | S | -47             |
| 15110     | 108  | 2.25 X 1/4 FB, STL | 1.91           | 4.09         | 7.82              | C           | 14.20     | 111             | 33.72        | A | 264             | 4.59       | S | -36             |
| 15110     | 109  | 2.25 X 1/4 FB, STL | 1.91           | 3.79         | 7.26              | C           | 16.12     | 117             | 33.39        | A | 242             | 4.34       | S | -31             |
| 15110     | 110  | 2.25 X 1/4 FB, STL | 1.91           | 4.61         | 8.81              | C           | 17.79     | 157             | 33.59        | A | 296             | 4.18       | S | -37             |
| 15110     | 111  | 2.25 X 1/4 FB, STL | 1.91           | 1.96         | 3.75              | C           | 12.79     | 48              | 37.11        | A | 139             | 5.67       | P | 21              |
| 15110     | 112  | 2.25 X 1/4 FB, STL | 1.91           | 1.96         | 3.75              | C           | 14.45     | 54              | 35.66        | A | 134             | 5.67       | P | 21              |
| 15110     | 113  | 2.25 X 1/4 FB, STL | 1.91           | 2.22         | 4.24              | C           | 16.12     | 68              | 35.19        | A | 149             | 5.54       | P | 23              |
| 15110     | 114  | 2.25 X 1/4 FB, STL | 1.91           | 1.96         | 3.75              | C           | 12.79     | 48              | 37.11        | A | 139             | 5.67       | S | -21             |
| 15110     | 115  | 2.25 X 1/4 FB, STL | 1.91           | 1.96         | 3.75              | C           | 14.45     | 54              | 35.66        | A | 134             | 5.67       | S | -21             |
| 15110     | 116  | 2.25 X 1/4 FB, STL | 1.91           | 2.22         | 4.24              | C           | 16.12     | 68              | 35.19        | A | 149             | 5.54       | S | -23             |
| 15110     |      |                    |                |              |                   |             |           |                 |              |   |                 |            |   |                 |
| 15111     | 117  | 2 X 1/4 FB, STL    | 1.70           | 11.61        | 19.73             | C           | 19.33     | 381             | 33.17        | A | 654             | 5.46       | P | 108             |
| 15111     | 118  | 2 X 1/4 FB, STL    | 1.70           | 11.61        | 19.73             | C           | 19.33     | 381             | 33.17        | A | 654             | 5.46       | S | -108            |
| 15111     | 119  | 2 X 1/4 FB, STL    | 1.70           | 4.17         | 7.08              | C           | 13.21     | 94              | 24.50        | A | 174             | 6.75       | P | 48              |
| 15111     | 120  | 2 X 1/4 FB, STL    | 1.70           | 2.01         | 3.41              | C           | 14.30     | 49              | 27.50        | A | 94              | 6.75       | P | 23              |
| 15111     | 121  | 2 X 1/4 FB, STL    | 1.70           | 2.17         | 3.68              | C           | 12.20     | 45              | 29.00        | A | 107             | 6.75       | P | 25              |
| 15111     | 122  | 2 X 1/4 FB, STL    | 1.70           | 0.83         | 1.42              | C           | 13.37     | 19              | 29.00        | A | 41              | 6.08       | P | 9               |
| 15111     | 123  | 2 X 1/4 FB, STL    | 1.70           | 2.17         | 3.68              | C           | 12.20     | 45              | 30.50        | A | 112             | 6.75       | P | 25              |
| 15111     | 124  | 2 X 1/4 FB, STL    | 1.70           | 0.83         | 1.42              | C           | 13.37     | 19              | 30.50        | A | 43              | 6.08       | P | 9               |
| 15111     | 125  | 2 X 1/4 FB, STL    | 1.70           | 2.01         | 3.41              | C           | 14.30     | 49              | 30.50        | A | 104             | 6.75       | P | 23              |
| 15111     | 126  | 2 X 1/4 FB, STL    | 1.70           | 0.65         | 1.10              | C           | 15.08     | 17              | 19.17        | A | 21              | 6.75       | P | 7               |
| 15111     | 127  | 2 X 1/4 FB, STL    | 1.70           | 8.58         | 14.58             | C           | 15.29     | 223             | 27.29        | A | 398             | 6.75       | P | 98              |
| 15111     | 128  | 2 X 1/4 FB, STL    | 1.70           | 4.17         | 7.08              | C           | 13.21     | 94              | 24.50        | A | 174             | 6.75       | S | -48             |
| 15111     | 129  | 2 X 1/4 FB, STL    | 1.70           | 2.01         | 3.41              | C           | 14.30     | 49              | 27.50        | A | 94              | 6.75       | S | -23             |
| 15111     | 130  | 2 X 1/4 FB, STL    | 1.70           | 2.17         | 3.68              | C           | 12.20     | 45              | 29.00        | A | 107             | 6.75       | S | -25             |
| 15111     | 131  | 2 X 1/4 FB, STL    | 1.70           | 0.83         | 1.42              | C           | 13.37     | 19              | 29.00        | A | 41              | 6.08       | S | -9              |
| 15111     | 132  | 2 X 1/4 FB, STL    | 1.70           | 2.17         | 3.68              | C           | 12.20     | 45              | 30.50        | A | 112             | 6.75       | S | -25             |
| 15111     | 133  | 2 X 1/4 FB, STL    | 1.70           | 0.83         | 1.42              | C           | 13.37     | 19              | 30.50        | A | 43              | 6.08       | S | -9              |
| 15111     | 134  | 2 X 1/4 FB, STL    | 1.70           | 2.01         | 3.41              | C           | 14.30     | 49              | 30.50        | A | 104             | 6.75       | S | -23             |

| SWBS # | ITEM | DESCRIPTION                | UNIT WEIGHT | NO. UNITS | CURRENT WEIGHT | E | VERTICAL |        | LONGITUDINAL |   |        | TRANSVERSE |   |        |
|--------|------|----------------------------|-------------|-----------|----------------|---|----------|--------|--------------|---|--------|------------|---|--------|
|        |      |                            |             |           |                |   | VCG      | MOMENT | LCG          | S | MOMENT | TCG        | S | MOMENT |
|        |      |                            | lb          | #         | lb             | A | ft       | ft*lb  | ft           |   | ft*lb  | ft         |   | ft*lb  |
| 15111  | 135  | 2 X 1/4 FB, STL            | 1.70        | 0.65      | 1.10           | C | 15.08    | 17     | 19.17        | A | 21     | 6.75       | S | -7     |
| 15111  | 136  | 2 X 1/4 FB, STL            | 1.70        | 8.58      | 14.58          | C | 15.29    | 223    | 27.29        | A | 398    | 6.75       | S | -98    |
| 15111  | 137  | 2 X 1/4 FB, STL            | 1.70        | 3.75      | 6.38           | C | 13.21    | 84     | 34.92        | A | 223    | 0.00       | C | 0      |
| 15111  | 138  | 2 X 1/4 FB, STL            | 1.70        | 3.80      | 6.46           | C | 13.18    | 85     | 34.92        | A | 226    | 1.67       | S | -11    |
| 15111  | 139  | 2 X 1/4 FB, STL            | 1.70        | 3.85      | 6.55           | C | 13.16    | 86     | 34.92        | A | 229    | 3.33       | S | -22    |
| 15111  | 140  | 2 X 1/4 FB, STL            | 1.70        | 0.68      | 1.16           | C | 18.53    | 22     | 36.08        | A | 42     | 0.00       | C | 0      |
| 15111  | 141  | 2 X 1/4 FB, STL            | 1.70        | 10.83     | 18.40          | C | 13.23    | 244    | 35.00        | A | 644    | 5.67       | P | 104    |
| 15111  | 142  | 2 X 1/4 FB, STL            | 1.70        | 10.83     | 18.40          | C | 13.23    | 244    | 35.00        | A | 644    | 5.67       | S | -104   |
| 15111  |      |                            |             |           |                |   |          |        |              |   |        |            |   |        |
| 15112  | 143  | 1.5 X 1/4 FB, STL          | 1.28        | 1.96      | 2.50           | C | 12.79    | 32     | 31.93        | A | 80     | 5.67       | P | 14     |
| 15112  | 144  | 1.5 X 1/4 FB, STL          | 1.28        | 1.96      | 2.50           | C | 14.45    | 36     | 31.74        | A | 79     | 5.67       | P | 14     |
| 15112  | 145  | 1.5 X 1/4 FB, STL          | 1.28        | 2.21      | 2.82           | C | 16.12    | 45     | 31.55        | A | 89     | 5.54       | P | 16     |
| 15112  | 146  | 1.5 X 1/4 FB, STL          | 1.28        | 2.38      | 3.03           | C | 17.79    | 54     | 31.39        | A | 95     | 5.46       | P | 17     |
| 15112  | 147  | 1.5 X 1/4 FB, STL          | 1.28        | 1.96      | 2.50           | C | 12.79    | 32     | 31.93        | A | 80     | 5.67       | S | -14    |
| 15112  | 148  | 1.5 X 1/4 FB, STL          | 1.28        | 1.96      | 2.50           | C | 14.45    | 36     | 31.74        | A | 79     | 5.67       | S | -14    |
| 15112  | 149  | 1.5 X 1/4 FB, STL          | 1.28        | 2.21      | 2.82           | C | 16.12    | 45     | 31.55        | A | 89     | 5.54       | S | -16    |
| 15112  | 150  | 1.5 X 1/4 FB, STL          | 1.28        | 2.38      | 3.03           | C | 17.79    | 54     | 31.39        | A | 95     | 5.46       | S | -17    |
| 15112  | 151  | 1.5 X 1/4 FB, STL          | 1.28        | 1.33      | 1.70           | C | 13.29    | 23     | 31.75        | A | 54     | 5.17       | P | 9      |
| 15112  | 152  | 1.5 X 1/4 FB, STL          | 1.28        | 1.33      | 1.70           | C | 13.29    | 23     | 31.75        | A | 54     | 5.17       | S | -9     |
| 15112  | 153  | 1.5 X 1/4 FB, STL          | 1.28        | 1.96      | 2.50           | C | 11.15    | 28     | 37.94        | A | 95     | 5.67       | P | 14     |
| 15112  | 154  | 1.5 X 1/4 FB, STL          | 1.28        | 1.96      | 2.50           | C | 11.15    | 28     | 37.94        | A | 95     | 5.67       | S | -14    |
| 15112  |      |                            |             |           |                |   |          |        |              |   |        |            |   |        |
| 15113  | 155  | 2 SCH 40 PIPE, ASTM A500 B | 3.65        | 7.50      | 27.38          | C | 15.08    | 413    | 26.00        | A | 712    | 0.00       | C | 0      |
|        |      |                            |             |           |                |   |          |        |              |   |        |            |   |        |
|        |      |                            |             |           |                |   |          |        |              |   |        |            |   |        |
| 16201  | 1    | 1 X 1 L, STL               | 0.80        | 55.94     | 44.75          | C | 10.62    | 475    | 41.90        | A | 1875   | 0.00       | C | 0      |
| 16201  |      |                            |             |           |                |   |          |        |              |   |        |            |   |        |
| 16202  | 2    | PLATE, 1/8 IN, STL         | 5.10        | 45.86     | 233.87         | C | 10.62    | 2484   | 41.90        | A | 9798   | 0.00       | C | 0      |
| 16202  |      |                            |             |           |                |   |          |        |              |   |        |            |   |        |
| 16203  | 3    | INTAKE LOUVER              | 15.00       | 2.00      | 30.00          | C | 10.98    | 329    | 41.90        | A | 1257   | 0.00       | C | 0      |
| 16203  |      |                            |             |           |                |   |          |        |              |   |        |            |   |        |
| 16204  | 4    | PLATE, 3/16 IN, STL        | 7.65        | 9.41      | 71.97          | C | 22.33    | 1607   | 32.88        | A | 2366   | 5.46       | P | 393    |
| 16204  | 5    | PLATE, 3/16 IN, STL        | 7.65        | 11.15     | 85.29          | C | 20.95    | 1787   | 33.07        | A | 2820   | 6.83       | P | 583    |
| 16204  | 6    | PLATE, 3/16 IN, STL        | 7.65        | 11.15     | 85.29          | C | 20.95    | 1787   | 33.07        | A | 2820   | 4.08       | P | 348    |
| 16204  | 7    | PLATE, 3/16 IN, STL        | 7.65        | 8.48      | 64.87          | C | 20.88    | 1354   | 31.17        | A | 2022   | 5.46       | P | 354    |
| 16204  | 8    | PLATE, 3/16 IN, STL        | 7.65        | 4.51      | 34.48          | C | 20.79    | 717    | 34.88        | A | 1203   | 5.46       | P | 188    |
| 16204  | 9    | PLATE, 3/16 IN, STL        | 7.65        | 9.41      | 71.97          | C | 22.33    | 1607   | 32.88        | A | 2366   | 5.46       | S | -393   |
| 16204  | 10   | PLATE, 3/16 IN, STL        | 7.65        | 11.15     | 85.29          | C | 20.95    | 1787   | 33.07        | A | 2820   | 6.83       | S | -583   |
| 16204  | 11   | PLATE, 3/16 IN, STL        | 7.65        | 11.15     | 85.29          | C | 20.95    | 1787   | 33.07        | A | 2820   | 4.08       | S | -348   |
| 16204  | 12   | PLATE, 3/16 IN, STL        | 7.65        | 8.48      | 64.87          | C | 20.88    | 1354   | 31.17        | A | 2022   | 5.46       | S | -354   |
| 16204  | 13   | PLATE, 3/16 IN, STL        | 7.65        | 4.51      | 34.48          | C | 20.79    | 717    | 34.88        | A | 1203   | 5.46       | S | -188   |

| SWBS # | ITEM | DESCRIPTION                                     | UNIT WEIGHT | NO. UNITS | CURRENT WEIGHT | E C A | VERTICAL |        | LONGITUDINAL |   |        | TRANSVERSE |   |        |
|--------|------|-------------------------------------------------|-------------|-----------|----------------|-------|----------|--------|--------------|---|--------|------------|---|--------|
|        |      |                                                 |             |           |                |       | VCG      | MOMENT | LCG          | S | MOMENT | TCG        | S | MOMENT |
|        |      |                                                 | lb          | #         | lb             |       | ft       | ft*lb  | ft           |   | ft*lb  | ft         |   | ft*lb  |
| 16204  |      |                                                 |             |           |                |       |          |        |              |   |        |            |   |        |
| 16205  | 14   | 2 X 1/4 FB, STL, ABS H36                        | 1.70        | 2.75      | 4.68           | C     | 20.83    | 97     | 31.25        | A | 146    | 5.46       | P | 26     |
| 16205  | 15   | 2 X 1/4 FB, STL, ABS H36                        | 1.70        | 2.99      | 5.09           | C     | 20.83    | 106    | 33.08        | A | 168    | 6.75       | P | 34     |
| 16205  | 16   | 2 X 1/4 FB, STL, ABS H36                        | 1.70        | 2.99      | 5.09           | C     | 20.83    | 106    | 33.08        | A | 168    | 4.17       | P | 21     |
| 16205  | 17   | 2 X 1/4 FB, STL, ABS H36                        | 1.70        | 2.75      | 4.68           | C     | 20.83    | 97     | 31.25        | A | 146    | 5.46       | S | -26    |
| 16205  | 18   | 2 X 1/4 FB, STL, ABS H36                        | 1.70        | 2.99      | 5.09           | C     | 20.83    | 106    | 33.08        | A | 168    | 6.75       | S | -34    |
| 16205  | 19   | 2 X 1/4 FB, STL, ABS H36                        | 1.70        | 2.99      | 5.09           | C     | 20.83    | 106    | 33.08        | A | 168    | 4.17       | S | -21    |
|        |      |                                                 |             |           |                |       |          |        |              |   |        |            |   |        |
| 16302  | 1    | SEA CHESTS                                      | 100.00      | 1.00      | 100.00         | E     | 5.00     | 500    | 24.17        | A | 2417   | 0.00       | C | 0      |
|        |      |                                                 |             |           |                |       |          |        |              |   |        |            |   |        |
| 16701  | 1    | DOOR, QAWT, 8 DOG, LH, 26 X 54                  | 255.00      | 1.00      | 255.00         | C     | 6.88     | 1753   | 12.62        | A | 3219   | 0.41       | P | 104    |
| 16701  | 2    | DOOR, QAWT, 8 DOG, LH, 26 X 54                  | 255.00      | 1.00      | 255.00         | C     | 6.88     | 1753   | 25.62        | A | 6534   | 0.41       | P | 104    |
| 16701  | 3    | DOOR, QAWT, 8 DOG, LH, 26 X 54                  | 255.00      | 1.00      | 255.00         | C     | 6.88     | 1753   | 34.64        | A | 8834   | 0.41       | P | 104    |
| 16701  |      |                                                 |             |           |                |       |          |        |              |   |        |            |   |        |
| 16702  | 4    | HATCH, FLUSH, HINGED, DOG TYPE C, BFHH0 15 X 24 | 1.00        | 46.00     | 46.00          | C     | 11.29    | 519    | 1.72         | A | 79     | 1.64       | S | -76    |
| 16702  | 5    | HATCH, FLUSH, HINGED, DOG TYPE C, BFHH0 15 X 24 | 1.00        | 46.00     | 46.00          | C     | 9.00     | 414    | 48.72        | A | 2241   | 7.60       | S | -349   |
| 16702  | 6    | HATCH, FLUSH, HINGED, DOG TYPE C, BFHH0 15 X 24 | 1.00        | 46.00     | 46.00          | C     | 9.00     | 414    | 51.28        | A | 2359   | 7.60       | S | -349   |
| 16702  |      |                                                 |             |           |                |       |          |        |              |   |        |            |   |        |
| 16703  | 7    | HATCH, FLUSH BOLTED, BMHC 15 X 23 F             | 1.00        | 98.00     | 98.00          | C     | 3.49     | 342    | 11.79        | A | 1156   | 2.24       | P | 220    |
| 16703  | 8    | HATCH, FLUSH BOLTED, BMHC 15 X 23 F             | 1.00        | 98.00     | 98.00          | C     | 3.49     | 342    | 27.41        | A | 2686   | 4.97       | S | -487   |
| 16703  | 9    | HATCH, FLUSH BOLTED, BMHC 15 X 23 F             | 1.00        | 98.00     | 98.00          | C     | 3.49     | 342    | 28.41        | A | 2784   | 6.59       | P | 645    |
| 16703  | 10   | HATCH, FLUSH BOLTED, BMHC 15 X 23 F             | 1.00        | 98.00     | 98.00          | C     | 3.49     | 342    | 30.46        | A | 2985   | 0.11       | P | 11     |
| 16703  |      |                                                 |             |           |                |       |          |        |              |   |        |            |   |        |
| 16704  | 11   | HATCH, YOKE HINGE, 24 X 24                      | 1.00        | 81.00     | 81.00          | C     | 11.27    | 913    | 11.41        | A | 924    | 2.76       | S | -223   |
| 16704  | 12   | HATCH, YOKE HINGE, 24 X 24                      | 1.00        | 81.00     | 81.00          | C     | 9.00     | 729    | 57.65        | A | 4669   | 3.33       | P | 270    |
| 16704  | 13   | HATCH, YOKE HINGE, 24 X 24                      | 1.00        | 81.00     | 81.00          | C     | 9.00     | 729    | 60.40        | A | 4892   | 3.33       | P | 270    |
|        |      |                                                 |             |           |                |       |          |        |              |   |        |            |   |        |
|        |      |                                                 |             |           |                |       |          |        |              |   |        |            |   |        |
| 16801  | 1    | DOOR, WEATHERTIGHT, 2 DOG, RH, 29.5 X 69.5      | 160.00      | 1.00      | 160.00         | C     | 14.85    | 2375   | 21.51        | A | 3441   | 81.25      | S | -13000 |
| 16801  |      |                                                 |             |           |                |       |          |        |              |   |        |            |   | 0      |
| 16802  | 2    | DOOR, WEATHERTIGHT, 2 DOG, LH, 29.5 X 69.5      | 160.00      | 1.00      | 160.00         | C     | 14.85    | 2375   | 21.51        | A | 3441   | 81.25      | P | 13000  |
| 16802  |      |                                                 |             |           |                |       |          |        |              |   |        |            |   | 0      |
| 16803  | 3    | DOOR, QAWT, 8 DOG, RH, 30 X 66                  | 301.00      | 1.00      | 301.00         | C     | 13.28    | 3998   | 35.21        | A | 10599  | 3.04       | P | 914    |
|        |      |                                                 |             |           |                |       |          |        |              |   |        |            |   | 0      |
|        |      |                                                 |             |           |                |       |          |        |              |   |        |            |   |        |
| 171    | 10   | MASTS, TOWERS, TETRAPODS                        | 200.00      | 1.00      | 200.00         | E     | 25.00    | 5000   | 29.17        | A | 5833   | 0.00       | C | 0      |
|        |      |                                                 |             |           |                |       |          |        |              |   |        |            |   |        |
| 18201  | 1    | PROPULSION PLANT FOUNDATIONS                    | 506.19      | 1.00      | 506.19         | E     | 2.00     | 1012   | 42.00        | A | 21260  | 0.00       | C | 0      |
| 18201  |      |                                                 |             |           |                |       |          |        |              |   |        |            |   |        |
| 18202  | 2    | PLATE, 5/16 IN, STL                             | 12.75       | 6.37      | 81.23          | C     | 1.69     | 137    | 38.00        | A | 3087   | 0.00       | C | 0      |
| 18202  | 3    | PLATE, 5/16 IN, STL                             | 12.75       | 3.67      | 46.85          | C     | 2.15     | 101    | 37.92        | A | 1776   | 0.00       | C | 0      |

| SWBS # | ITEM | DESCRIPTION                                | UNIT WEIGHT | NO. UNITS | CURRENT WEIGHT | E<br>C<br>A | VERTICAL |        | LONGITUDINAL |   |        | TRANSVERSE |   |        |
|--------|------|--------------------------------------------|-------------|-----------|----------------|-------------|----------|--------|--------------|---|--------|------------|---|--------|
|        |      |                                            |             |           |                |             | VCG      | MOMENT | LCG          | S | MOMENT | TCG        | S | MOMENT |
|        |      |                                            | lb          | #         | lb             |             | ft       | ft*lb  | ft           |   | ft*lb  | ft         |   | ft*lb  |
| 18202  |      |                                            |             |           |                |             |          |        |              |   |        |            |   |        |
| 18203  | 4    | PLATE, 1/4 IN, STL                         | 10.20       | 2.68      | 27.32          | C           | 1.98     | 54     | 42.50        | A | 1161   | 0.00       | C | 0      |
| 18203  | 5    | PLATE, 1/4 IN, STL                         | 10.20       | 3.60      | 36.77          | C           | 1.81     | 67     | 44.00        | A | 1618   | 0.00       | C | 0      |
| 18203  | 6    | PLATE, 1/4 IN, STL                         | 10.20       | 12.10     | 123.46         | C           | 1.83     | 225    | 43.21        | A | 5334   | 0.00       | C | 0      |
| 18203  |      |                                            |             |           |                |             |          |        |              |   |        |            |   |        |
| 18204  | 7    | PLATE, 5/32 IN, STL                        | 6.38        | 7.05      | 44.97          | C           | 1.67     | 75     | 37.93        | A | 1706   | 0.00       | C | 0      |
| 18204  |      |                                            |             |           |                |             |          |        |              |   |        |            |   |        |
| 18205  | 8    | PLATE, 3/8 IN, STL                         | 15.30       | 4.20      | 64.20          | C           | 2.62     | 168    | 40.56        | A | 2604   | 0.00       | C | 0      |
| 18205  |      |                                            |             |           |                |             |          |        |              |   |        |            |   |        |
|        |      |                                            |             |           |                |             |          |        |              |   |        |            |   |        |
| 18301  | 1    | ELECTRICAL PLANT FOUNDATIONS               | 1208.68     | 1.00      | 1208.68        | E           | 2.00     | 2417   | 40.00        | A | 48347  | 0.00       | C | 0      |
| 18301  |      |                                            |             |           |                |             |          |        |              |   |        |            |   |        |
| 18302  | 2    | PLATE, 1/4 IN, STL                         | 10.20       | 8.40      | 85.67          | C           | 2.89     | 248    | 43.94        | A | 3764   | 0.00       | C | 0      |
| 18302  |      |                                            |             |           |                |             |          |        |              |   |        |            |   |        |
| 18303  | 3    | PLATE, 5/16 IN, STL                        | 12.75       | 8.27      | 105.39         | C           | 3.61     | 380    | 44.00        | A | 4637   | 0.00       | C | 0      |
| 18303  |      |                                            |             |           |                |             |          |        |              |   |        |            |   |        |
|        |      |                                            |             |           |                |             |          |        |              |   |        |            |   |        |
| 18401  | 1    | COMMAND AND SURVEILLANCE FOUNDATIONS (13%) | 516.22      | 1.00      | 516.22         | E           | 16.75    | 8644   | 29.17        | A | 15056  | 0.00       | C | 0      |
|        |      |                                            |             |           |                |             |          |        |              |   |        |            |   |        |
| 18501  | 1    | PLATE, 1/2 IN, STL                         | 20.40       | 18.43     | 375.96         | C           | 9.02     | 3391   | 47.00        | A | 17670  | 0.00       | C | 0      |
| 18501  | 2    | PLATE, 1/2 IN, STL                         | 20.40       | 1.72      | 35.05          | C           | 8.02     | 281    | 47.00        | A | 1647   | 0.00       | C | 0      |
| 18501  | 3    | PLATE, 1/2 IN, STL                         | 20.40       | 1.72      | 35.05          | C           | 8.02     | 281    | 47.00        | A | 1647   | 0.00       | C | 0      |
| 18501  | 4    | PLATE, 1/2 IN, STL                         | 20.40       | 7.53      | 153.65         | C           | 12.01    | 1846   | 47.00        | A | 7221   | 0.00       | C | 0      |
| 18501  | 5    | PLATE, 1/2 IN, STL                         | 20.40       | 14.50     | 295.85         | C           | 9.22     | 2728   | 47.00        | A | 13905  | 0.00       | C | 0      |
| 18501  |      |                                            |             |           |                |             |          |        |              |   |        |            |   |        |
| 18502  | 6    | 12 X 4 X 19 MT, STL                        | 14.14       | 12.27     | 173.47         | C           | 8.40     | 1458   | 47.02        | A | 8156   | 0.00       | C | 0      |
| 18502  |      |                                            |             |           |                |             |          |        |              |   |        |            |   |        |
|        |      |                                            |             |           |                |             |          |        |              |   |        |            |   |        |
| 18501  | 1    | AUXILIARY SYSTEMS FOUNDATIONS (10%)        | 1729.29     | 1.00      | 1729.29        | E           | 5.58     | 9652   | 40.00        | A | 69172  | 0.00       | C | 0      |
|        |      |                                            |             |           |                |             |          |        |              |   |        |            |   |        |
| 18601  | 1    | OUTFIT AND FURNISHING FOUNDATIONS (1%)     | 215.71      | 1.00      | 215.71         | E           | 8.65     | 1866   | 30.50        | A | 6579   | 0.00       | C | 0      |
|        |      |                                            |             |           |                |             |          |        |              |   |        |            |   |        |
|        |      |                                            |             |           |                |             |          |        |              |   |        |            |   |        |
| 19601  |      | WELDING                                    | 1437.00     | 1.00      | 1437.00        | E           | 6.97     | 10022  | 31.75        | A | 45621  | 0.01       | P | 14     |
|        |      |                                            |             |           |                |             |          |        |              |   |        |            |   |        |
| 19701  | 10   | MILL TOLERANCE                             | 1915.99     | 1.00      | 1915.99        | E           | 6.97     | 13363  | 31.75        | A | 60827  | 0.01       | P | 19     |
|        |      |                                            |             |           |                |             |          |        |              |   |        |            |   |        |
| 198    | 10   |                                            |             |           |                |             |          |        |              |   |        |            |   |        |
| NA     | 10   | BRACKETS & MISC. STRUCTURE                 | 1652.98     | 1.00      | 1652.98        | E           | 6.42     | 10612  | 31.11        | A | 51424  | 0.00       | C | 0      |
|        |      |                                            |             |           |                |             |          |        |              |   |        |            |   |        |
|        |      |                                            |             |           |                |             |          |        |              |   |        |            |   |        |

| SWBS # | ITEM | DESCRIPTION                                     | UNIT WEIGHT<br>lb | NO. UNITS<br># | CURRENT WEIGHT<br>lb | E<br>C<br>A | VERTICAL  |                 | LONGITUDINAL |   |                 | TRANSVERSE |   |                 |
|--------|------|-------------------------------------------------|-------------------|----------------|----------------------|-------------|-----------|-----------------|--------------|---|-----------------|------------|---|-----------------|
|        |      |                                                 |                   |                |                      |             | VCG<br>ft | MOMENT<br>ft*lb | LCG<br>ft    | S | MOMENT<br>ft*lb | TCG<br>ft  | S | MOMENT<br>ft*lb |
|        |      |                                                 |                   |                |                      |             |           |                 |              |   |                 |            |   |                 |
| 23301  | 1    | MAIN PROPULSION DIESEL ENGINE, PORT             | 4299.00           | 1.00           | 4299                 | E           | 4.06      | 17461           | 39.83        | A | 171229          | 3.49       | P | 14993           |
| 23301  | 2    | MAIN PROPULSION DIESEL ENGINE, STBD             | 4299.00           | 1.00           | 4299                 | E           | 4.06      | 17461           | 39.83        | A | 171229          | 3.49       | S | -14993          |
|        |      |                                                 |                   |                |                      |             |           |                 |              |   |                 |            |   |                 |
| 24101  | 1    | MARINE GEAR, PORT                               | 758.00            | 1.00           | 758                  | E           | 2.76      | 2088            | 43.43        | A | 32919           | 3.50       | P | 2653            |
| 24101  | 2    | MARINE GEAR, STBD                               | 758.00            | 1.00           | 758                  | E           | 2.76      | 2088            | 43.43        | A | 32919           | 3.50       | S | -2653           |
|        |      |                                                 |                   |                |                      |             |           |                 |              |   |                 |            |   |                 |
| 24201  | 1    | PROPULSION CLUTCHES AND COUPLINGS               | 70.00             | 2.00           | 140                  | C           | 3.80      | 532             | 48.00        | A | 6720            | 0.00       | C | 0               |
|        |      |                                                 |                   |                |                      |             |           |                 |              |   |                 |            |   |                 |
| 24301  | 1    | SHAFT                                           | 1120.00           | 2.00           | 2240                 | C           | 2.13      | 4763            | 53.06        | A | 118853          | 0.00       | C | 0               |
|        |      |                                                 |                   |                |                      |             |           |                 |              |   |                 |            |   |                 |
| 24401  | 1    | SHAFT BEARING                                   | 76.00             | 2.00           | 152                  | E           | 2.23      | 339             | 49.22        | A | 7481            | 0.00       | C | 0               |
|        |      |                                                 |                   |                |                      |             |           |                 |              |   |                 |            |   |                 |
| 24501  | 1    | PROPELLER                                       | 307.00            | 2.00           | 614                  | E           | 1.92      | 1176            | 60.94        | A | 37419           | 0.00       | C | 0               |
|        |      |                                                 |                   |                |                      |             |           |                 |              |   |                 |            |   |                 |
| 25101  | 1    | COMBUSTION AIR SYSTEM                           | 300.00            | 1.00           | 300                  | E           | 8.00      | 2400            | 42.50        | A | 12750           | 0.00       | C | 0               |
|        |      |                                                 |                   |                |                      |             |           |                 |              |   |                 |            |   |                 |
| 25201  | 1    | PROPULSION CONTROL SYSTEM                       | 150.00            | 1.00           | 150                  | E           | 3.67      | 551             | 40.00        | A | 6000            | 0.00       | C | 0               |
|        |      |                                                 |                   |                |                      |             |           |                 |              |   |                 |            |   |                 |
| 25601  | 1    | CIRCULATING & COOLING SEA WATER SYSTEM          | 2492.03           | 0.00           | 0                    | E           | 1.83      | 0               | 38.79        | A | 0               | 0.00       | C | 0               |
|        |      |                                                 |                   |                |                      |             |           |                 |              |   |                 |            |   |                 |
| 25901  | 1    | SILENCER, MPDE                                  | 108.00            | 2.00           | 216                  | E           | 19.68     | 4251            | 32.39        | A | 6997            | 0.00       | C | 0               |
| 25902  | 2    | EXHAUST SYSTEM                                  | 904.00            | 1.00           | 904                  | E           | 11.00     | 9944            | 38.00        | A | 34352           | 0.00       | C | 0               |
|        |      |                                                 |                   |                |                      |             |           |                 |              |   |                 |            |   |                 |
| 26101  | 1    | FUEL SERVICE SYSTEM                             | 114.00            | 1.00           | 114                  | E           | 2.00      | 228             | 38.00        | A | 4332            | 0.00       | C | 0               |
|        |      |                                                 |                   |                |                      |             |           |                 |              |   |                 |            |   |                 |
| 26201  | 1    | MAIN PROPULSION LUBE OIL SYSTEM                 | 108.00            | 1.00           | 108                  | E           | 3.67      | 396             | 40.00        | A | 4320            | 0.00       | C | 0               |
|        |      |                                                 |                   |                |                      |             |           |                 |              |   |                 |            |   |                 |
| 29801  | 1    | PROPULSION PLANT OPERATING FLUIDS               | 693.45            | 1.00           | 693                  | E           | 3.67      | 2543            | 40.00        | A | 27738           | 0.00       | C | 0               |
| 29801  |      |                                                 |                   |                |                      |             |           |                 |              |   |                 |            |   |                 |
| 29802  | 2    | MPDE OIL, PORT                                  | 54.25             | 1.00           | 54                   | E           | 4.06      | 220             | 39.83        | A | 2161            | 3.49       | P | 189             |
| 29802  | 3    | MPDE OIL, STBD                                  | 54.25             | 1.00           | 54                   | E           | 4.06      | 220             | 39.83        | A | 2161            | 3.49       | S | -189            |
| 29802  |      |                                                 |                   |                |                      |             |           |                 |              |   |                 |            |   |                 |
| 29803  | 4    | MPDE COOLANT, PORT                              | 120.00            | 1.00           | 120                  | E           | 4.06      | 487             | 39.83        | A | 4780            | 3.49       | P | 419             |
| 29803  | 5    | MPDE COOLANT, STBD                              | 120.00            | 1.00           | 120                  | E           | 4.06      | 487             | 39.83        | A | 4780            | 3.49       | S | -419            |
| 29803  |      |                                                 |                   |                |                      |             |           |                 |              |   |                 |            |   |                 |
| 29804  | 6    | MARINE GEAR OIL, PORT                           | 39.03             | 1.00           | 39                   | E           | 2.76      | 108             | 43.43        | A | 1695            | 3.50       | P | 137             |
| 29804  | 7    | MARINE GEAR OIL, STBD                           | 39.03             | 1.00           | 39                   | E           | 2.76      | 108             | 43.43        | A | 1695            | 3.50       | S | -137            |
|        |      |                                                 |                   |                |                      |             |           |                 |              |   |                 |            |   |                 |
| 29901  | 1    | PROPULSION PLANT REPAIR PARTS AND SPECIAL TOOLS | 896.00            | 1.00           | 896                  | E           | 3.67      | 3285            | 40.00        | A | 35840           | 0.00       | C | 0               |
|        |      |                                                 |                   |                |                      |             |           |                 |              |   |                 |            |   |                 |

| SWBS # | ITEM | DESCRIPTION                                 | UNIT WEIGHT<br># | NO. UNITS<br>lb | CURRENT WEIGHT<br>lb | E<br>C<br>A | VERTICAL  |                 | LONGITUDINAL |   |                 | TRANSVERSE |   |                 |
|--------|------|---------------------------------------------|------------------|-----------------|----------------------|-------------|-----------|-----------------|--------------|---|-----------------|------------|---|-----------------|
|        |      |                                             |                  |                 |                      |             | VCG<br>ft | MOMENT<br>ft*lb | LCG<br>ft    | S | MOMENT<br>ft*lb | TCG<br>ft  | S | MOMENT<br>ft*lb |
| 31101  | 1    | SSDG DRY WEIGHT, PORT                       | 1.00             | 2885.00         | 2885                 | E           | 5.30      | 15291           | 43.86        | A | 126550          | 7.48       | P | 21589           |
| 31102  | 2    | SSDG DRY WEIGHT, STBD                       | 1.00             | 2885.00         | 2885                 | E           | 5.30      | 15291           | 44.04        | A | 127062          | 7.36       | S | -21222          |
|        |      |                                             |                  |                 |                      |             |           |                 |              |   |                 |            |   |                 |
| 31301  | 1    | BATTERIES AND SERVICE FACILITIES            | 1.00             | 1120.00         | 1120                 | E           | 3.50      | 3920            | 36.00        | A | 40320           | 0.00       | C | 0               |
|        |      |                                             |                  |                 |                      |             |           |                 |              |   |                 |            |   |                 |
| 32001  | 1    | POWER DISTRIBUTION SYSTEM                   | 1.00             | 2274.00         | 2274                 | E           | 8.50      | 19329           | 35.00        | A | 79590           | 0.00       | C | 0               |
|        |      |                                             |                  |                 |                      |             |           |                 |              |   |                 |            |   |                 |
| 33001  | 1    | LIGHTING SYSTEM                             | 1.00             | 694.00          | 694                  | E           | 11.00     | 7634            | 32.40        | A | 22486           | 0.00       | C | 0               |
|        |      |                                             |                  |                 |                      |             |           |                 |              |   |                 |            |   |                 |
| 34201  | 1    | SILENCER, SSDG                              | 2.00             | 54.00           | 108                  | E           | 7.65      | 826             | 39.83        | A | 4302            | 0.00       | C | 0               |
| 34201  |      |                                             |                  |                 |                      |             |           |                 |              |   |                 |            |   |                 |
| 34202  | 2    | DIESEL SUPPORT SYSTEM                       | 1.00             | 1012.00         | 1012                 | E           | 3.16      | 3198            | 32.40        | A | 32789           | 0.00       | C | 0               |
|        |      |                                             |                  |                 |                      |             |           |                 |              |   |                 |            |   |                 |
| 39801  | 1    | ELECTRIC PLANT OPERATING FLUIDS             | 1.00             | 151.00          | 151                  | E           | 5.50      | 831             | 43.08        | A | 6506            | 0.00       | C | 0               |
| 39801  |      |                                             |                  |                 |                      |             |           |                 |              |   |                 |            |   |                 |
| 39802  | 2    | SSDG WET OIL, PORT                          | 1.00             | 38.50           | 39                   | E           | 5.30      | 204             | 43.86        | A | 1689            | 7.48       | P | 288             |
| 39802  | 3    | SSDG WET OIL, STBD                          | 1.00             | 38.50           | 39                   | E           | 5.30      | 204             | 44.04        | A | 1696            | 7.36       | S | -283            |
| 39802  |      |                                             |                  |                 |                      |             |           |                 |              |   |                 |            |   |                 |
| 39803  | 4    | SSDG COOLANT, PORT                          | 1.00             | 110.00          | 110                  | E           | 5.30      | 583             | 43.86        | A | 4825            | 7.48       | P | 823             |
| 39803  | 5    | SSDG COOLANT, STBD                          | 1.00             | 110.00          | 110                  | E           | 5.30      | 583             | 44.04        | A | 4845            | 7.36       | S | -809            |
|        |      |                                             |                  |                 |                      |             |           |                 |              |   |                 |            |   |                 |
| 39901  | 1    | ELECTRIC PLANT REPAIR PARTS & SPECIAL TOOLS | 1.00             | 224.00          | 224                  | E           | 6.76      | 1514            | 32.40        | A | 7258            | 0.00       | C | 0               |
|        |      |                                             |                  |                 |                      |             |           |                 |              |   |                 |            |   |                 |

| SWBS # | ITEM | DESCRIPTION                               | UNIT WEIGHT<br>lb | NO. UNITS<br># | CURRENT WEIGHT<br>lb | E<br>C<br>A | VERTICAL |        | LONGITUDINAL |   |        | TRANSVERSE |   |        |
|--------|------|-------------------------------------------|-------------------|----------------|----------------------|-------------|----------|--------|--------------|---|--------|------------|---|--------|
|        |      |                                           |                   |                |                      |             | VCG      | MOMENT | LCG          | S | MOMENT | TCG        | S | MOMENT |
|        |      |                                           |                   |                |                      |             | ft       | ft*lb  | ft           |   | ft*lb  | ft         |   | ft*lb  |
| 41101  | 1    | DATA DISPLAY GROUP                        | 448.00            | 1.00           | 448                  | E           | 14.50    | 6496   | 18.00        | A | 8064   | 0.00       | C | 0      |
| 42101  | 1    | NON-ELECTRICAL/ELECTRONIC NAVIGATION AIDS | 217.00            | 1.00           | 217                  | E           | 16.32    | 3541   | 18.00        | A | 3906   | 0.00       | C | 0      |
| 42201  | 1    | ELECTRICAL NAVIGATION AIDS                | 448.00            | 1.00           | 448                  | E           | 16.32    | 7311   | 18.00        | A | 8064   | 0.00       | C | 0      |
| 42601  | 1    | ELECTRICAL NAVIGATION SYSTEMS             | 448.00            | 1.00           | 448                  | E           | 14.50    | 6496   | 18.00        | A | 8064   | 0.00       | C | 0      |
| 43401  | 1    | ENTERTAINMENT & TRAINING SYSTEMS          | 57.93             | 1.00           | 58                   | E           | 9.33     | 540    | 25.00        | A | 1448   | 0.00       | C | 0      |
| 43601  | 1    | ALARM, SAFETY, AND WARNING SYSTEMS        | 448.00            | 1.00           | 448                  | E           | 7.77     | 3481   | 25.00        | A | 11200  | 0.00       | C | 0      |
| 43701  | 1    | INDICATING, ORDER, AND METERING SYSTEMS   | 224.00            | 1.00           | 224                  | E           | 8.04     | 1801   | 25.00        | A | 5600   | 0.00       | C | 0      |
| 43801  | 1    | INTEGRATED CONTROL SYSTEM                 | 560.00            | 1.00           | 560                  | E           | 9.89     | 5538   | 25.00        | A | 14000  | 0.00       | C | 0      |
| 44101  | 1    | RADIO SYSTEMS                             | 448.00            | 1.00           | 448                  | E           | 10.26    | 4596   | 25.00        | A | 11200  | 0.00       | C | 0      |
| 44301  | 1    | VISUAL AND AUDIBLE COMMUNICATION SYSTEMS  | 224.00            | 1.00           | 224                  | E           | 19.12    | 4283   | 25.00        | A | 5600   | 0.00       | C | 0      |
| 44601  | 1    | SECURITY EQUIPMENT                        |                   |                |                      |             |          | 0      |              |   | 0      |            |   | 0      |
| 45101  | 1    | SURFACE SEARCH RADAR                      | 448.00            | 1.00           | 448                  | E           | 50.02    | 22409  | 22.00        | A | 9856   | 0.00       | C | 0      |

| SWBS # | ITEM | DESCRIPTION                                         | UNIT WEIGHT | NO. UNITS | CURRENT WEIGHT | E<br>C<br>A | VERTICAL |        | LONGITUDINAL |   |        | TRANSVERSE |   |        |
|--------|------|-----------------------------------------------------|-------------|-----------|----------------|-------------|----------|--------|--------------|---|--------|------------|---|--------|
|        |      |                                                     |             |           |                |             | VCG      | MOMENT | LCG          | S | MOMENT | TCG        | S | MOMENT |
|        |      |                                                     |             |           |                |             | ft       | ft*lb  | ft           |   | ft*lb  | ft         |   | ft*lb  |
|        |      |                                                     | lb          | #         | lb             |             |          |        |              |   |        |            |   |        |
| 51101  | 1    | COMPARTMENT HEATING SYSTEM                          | 183.74      | 1.00      | 184            | E           | 7.89     | 1450   | 32.40        | A | 5953   | 0.00       | C | 0      |
| 51201  | 1    | VENTILATION SYSTEM                                  | 407.06      | 1.00      | 407            | E           | 12.00    | 4885   | 26.00        | A | 10584  | 0.00       | C | 0      |
| 51301  | 1    | MACHINERY SPACE VENTILATION SYSTEM                  | 694.40      | 1.00      | 694            | E           | 21.00    | 14582  | 40.00        | A | 27776  | 0.00       | C | 0      |
| 51401  | 1    | AIR CONDITIONING SYSTEM                             | 282.68      | 1.00      | 283            | E           | 12.00    | 3392   | 26.00        | A | 7350   | 0.00       | C | 0      |
| 51601  | 1    | REFRIGERATION SYSTEM                                | 604.80      | 1.00      | 605            | E           | 6.00     | 3629   | 15.00        | A | 9072   | 0.00       | C | 0      |
| 52301  | 1    | WASHDOWN SYSTEM                                     | 192.11      | 1.00      | 192            | E           | 10.00    | 1921   | 42.00        | A | 8069   | 0.00       | C | 0      |
| 52601  | 1    | SCUPPERS AND DECK DRAINS                            | 640.71      | 1.00      | 641            | E           | 10.31    | 6606   | 32.40        | A | 20759  | 0.00       | C | 0      |
| 52901  | 1    | DRAINAGE AND BALLASTING SYSTEM                      | 602.96      | 1.00      | 603            | E           | 3.81     | 2297   | 32.40        | A | 19536  | 0.00       | C | 0      |
| 53201  | 1    | KEEL COOLER FOR ENGINE JACKET WATER                 | 200.00      | 2.00      | 400            | E           | 1.83     | 732    | 38.79        | A | 15516  | 0.00       | C | 0      |
| 53202  | 2    | KEEL COOLER FOR ENGINE AFTER COOLER AND MARINE GEAR | 209.00      | 2.00      | 418            | E           | 2.45     | 1025   | 46.19        | A | 19308  | 0.00       | C | 0      |
| 53203  | 3    | KEEL COOLER FOR GENERATOR                           | 200.00      | 2.00      | 400            | E           | 0.54     | 218    | 38.01        | A | 15206  | 0.00       | C | 0      |
| 53204  | 4    | COOLING WATER                                       | 487.20      | 2.00      | 974            | E           | 0.73     | 715    | 45.37        | A | 44209  | 0.00       | C | 0      |
| 53301  | 1    | POTABLE WATER                                       | 1351.72     | 1.00      | 1352           | E           | 4.00     | 5407   | 23.00        | A | 31090  | 0.00       | C | 0      |
| 55501  | 1    | FIRE EXTINGUISHING SYSTEMS                          | 850.12      | 1.00      | 850            | E           | 6.00     | 5101   | 38.00        | A | 32305  | 0.00       | C | 0      |
| 55601  | 1    | HYDRAULIC SYSTEM                                    | 1965.00     | 1.00      | 1965           | E           | 6.00     | 11790  | 52.00        | A | 102180 | 0.00       | C | 0      |
| 56101  | 1    | STEERING CONTROL SYSTEM                             | 2240.00     | 1.00      | 2240           | E           | 6.00     | 13440  | 62.00        | A | 138880 | 0.00       | C | 0      |
| 56201  | 1    | RUDDER                                              | 336.00      | 2.00      | 672            | E           | 3.00     | 2016   | 63.50        | A | 42672  | 0.00       | C | 0      |
| 58101  | 1    | ANCHOR                                              | 32.00       | 1.00      | 32             | E           | 12.36    | 395    | 5.94         | A | 190    | 6.73       | S | -215   |
| 58101  |      |                                                     |             |           |                |             |          |        |              |   |        |            |   |        |
| 58102  | 2    | ANCHOR LINE REEL                                    | 45.88       | 1.00      | 46             | C           | 14.08    | 646    | 16.08        | A | 738    | 1.92       | P | 88     |
| 58102  |      |                                                     |             |           |                |             |          |        |              |   |        |            |   |        |
| 58103  | 3    | ANCHOR ROPE LINE                                    | 2.12        | 275.00    | 583            | C           | 9.72     | 5666   | 1.47         | A | 857    | 3.25       | P | 1897   |
| 58103  |      |                                                     |             |           |                |             |          |        |              |   |        |            |   |        |
| 58104  | 4    | ANCHOR CHAIN AND SHACKLE                            | 2.50        | 25.00     | 63             | C           | 9.72     | 607    | 1.47         | A | 92     | 3.25       | P | 203    |
| 58104  |      |                                                     |             |           |                |             |          |        |              |   |        |            |   |        |
| 58105  | 5    | ELECTRIC CAPSTAN                                    | 88.00       | 1.00      | 88             | C           | 11.62    | 1023   | 15.11        | A | 1330   | 0.36       | P | 31     |
| 58201  | 1    | MOORING LINES                                       | 2.12        | 360.00    | 763            | C           | 13.34    | 10181  | 14.95        | A | 11412  | 0.40       | S | -308   |
| 58301  | 1    | LIFERAFT, VIKING 12, LOW PROFILE                    | 135.00      | 1.00      | 135            | E           | 19.94    | 2691   | 22.73        | A | 3068   | 4.00       | P | 540    |
| 58301  |      |                                                     |             |           |                |             |          |        |              |   |        |            |   |        |
| 58302  | 2    | LIFE RING, 30 IN                                    | 5.50        | 2.00      | 11             | E           | 13.39    | 147    | 8.18         | A | 90     | 0.00       | C | 0      |
| 58302  | 3    | LIFE RING, 30 IN                                    | 5.50        | 2.00      | 11             | E           | 12.99    | 143    | 25.01        | A | 275    | 0.00       | C | 0      |
| 58302  | 4    | LIFE RING, 30 IN                                    | 5.50        | 2.00      | 11             | E           | 11.67    | 128    | 63.93        | A | 703    | 0.00       | C | 0      |

|       |   |                                          |         |      |      |   |       |        |       |   |        |      |   |   |
|-------|---|------------------------------------------|---------|------|------|---|-------|--------|-------|---|--------|------|---|---|
|       |   |                                          |         |      |      |   |       |        |       |   |        |      |   |   |
| 58901 | 1 | WORKING DECK CRANE                       | 8400.00 | 1.00 | 8400 | E | 15.00 | 126000 | 47.00 | A | 394800 | 0.00 | C | 0 |
|       |   |                                          |         |      |      |   |       |        |       |   |        |      |   |   |
| 58902 | 2 | CROSS DECK WINCH                         | 370.00  | 2.00 | 740  | E | 9.81  | 7260   | 56.51 | A | 41814  | 0.00 | C | 0 |
|       |   |                                          |         |      |      |   |       |        |       |   |        |      |   |   |
| 58903 | 3 | CHAIN STOPPER                            | 900.00  | 2.00 | 1800 | E | 9.30  | 16740  | 47.50 | A | 85500  | 0.00 | C | 0 |
|       |   |                                          |         |      |      |   |       |        |       |   |        |      |   |   |
| 58901 | 1 | AUXILIARY SYSTEMS OPERATING FLUIDS       |         |      | 0    | E | 2.55  | 0      | 44.00 | A | 0      | 0.00 | C | 0 |
|       |   |                                          |         |      |      |   |       |        |       |   |        |      |   |   |
| 59301 | 1 | ENVIRONMENTAL POLLUTION CONTROL SYSTEMS  | 1544.83 | 1.00 | 1545 | E | 4.50  | 6952   | 42.00 | A | 64883  | 0.00 | C | 0 |
|       |   |                                          |         |      |      |   |       |        |       |   |        |      |   |   |
| 59901 | 1 | AUXILIARY SYSTEMS REPAIR PARTS AND TOOLS | 470.40  | 1.00 | 470  | E | 5.00  | 2352   | 36.00 | A | 16934  | 0.00 | C | 0 |
|       |   |                                          |         |      |      |   |       |        |       |   |        |      |   |   |
|       |   |                                          |         |      |      |   |       |        |       |   |        |      |   |   |

| SWBS # | ITEM | DESCRIPTION                                         | UNIT WEIGHT | NO. UNITS | CURRENT WEIGHT | E C A | VERTICAL |        | LONGITUDINAL |   |        | TRANSVERSE |   |        |
|--------|------|-----------------------------------------------------|-------------|-----------|----------------|-------|----------|--------|--------------|---|--------|------------|---|--------|
|        |      |                                                     |             |           |                |       | VCG      | MOMENT | LCG          | S | MOMENT | TCG        | S | MOMENT |
|        |      |                                                     | lb          | #         | lb             |       | ft       | ft*lb  | ft           |   | ft*lb  | ft         |   | ft*lb  |
|        |      |                                                     |             |           |                |       |          |        |              |   |        |            |   |        |
| 61101  | 1    | CHOCK                                               | 69.93       | 1.00      | 70             | E     | 11.75    | 822    | 0.62         | F | -43    | 0.00       | C | 0      |
| 61101  | 2    | CHOCK                                               | 69.93       | 2.00      | 140            | E     | 11.75    | 1643   | 2.37         | A | 331    | 0.00       | C | 0      |
| 61101  | 3    | CHOCK                                               | 69.93       | 2.00      | 140            | E     | 11.75    | 1643   | 24.81        | A | 3470   | 0.00       | C | 0      |
| 61101  | 4    | CHOCK                                               | 69.93       | 2.00      | 140            | E     | 9.60     | 1343   | 58.98        | A | 8250   | 0.00       | C | 0      |
| 61101  | 5    | CHOCK                                               | 69.93       | 1.00      | 70             | E     | 9.60     | 671    | 67.97        | A | 4753   | 0.00       | C | 0      |
| 61101  |      |                                                     |             |           |                |       |          |        |              |   |        |            |   |        |
| 61102  | 6    | DOUBLE BOLLARD                                      | 100.00      | 1.00      | 100            | E     | 11.62    | 1162   | 1.25         | A | 125    | 0.00       | C | 0      |
| 61102  | 7    | DOUBLE BOLLARD                                      | 100.00      | 2.00      | 200            | E     | 11.45    | 2291   | 14.59        | A | 2918   | 0.00       | C | 0      |
| 61102  | 8    | DOUBLE BOLLARD                                      | 100.00      | 2.00      | 200            | E     | 9.44     | 1888   | 64.93        | A | 12986  | 0.00       | C | 0      |
| 61102  |      |                                                     |             |           |                |       |          |        |              |   |        |            |   |        |
| 61103  | 9    | FENDERS                                             | 12.00       | 6.00      | 72             | E     | 6.72     | 484    | 50.45        | A | 3632   | 2.76       | S | -199   |
|        |      |                                                     |             |           |                |       |          |        |              |   |        |            |   |        |
| 61201  | 1    | PIPE, 1 1/4" NPS, SCHED 40                          | 2.27        | 24.55     | 56             | C     | 14.67    | 818    | 3.75         | A | 210    | 0.00       | C | 0      |
| 61201  | 2    | PIPE, 1 1/4" NPS, SCHED 40                          | 2.27        | 41.05     | 93             | C     | 14.67    | 1368   | 23.56        | A | 2198   | 0.00       | C | 0      |
| 61201  | 3    | PIPE, 1 1/4" NPS, SCHED 40                          | 2.27        | 31.06     | 71             | C     | 12.57    | 887    | 64.76        | A | 4572   | 0.00       | C | 0      |
| 61201  | 4    | PIPE, 1 1/4" NPS, SCHED 40                          | 2.27        | 42.07     | 96             | C     | 14.49    | 1385   | 24.36        | A | 2330   | 0.00       | C | 0      |
| 61201  |      |                                                     |             |           |                |       |          |        |              |   |        |            |   |        |
| 61202  | 5    | PIPE, 3/4" NPS, SCHED 40                            | 1.13        | 24.55     | 28             | C     | 13.67    | 379    | 3.75         | A | 104    | 0.00       | C | 0      |
| 61202  | 6    | PIPE, 3/4" NPS, SCHED 40                            | 1.13        | 41.05     | 46             | C     | 13.67    | 634    | 23.56        | A | 1094   | 0.00       | C | 0      |
| 61202  | 7    | PIPE, 3/4" NPS, SCHED 40                            | 1.13        | 31.06     | 35             | C     | 11.57    | 406    | 64.76        | A | 2275   | 0.00       | C | 0      |
| 61202  | 8    | PIPE, 3/4" NPS, SCHED 40                            | 1.13        | 20.85     | 24             | C     | 10.57    | 249    | 64.76        | A | 1527   | 0.00       | C | 0      |
| 61202  |      |                                                     |             |           |                |       |          |        |              |   |        |            |   |        |
| 61203  | 9    | PIPE, 1 1/4" NPS, SCHED 40                          | 2.27        | 36.00     | 82             | C     | 13.73    | 1123   | 15.13        | A | 1238   | 0.00       | C | 0      |
| 61203  | 10   | PIPE, 1 1/4" NPS, SCHED 40                          | 2.27        | 75.00     | 170            | C     | 10.72    | 1828   | 55.78        | A | 9509   | 0.00       | C | 0      |
| 61203  |      |                                                     |             |           |                |       |          |        |              |   |        |            |   |        |
| 61204  | 11   | PLATE, 7.65#                                        | 7.65        | 1.55      | 12             | C     | 9.13     | 108    | 55.78        | A | 661    | 0.00       | C | 0      |
| 61204  |      |                                                     |             |           |                |       |          |        |              |   |        |            |   |        |
| 61205  | 12   | D HOOK                                              | 0.15        | 74.00     | 11             | C     | 11.67    | 130    | 38.02        | A | 422    | 0.00       | C | 0      |
| 61205  |      |                                                     |             |           |                |       |          |        |              |   |        |            |   |        |
| 61206  | 13   | 1/4" CHAIN                                          | 0.75        | 123.17    | 92             | C     | 11.67    | 1078   | 38.02        | A | 3512   | 0.00       | C | 0      |
| 61206  |      |                                                     |             |           |                |       |          |        |              |   |        |            |   |        |
| 61207  | 14   | ROD 5/16" DIA (STAPLE)                              | 0.10        | 74.00     | 7              | C     | 11.67    | 86     | 38.02        | A | 281    | 0.00       | C | 0      |
| 61207  |      |                                                     |             |           |                |       |          |        |              |   |        |            |   |        |
| 61208  | 15   | SHEET, 0.188", 3" DIA                               | 0.50        | 17.00     | 9              | C     | 14.49    | 123    | 24.80        | A | 211    | 0.00       | C | 0      |
| 61208  |      |                                                     |             |           |                |       |          |        |              |   |        |            |   |        |
| 61209  | 16   | SHACKLE, ANCOR W/ SCREW PIN 3/8" DIA                | 0.30        | 74.00     | 22             | C     | 11.67    | 259    | 38.02        | A | 844    | 0.00       | C | 0      |
| 61209  |      |                                                     |             |           |                |       |          |        |              |   |        |            |   |        |
| 61210  | 17   | COMPRESSION SLEEVE - N-TYPE (OVAL) 1/16" CABLE SIZE | 0.00        | 74.00     | 0              | C     | 11.67    | 1      | 38.02        | A | 4      | 0.00       | C | 0      |
| 61210  |      |                                                     |             |           |                |       |          |        |              |   |        |            |   |        |
|        |      |                                                     |             |           |                |       |          |        |              |   |        |            |   |        |
| 62101  | 1    | JOINER PANEL, ALUMINUM HONEYCOMB, 5/8"              | 1.30        | 50.12     | 65             | C     | 7.44     | 484    | 19.51        | A | 1269   | 6.75       | S | -439   |
| 62101  | 2    | JOINER PANEL, ALUMINUM HONEYCOMB, 5/8"              | 1.30        | 48.75     | 63             | C     | 7.26     | 459    | 22.76        | A | 1441   | 3.42       | S | -216   |
| 62101  | 3    | JOINER PANEL, ALUMINUM HONEYCOMB, 5/8"              | 1.30        | 81.13     | 105            | C     | 6.58     | 693    | 21.57        | A | 2272   | 2.22       | S | -234   |
| 62101  | 4    | JOINER PANEL, ALUMINUM HONEYCOMB, 5/8"              | 1.30        | 48.66     | 63             | C     | 7.26     | 458    | 22.75        | A | 1438   | 1.67       | P | 105    |
| 62101  | 5    | JOINER PANEL, ALUMINUM HONEYCOMB, 5/8"              | 1.30        | 63.47     | 82             | C     | 7.41     | 611    | 19.51        | A | 1608   | 5.86       | P | 483    |
| 62101  | 6    | JOINER PANEL, ALUMINUM HONEYCOMB, 5/8"              | 1.30        | 80.85     | 105            | C     | 7.40     | 777    | 30.50        | A | 3202   | 8.88       | S | -933   |
| 62101  | 7    | JOINER PANEL, ALUMINUM HONEYCOMB, 5/8"              | 1.30        | 53.92     | 70             | C     | 7.44     | 521    | 31.07        | A | 2176   | 5.30       | S | -371   |

| SWBS # | ITEM | DESCRIPTION                                                              | UNIT WEIGHT | NO. UNITS | CURRENT WEIGHT | E<br>C<br>A | VERTICAL  |                 | LONGITUDINAL |   |                 | TRANSVERSE |   |                 |
|--------|------|--------------------------------------------------------------------------|-------------|-----------|----------------|-------------|-----------|-----------------|--------------|---|-----------------|------------|---|-----------------|
|        |      |                                                                          |             |           |                |             | VCG<br>ft | MOMENT<br>ft*lb | LCG<br>ft    | S | MOMENT<br>ft*lb | TCG<br>ft  | S | MOMENT<br>ft*lb |
|        |      |                                                                          |             |           |                |             |           |                 |              |   |                 |            |   |                 |
| 62101  | 8    | JOINER PANEL, ALUMINUM HONEYCOMB, 5/8"                                   | 1.30        | 67.50     | 88             | C           | 7.25      | 636             | 30.50        | A | 2673            | 1.67       | S | -146            |
| 62101  | 9    | JOINER PANEL, ALUMINUM HONEYCOMB, 5/8"                                   | 1.30        | 63.68     | 83             | C           | 7.04      | 582             | 30.43        | A | 2516            | 1.67       | P | 138             |
| 62101  | 10   | JOINER PANEL, ALUMINUM HONEYCOMB, 5/8"                                   | 1.30        | 80.85     | 105            | C           | 7.40      | 777             | 30.50        | A | 3202            | 8.88       | P | 933             |
| 62101  |      |                                                                          |             |           |                |             |           |                 |              |   |                 |            |   |                 |
| 62102  | 2    | STAINLESS STEEL SHOWER PARTITIONS, FIXTURE, BASE & CURTAINS (LARGE HEAD) | 85.00       | 1.00      | 85             | C           | 6.23      | 530             | 29.46        | A | 2504            | 6.35       | S | -540            |
|        |      |                                                                          |             |           |                |             |           |                 |              |   |                 |            |   |                 |
| 62201  | 1    | ALUMINUM TREAD PLATE                                                     | 2.65        | 14.32     | 38             | C           | 6.00      | 228             | 2.78         | A | 105             | 0.00       | C | 0               |
| 62201  | 2    | ALUMINUM TREAD PLATE                                                     | 2.65        | 144.89    | 384            | C           | 1.97      | 756             | 41.49        | A | 15931           | 0.00       | C | 0               |
| 62201  | 3    | ALUMINUM TREAD PLATE                                                     | 2.65        | 90.35     | 239            | C           | 4.35      | 1041            | 52.63        | A | 12600           | 0.00       | C | 0               |
| 62201  | 4    | ALUMINUM TREAD PLATE                                                     | 2.65        | 11.91     | 32             | C           | 4.72      | 149             | 55.35        | A | 1747            | 0.00       | C | 0               |
| 62201  | 5    | ALUMINUM TREAD PLATE                                                     | 2.65        | 59.09     | 157            | C           | 5.07      | 794             | 57.13        | A | 8947            | 0.00       | C | 0               |
| 62201  | 6    | ALUMINUM TREAD PLATE                                                     | 2.65        | 87.31     | 231            | C           | 5.30      | 1226            | 62.09        | A | 14367           | 0.00       | C | 0               |
| 62201  |      |                                                                          |             |           |                |             |           |                 |              |   |                 |            |   |                 |
| 62202  | 7    | FLOORPLATE SUPPORT STRUCTURE                                             | 29.97       | 1.00      | 30             | C           | 5.00      | 150             | 2.78         | A | 83              | 0.00       | C | 0               |
| 62202  | 8    | FLOORPLATE SUPPORT STRUCTURE                                             | 303.32      | 1.00      | 303            | C           | 0.97      | 294             | 41.49        | A | 12586           | 0.00       | C | 0               |
| 62202  | 9    | FLOORPLATE SUPPORT STRUCTURE                                             | 189.15      | 1.00      | 189            | C           | 3.35      | 633             | 52.63        | A | 9954            | 0.00       | C | 0               |
| 62202  | 10   | FLOORPLATE SUPPORT STRUCTURE                                             | 24.93       | 1.00      | 25             | C           | 3.72      | 93              | 55.35        | A | 1380            | 0.00       | C | 0               |
| 62202  | 11   | FLOORPLATE SUPPORT STRUCTURE                                             | 123.71      | 1.00      | 124            | C           | 4.07      | 504             | 57.13        | A | 7068            | 0.00       | C | 0               |
| 62202  | 12   | FLOORPLATE SUPPORT STRUCTURE                                             | 182.78      | 1.00      | 183            | C           | 4.30      | 786             | 62.09        | A | 11350           | 0.00       | C | 0               |
| 62202  |      |                                                                          |             |           |                |             |           |                 |              |   |                 |            |   |                 |
| 62203  | 13   | 2 X 1/4 FB, AL, 6061                                                     | 0.60        | 73.23     | 44             | C           | 3.41      | 150             | 21.21        | A | 932             | 0.25       | P | 11              |
| 62203  |      |                                                                          |             |           |                |             |           | 0               |              |   |                 |            |   | 0               |
| 62204  | 14   | 2 X 2 X 3/16 ANGLE, AL, 6061                                             | 0.85        | 25.36     | 22             | C           | 3.44      | 74              | 21.98        | A | 474             | 0.00       | C | 0               |
| 62204  | 15   | 2 X 2 X 3/16 ANGLE, AL, 6061                                             | 0.85        | 132.90    | 113            | C           | 3.44      | 389             | 19.85        | A | 2242            | 0.05       | S | -5              |
| 62204  |      |                                                                          |             |           |                |             |           | 0               |              |   |                 |            |   | 0               |
| 62204  | 16   | 0.188 PLATE, AL, ASTM B632                                               | 2.85        | 165.26    | 471            | C           | 3.50      | 1647            | 20.91        | A | 9848            | 0.02       | S | -11             |
|        |      |                                                                          |             |           |                |             |           |                 |              |   |                 |            |   |                 |
| 62301  | 1    | VERTICAL LADDER                                                          | 2.59        | 4.81      | 12             | C           | 8.51      | 106             | 3.66         | A | 46              | 1.72       | S | -21             |
| 62301  | 2    | VERTICAL LADDER                                                          | 2.59        | 7.41      | 19             | C           | 7.21      | 138             | 12.82        | A | 246             | 2.76       | S | -53             |
| 62301  | 3    | VERTICAL LADDER                                                          | 2.59        | 4.64      | 12             | C           | 5.50      | 66              | 49.85        | A | 598             | 6.92       | S | -83             |
| 62301  | 4    | VERTICAL LADDER                                                          | 2.59        | 4.64      | 12             | C           | 5.50      | 66              | 50.15        | A | 602             | 6.92       | S | -83             |
| 62301  | 5    | VERTICAL LADDER                                                          | 2.59        | 4.00      | 10             | C           | 6.89      | 71              | 60.34        | A | 624             | 0.00       | P | 0               |
| 62301  | 6    | VERTICAL LADDER                                                          | 2.59        | 4.00      | 10             | C           | 6.89      | 71              | 60.34        | A | 624             | 0.00       | P | 0               |
| 62301  | 7    | VERTICAL LADDER                                                          | 2.59        | 8.75      | 23             | C           | 15.88     | 359             | 26.26        | A | 594             | 5.01       | P | 113             |
| 62301  | 8    | VERTICAL LADDER FOUNDATION                                               | 9.89        | 1.00      | 10             | C           | 9.25      | 92              | 38.76        | A | 384             | 0.27       | P | 3               |
| 62301  |      |                                                                          |             |           |                |             |           |                 |              |   |                 |            |   |                 |
| 62302  | 9    | INCLINED LADDER                                                          | 14.73       | 9.00      | 133            | C           | 8.42      | 1117            | 20.29        | A | 2690            | 0.00       | C | 0               |
| 62302  | 10   | INCLINED LADDER                                                          | 14.73       | 6.92      | 102            | C           | 7.41      | 755             | 32.00        | A | 3260            | 3.03       | P | 309             |
| 62302  | 11   | INCLINED LADDER FOUNDATION                                               | 23.45       | 1.00      | 23             | C           | 7.98      | 187             | 25.39        | A | 595             | 1.32       | P | 31              |
| 62302  |      |                                                                          |             |           |                |             |           |                 |              |   |                 |            |   |                 |
| 62303  | 12   | WORKING DECK STEP UP                                                     | 91.43       | 2.00      | 183            | C           | 10.29     | 1882            | 35.95        | A | 6574            | 0.00       | C | 0               |
| 62303  | 13   | WORKING DECK STEP UP FOUNDATION                                          | 18.29       | 1.00      | 18             | C           | 10.29     | 188             | 35.95        | A | 657             | 0.00       | C | 0               |
| 62303  |      |                                                                          |             |           |                |             |           |                 |              |   |                 |            |   |                 |
| 62304  | 14   | MACHINERY ROOM STEP UP                                                   | 65.00       | 1.00      | 65             | C           | 3.05      | 199             | 35.80        | A | 2327            | 0.00       | C | 0               |
| 62304  | 15   | MACHINERY ROOM STEP UP FOUNDATION                                        | 6.50        | 1.00      | 7              | C           | 3.05      | 20              | 35.80        | A | 233             | 0.00       | C | 0               |
| 62304  |      |                                                                          |             |           |                |             |           |                 |              |   |                 |            |   |                 |
| 62305  | 16   | WORKING DECK TO PILOT HOUSE STEP UP                                      | 165.00      | 1.00      | 165            | C           | 9.64      | 1591            | 36.22        | A | 5976            | 3.25       | P | 537             |
| 62305  | 17   | FOSCLE DECK TO PILOT HOUSE STEP UP                                       | 53.00       | 2.00      | 106            | C           | 11.49     | 1218            | 21.92        | A | 2323            | 0.00       | C | 0               |

| SWBS # | ITEM | DESCRIPTION                           | UNIT WEIGHT | NO. UNITS | CURRENT WEIGHT | E C A | VERTICAL |        | LONGITUDINAL |   |        | TRANSVERSE |   |        |
|--------|------|---------------------------------------|-------------|-----------|----------------|-------|----------|--------|--------------|---|--------|------------|---|--------|
|        |      |                                       |             |           |                |       | VCG      | MOMENT | LCG          | S | MOMENT | TCG        | S | MOMENT |
|        |      |                                       |             |           |                |       |          |        |              |   |        |            |   |        |
| 62305  | 18   | PILOUT HOUSE STEP UP FOUNDATION       | 27.10       | 1.00      | 27             | C     | 10.38    | 281    | 30.48        | A | 826    | 1.95       | P | 53     |
| 62401  | 1    | DOOR, METAL JOINER, TY-E, LH, 26 X 78 | 64.00       | 1.00      | 64             | C     | 6.93     | 443    | 24.42        | A | 1563   | 1.68       | P | 107    |
| 62401  |      |                                       |             |           |                |       |          |        |              |   |        |            |   |        |
| 62402  | 2    | DOOR, METAL JOINER, TY-E, RH, 26 X 78 | 64.00       | 1.00      | 64             | C     | 6.93     | 443    | 24.36        | A | 1559   | 22.27      | S | -1425  |
| 62402  | 3    | DOOR, METAL JOINER, TY-E, RH, 26 X 78 | 64.00       | 1.00      | 64             | C     | 6.93     | 443    | 27.57        | A | 1765   | 1.68       | P | 107    |
| 62402  | 4    | DOOR, METAL JOINER, TY-E, RH, 26 X 78 | 64.00       | 1.00      | 64             | C     | 6.93     | 443    | 29.56        | A | 1892   | 6.93       | S | -443   |
| 62402  | 5    | DOOR, METAL JOINER, TY-E, RH, 26 X 78 | 64.00       | 1.00      | 64             | C     | 6.93     | 443    | 33.38        | A | 2136   | 1.68       | S | -107   |
| 62402  |      |                                       |             |           |                |       |          |        |              |   |        |            |   |        |
| 62403  | 6    | DOOR, METAL JOINER, TY-E, RH, 20 X 64 | 45.00       | 1.00      | 45             | C     | 7.44     | 335    | 22.86        | A | 1029   | 2.29       | S | -103   |
|        |      |                                       |             |           |                |       |          |        |              |   |        |            |   |        |
| 62501  | 1    | WINDOW, FRONT CENTRAL PORT            | 7.48        | 10.93     | 82             | C     | 16.74    | 1369   | 15.49        | A | 1266   | 1.85       | P | 151    |
| 62501  | 2    | WINDOW, FRONT CENTRAL STARBOARD       | 7.48        | 10.93     | 82             | C     | 16.74    | 1369   | 15.49        | A | 1266   | 1.85       | S | -151   |
| 62501  | 3    | WINDOW, FRONT OUTBOARD PORT           | 9.59        | 11.84     | 114            | C     | 16.74    | 1901   | 16.99        | A | 1928   | 5.10       | P | 579    |
| 62501  | 4    | WINDOW, FRONT OUTBOARD STARBOARD      | 9.59        | 11.84     | 114            | C     | 16.74    | 1901   | 16.99        | A | 1928   | 5.10       | S | -579   |
| 62501  | 5    | WINDOW, PORT 1                        | 2.54        | 7.29      | 19             | C     | 16.79    | 311    | 19.24        | A | 356    | 6.83       | P | 127    |
| 62501  | 6    | WINDOW, PORT 2                        | 6.25        | 7.29      | 46             | C     | 16.79    | 765    | 24.50        | A | 1116   | 6.83       | P | 311    |
| 62501  | 7    | WINDOW, PORT 3                        | 6.25        | 7.29      | 46             | C     | 16.79    | 765    | 27.50        | A | 1252   | 6.83       | P | 311    |
| 62501  | 8    | WINDOW, PORT 4                        | 4.38        | 7.29      | 32             | C     | 16.79    | 535    | 30.13        | A | 960    | 6.83       | P | 218    |
| 62501  | 9    | WINDOW, STARBOARD 1                   | 2.54        | 7.29      | 19             | C     | 16.79    | 311    | 19.24        | A | 356    | 6.83       | S | -127   |
| 62501  | 10   | WINDOW, STARBOARD 2                   | 6.25        | 7.29      | 46             | C     | 16.79    | 765    | 24.50        | A | 1116   | 6.83       | S | -311   |
| 62501  | 11   | WINDOW, STARBOARD 3                   | 6.25        | 7.29      | 46             | C     | 16.79    | 765    | 27.50        | A | 1252   | 6.83       | S | -311   |
| 62501  | 12   | WINDOW, STARBOARD 4                   | 4.38        | 7.29      | 32             | C     | 16.79    | 535    | 30.13        | A | 960    | 6.83       | S | -218   |
| 62501  | 13   | WINDOW, BACK PORT                     | 5.21        | 7.29      | 38             | C     | 16.74    | 635    | 35.51        | A | 1348   | 2.88       | P | 109    |
| 62501  | 14   | WINDOW, BACK STARBOARD                | 12.71       | 10.02     | 127            | C     | 16.74    | 2132   | 35.51        | A | 4522   | 1.21       | S | -154   |
| 62501  |      |                                       |             |           |                |       |          |        |              |   |        |            |   |        |
| 62502  | 15   | WIPER, FRONT CENTRAL PORT             | 11.00       | 1.00      | 11             | E     | 16.74    | 1369   | 15.49        | A | 1266   | 1.85       | P | 20     |
| 62502  | 16   | WIPER, FRONT CENTRAL STARBOARD        | 11.00       | 1.00      | 11             | E     | 16.74    | 1369   | 15.49        | A | 1266   | 1.85       | S | -20    |
| 62502  | 17   | WIPER, FRONT OUTBOARD PORT            | 11.00       | 1.00      | 11             | E     | 16.74    | 1901   | 16.99        | A | 1928   | 5.10       | P | 56     |
| 62502  | 18   | WIPER, FRONT OUTBOARD STARBOARD       | 11.00       | 1.00      | 11             | E     | 16.74    | 1901   | 16.99        | A | 1928   | 5.10       | S | -56    |
| 62502  | 19   | WIPER, BACK PORT                      | 11.00       | 1.00      | 11             | E     | 16.74    | 635    | 35.51        | A | 1348   | 2.88       | P | 32     |
| 62502  | 20   | WIPER, BACK STARBOARD                 | 11.00       | 1.00      | 11             | E     | 16.74    | 2132   | 35.51        | A | 4522   | 1.21       | S | -13    |
| 62502  |      |                                       |             |           |                |       |          |        |              |   |        |            |   |        |
| 62503  | 21   | HEATER, FRONT CENTRAL PORT            | 2.00        | 11.08     | 22             | E     | 16.74    | 1369   | 15.49        | A | 1266   | 1.85       | P | 41     |
| 62503  | 22   | HEATER, FRONT CENTRAL STARBOARD       | 2.00        | 8.49      | 17             | E     | 16.74    | 1369   | 15.49        | A | 1266   | 1.85       | S | -31    |
| 62503  | 23   | HEATER, FRONT OUTBOARD PORT           | 2.00        | 8.49      | 17             | E     | 16.74    | 1901   | 16.99        | A | 1928   | 5.10       | P | 87     |
| 62503  | 24   | HEATER, FRONT OUTBOARD STARBOARD      | 2.00        | 4.26      | 9              | E     | 16.74    | 1901   | 16.99        | A | 1928   | 5.10       | S | -43    |
| 62503  | 25   | HEATER, PORT 1                        | 2.00        | 6.24      | 12             | E     | 16.79    | 311    | 19.24        | A | 356    | 6.83       | P | 85     |
| 62503  | 26   | HEATER, PORT 2                        | 2.00        | 6.24      | 12             | E     | 16.79    | 765    | 24.50        | A | 1116   | 6.83       | P | 85     |
| 62503  | 27   | HEATER, PORT 3                        | 2.00        | 6.24      | 12             | E     | 16.79    | 765    | 27.50        | A | 1252   | 6.83       | P | 85     |
| 62503  | 28   | HEATER, PORT 4                        | 2.00        | 7.11      | 14             | E     | 16.79    | 535    | 30.13        | A | 960    | 6.83       | P | 97     |
| 62503  | 29   | HEATER, STARBOARD 1                   | 2.00        | 4.26      | 9              | E     | 16.79    | 311    | 19.24        | A | 356    | 6.83       | S | -58    |
| 62503  | 30   | HEATER, STARBOARD 2                   | 2.00        | 6.24      | 12             | E     | 16.79    | 765    | 24.50        | A | 1116   | 6.83       | S | -85    |
| 62503  | 31   | HEATER, STARBOARD 3                   | 2.00        | 6.24      | 12             | E     | 16.79    | 765    | 27.50        | A | 1252   | 6.83       | S | -85    |
| 62503  | 32   | HEATER, STARBOARD 4                   | 2.00        | 6.24      | 12             | E     | 16.79    | 535    | 30.13        | A | 960    | 6.83       | S | -85    |
| 62503  | 33   | HEATER, BACK PORT                     | 2.00        | 7.11      | 14             | E     | 16.74    | 635    | 35.51        | A | 1348   | 2.88       | P | 41     |
| 62503  | 34   | HEATER, BACK STARBOARD                | 2.00        | 7.11      | 14             | E     | 16.74    | 2132   | 35.51        | A | 4522   | 1.21       | S | -17    |
|        |      |                                       |             |           |                |       |          |        |              |   |        |            |   |        |

| SWBS # | ITEM | DESCRIPTION                                                                    | UNIT WEIGHT | NO. UNITS | CURRENT WEIGHT | E C A | VERTICAL |        | LONGITUDINAL |   |        | TRANSVERSE |   |        |
|--------|------|--------------------------------------------------------------------------------|-------------|-----------|----------------|-------|----------|--------|--------------|---|--------|------------|---|--------|
|        |      |                                                                                |             |           |                |       | VCG      | MOMENT | LCG          | S | MOMENT | TCG        | S | MOMENT |
|        |      |                                                                                | lb          | #         | lb             |       | ft       | ft*lb  | ft           |   | ft*lb  | ft         |   | ft*lb  |
| 63101  | 1    | PAINT                                                                          | 1528.07     | 1.00      | 1528           | C     | 6.56     | 10026  | 32.21        | A | 49220  | 0.00       | C | 0      |
|        |      |                                                                                |             |           |                |       |          |        |              |   |        |            |   |        |
| 63301  | 1    | CATHODIC PROTECTION                                                            | 300.00      | 1.00      | 300            | E     | 3.00     | 900    | 32.40        | A | 9720   | 0.00       | C | 0      |
|        |      |                                                                                |             |           |                |       |          |        |              |   |        |            |   |        |
| 63401  | 1    | POLYVINAL CHLORIDE SHEETING, MIL-M-15562-F TYPE I, 0.125" THICK                | 1.00        | 63.71     | 64             | C     | 3.50     | 223    | 10.00        | A | 637    | 0.00       | C | 0      |
| 63401  | 2    | POLYVINAL CHLORIDE SHEETING, MIL-M-15562-F TYPE I, 0.125" THICK                | 1.00        | 138.60    | 139            | C     | 3.50     | 485    | 17.91        | A | 2482   | 0.00       | C | 0      |
| 63401  | 3    | POLYVINAL CHLORIDE SHEETING, MIL-M-15562-F TYPE I, 0.125" THICK                | 1.00        | 50.11     | 50             | C     | 3.50     | 175    | 22.80        | A | 1143   | 5.53       | S | -277   |
| 63401  | 4    | POLYVINAL CHLORIDE SHEETING, MIL-M-15562-F TYPE I, 0.125" THICK                | 1.00        | 50.11     | 50             | C     | 3.50     | 175    | 22.80        | A | 1143   | 5.53       | P | 277    |
| 63401  | 5    | POLYVINAL CHLORIDE SHEETING, MIL-M-15562-F TYPE I, 0.125" THICK                | 1.00        | 30.00     | 30             | C     | 3.50     | 105    | 30.50        | A | 915    | 0.00       | C | 0      |
| 63401  | 6    | POLYVINAL CHLORIDE SHEETING, MIL-M-15562-F TYPE I, 0.125" THICK                | 1.00        | 199.08    | 199            | C     | 12.54    | 2496   | 26.25        | A | 5226   | 0.18       | S | -36    |
| 63401  |      |                                                                                |             |           |                |       |          |        |              |   |        |            |   |        |
| 63402  | 7    | POLYVINAL CHLORIDE SHEETING, MIL-M-15562-F TYPE III, 0.187" THICK              | 1.50        | 59.58     | 89             | C     | 3.50     | 312    | 30.50        | A | 2718   | 4.98       | S | -444   |
| 63402  |      |                                                                                |             |           |                |       |          |        |              |   |        |            |   |        |
| 63403  | 8    | BASE, WALL, VINYL, 0.08" GAUGE, 4"                                             | 0.26        | 72.07     | 19             | C     | 12.54    | 235    | 26.25        | A | 492    | 0.18       | S | -3     |
| 63403  |      |                                                                                |             |           |                |       |          |        |              |   |        |            |   |        |
| 63404  | 9    | MAT, FLOOR, STANDING, UNICELLUAR FOAM BASE, 30" X 60" X 1/2"                   | 23.00       | 1.00      | 23             | C     | 12.54    | 288    | 32.09        | A | 738    | 1.16       | S | -27    |
| 63404  |      |                                                                                |             |           |                |       |          |        |              |   |        |            |   |        |
| 63405  | 10   | MAT, FLOOR, STANDING, UNICELLUAR FOAM BASE, 30" X 36" X 1/2"                   | 13.80       | 2.00      | 28             | C     | 12.54    | 346    | 21.07        | A | 581    | 0.00       | C | 0      |
| 63405  |      |                                                                                |             |           |                |       |          |        |              |   |        |            |   |        |
| 63406  | 11   | SAFETY WALK, NON-SLIP FABRIC CLEATS, TYPE "B" MED, GRIT, 6" X 24", MIL-D-17951 | 0.50        | 4.00      | 2              | C     | 9.00     | 18     | 37.36        | A | 75     | 0.00       | C | 0      |
| 63406  | 12   | SAFETY WALK, NON-SLIP FABRIC CLEATS, TYPE "B" MED, GRIT, 6" X 24", MIL-D-17951 | 0.50        | 4.00      | 2              | C     | 9.00     | 18     | 39.08        | A | 78     | 0.00       | C | 0      |
| 63406  | 13   | SAFETY WALK, NON-SLIP FABRIC CLEATS, TYPE "B" MED, GRIT, 6" X 24", MIL-D-17951 | 0.50        | 2.00      | 1              | C     | 9.00     | 9      | 47.36        | A | 47     | 9.02       | S | -9     |
| 63406  | 14   | SAFETY WALK, NON-SLIP FABRIC CLEATS, TYPE "B" MED, GRIT, 6" X 24", MIL-D-17951 | 0.50        | 2.00      | 1              | C     | 9.00     | 9      | 56.35        | A | 56     | 0.00       | C | 0      |
| 63406  | 15   | SAFETY WALK, NON-SLIP FABRIC CLEATS, TYPE "B" MED, GRIT, 6" X 24", MIL-D-17951 | 0.50        | 4.00      | 2              | C     | 9.00     | 18     | 51.34        | A | 103    | 6.96       | S | -14    |
| 63406  | 16   | SAFETY WALK, NON-SLIP FABRIC CLEATS, TYPE "B" MED, GRIT, 6" X 24", MIL-D-17951 | 0.50        | 4.00      | 2              | C     | 9.00     | 18     | 60.42        | A | 121    | 0.00       | C | 0      |
| 63406  | 17   | SAFETY WALK, NON-SLIP FABRIC CLEATS, TYPE "B" MED, GRIT, 6" X 24", MIL-D-17951 | 0.50        | 4.00      | 2              | C     | 11.05    | 22     | 22.00        | A | 44     | 0.00       | C | 0      |
| 63406  | 18   | SAFETY WALK, NON-SLIP FABRIC CLEATS, TYPE "B" MED, GRIT, 6" X 24", MIL-D-17951 | 0.50        | 4.00      | 2              | C     | 11.15    | 22     | 11.50        | A | 23     | 2.78       | S | -6     |
| 63406  | 19   | SAFETY WALK, NON-SLIP FABRIC CLEATS, TYPE "B" MED, GRIT, 6" X 24", MIL-D-17951 | 0.50        | 2.00      | 1              | C     | 11.17    | 11     | 4.26         | A | 4      | 1.74       | S | -2     |
| 63406  | 20   | SAFETY WALK, NON-SLIP FABRIC CLEATS, TYPE "B" MED, GRIT, 6" X 24", MIL-D-17951 | 0.50        | 2.00      | 1              | C     | 9.00     | 9      | 38.14        | A | 38     | 3.25       | P | 3      |
| 63406  |      |                                                                                |             |           |                |       |          |        |              |   |        |            |   |        |
| 63407  | 21   | DECK COVERING, ONE-STEP, SEAMLESS, EPOXY BASE, MIL-D-24613 TYPE III            | 0.04        | 35.44     | 1              | C     | 3.50     | 4      | 28.68        | A | 37     | 5.50       | S | -7     |
| 63407  | 22   | DECK COVERING, ONE-STEP, SEAMLESS, EPOXY BASE, MIL-D-24613 TYPE III            | 0.04        | 24.14     | 1              | C     | 3.50     | 3      | 33.18        | A | 29     | 5.50       | S | -5     |
| 63407  |      |                                                                                |             |           |                |       |          |        |              |   |        |            |   |        |
| 63408  | 23   | TAPE, ELECTRIC, DOUBLE SIDED, 4" X 20 MIL                                      | 0.11        | 31.24     | 3              | C     | 3.50     | 12     | 30.50        | A | 105    | 4.98       | P | 17     |
| 63408  |      |                                                                                |             |           |                |       |          |        |              |   |        |            |   |        |
| 63409  | 24   | FUSION STRIP, WITH ITEM 1, 500 FT ROLL                                         | 0.01        | 35.18     | 0              | C     | 3.50     | 1      | 10.00        | A | 2      | 0.00       | C | 0      |
| 63409  | 25   | FUSION STRIP, WITH ITEM 1, 500 FT ROLL                                         | 0.01        | 58.28     | 0              | C     | 3.50     | 1      | 17.91        | A | 5      | 0.00       | C | 0      |
| 63409  | 26   | FUSION STRIP, WITH ITEM 1, 500 FT ROLL                                         | 0.01        | 28.31     | 0              | C     | 3.50     | 1      | 22.80        | A | 3      | 5.53       | P | 1      |
| 63409  | 27   | FUSION STRIP, WITH ITEM 1, 500 FT ROLL                                         | 0.01        | 28.31     | 0              | C     | 3.50     | 1      | 22.80        | A | 3      | 5.53       | S | -1     |
| 63409  | 28   | FUSION STRIP, WITH ITEM 1, 500 FT ROLL                                         | 0.01        | 24.67     | 0              | C     | 3.50     | 0      | 30.50        | A | 4      | 0.00       | C | 0      |
| 63409  | 29   | FUSION STRIP, WITH ITEM 1, 500 FT ROLL                                         | 0.01        | 72.07     | 0              | C     | 12.54    | 5      | 26.25        | A | 10     | 0.18       | S | 0      |
| 63409  |      |                                                                                |             |           |                |       |          | 0      |              |   | 0      |            |   | 0      |
| 63410  | 30   | SEALER, WITH ITEM 8, EPOXY TYPE                                                | 0.04        | 35.44     | 1              | C     | 3.50     | 4      | 28.68        | A | 37     | 5.50       | S | -7     |
| 63410  | 31   | SEALER, WITH ITEM 8, EPOXY TYPE                                                | 0.04        | 24.14     | 1              | C     | 3.50     | 3      | 33.18        | A | 29     | 5.50       | S | -5     |
| 63410  |      |                                                                                |             |           |                |       |          |        |              |   |        |            |   |        |
| 63501  | 1    | THERMAL INSULATION                                                             | 1058.00     | 1.00      | 1058           | C     | 9.10     | 9628   | 28.82        | A | 30492  | 0.02       | P | 21     |
| 63501  |      |                                                                                |             |           |                |       |          |        |              |   |        |            |   |        |

| SWBS # | ITEM | DESCRIPTION                                             | UNIT WEIGHT | NO. UNITS | CURRENT WEIGHT | E C A | VERTICAL |        | LONGITUDINAL |   |        | TRANSVERSE |   |        |
|--------|------|---------------------------------------------------------|-------------|-----------|----------------|-------|----------|--------|--------------|---|--------|------------|---|--------|
|        |      |                                                         |             |           |                |       | VCG      | MOMENT | LCG          | S | MOMENT | TCG        | S | MOMENT |
|        |      |                                                         | lb          | #         | lb             |       | ft       | ft*lb  | ft           |   | ft*lb  | ft         |   | ft*lb  |
| 63502  | 2    | FIRE PROTECTION                                         | 1992.00     | 1.00      | 1992           | C     | 7.93     | 15797  | 33.16        | A | 66055  | 0.01       | P | 20     |
| 63502  |      |                                                         |             |           |                |       |          |        |              |   |        |            |   |        |
| 63503  | 3    | ACOUSTIC INSULATION                                     | 192.13      | 1.00      | 192            | C     | 7.48     | 1438   | 35.00        | A | 6725   | 0.00       | C | 0      |
| 63503  |      |                                                         |             |           |                |       |          |        |              |   |        |            |   |        |
| 63504  | 4    | FASTENERS, SECUREMENT, INSULATION STOP                  | 1000.00     | 1.00      | 1000           | C     | 8.53     | 8530   | 31.97        | A | 31970  | 0.05       | P | 50     |
| 63701  | 1    | SHEATHING                                               | 696.00      | 1.00      | 696            | C     | 9.39     | 6535   | 33.37        | A | 23226  | 0.06       | S | -42    |
| 64101  | 1    | PORT BERTHING BUNKS                                     | 300.00      | 1.00      | 300            | C     | 7.18     | 2153   | 22.66        | A | 6799   | 8.10       | P | 2430   |
| 64101  | 2    | PORT BERTHING LOCKERS (2)                               | 62.00       | 2.00      | 124            | C     | 6.77     | 840    | 20.71        | A | 2567   | 2.87       | P | 356    |
| 64101  | 3    | STBD BERTHING BUNKS                                     | 300.00      | 1.00      | 300            | C     | 7.18     | 2153   | 22.67        | A | 6800   | 8.05       | S | -2416  |
| 64101  | 4    | STBD BERTHING LOCKERS (2)                               | 62.00       | 2.00      | 124            | C     | 6.77     | 840    | 20.65        | A | 2560   | 4.62       | S | -573   |
| 64201  | 1    | FWD BERTHING BUNKS (4)                                  | 700.00      | 1.00      | 700            | C     | 7.29     | 5100   | 9.69         | A | 6781   | 0.00       | C | 0      |
| 64202  | 2    | CREW BERTHING LOCKERS (4)                               | 62.00       | 4.00      | 248            | C     | 6.51     | 1615   | 5.07         | A | 1256   | 0.00       | C | 0      |
| 64302  | 1    | B-3F-1 BENCH W/ BACKREST, STOWAGE UNDER                 | 301.00      | 1.00      | 301            | C     | 5.28     | 1591   | 18.67        | A | 5620   | 5.72       | P | 1721   |
| 64302  | 2    | B-3F-1 BENCH W/ BACKREST, STOWAGE UNDER                 | 301.00      | 1.00      | 301            | C     | 5.28     | 1591   | 13.85        | A | 4170   | 5.48       | P | 1649   |
| 64302  | 3    | B-4B-2 TABLE, MESS, CREW                                | 90.00       | 1.00      | 90             | C     | 6.08     | 547    | 16.25        | A | 1462   | 5.54       | P | 499    |
| 64401  | 1    | LAVATORY UNIT, SINK B-2A-2 (LARGE HEAD)                 | 43.10       | 1.00      | 43             | C     | 5.65     | 244    | 27.04        | A | 1165   | 2.34       | S | -101   |
| 64401  | 2    | LAVATORY UNIT, SINK B-2A-2 (SMALL HEAD)                 | 43.10       | 1.00      | 43             | C     | 5.65     | 244    | 34.28        | A | 1477   | 4.62       | S | -199   |
| 64402  | 3    | TOILET (LARGE HEAD)                                     | 30.00       | 1.00      | 30             | C     | 3.86     | 116    | 27.29        | A | 819    | 7.11       | S | -213   |
| 64402  | 4    | TOILET (SMALL HEAD)                                     | 30.00       | 1.00      | 30             | C     | 3.86     | 116    | 32.61        | A | 978    | 7.11       | S | -213   |
| 65101  | 1    | GALLEY EQUIP, CABINET, COUNTERTOP, FIDGE & OVERHEAD CAB | 950.00      | 1.00      | 950            | C     | 6.52     | 6197   | 16.48        | A | 15651  | 5.56       | S | -5282  |
| 65102  | 2    | ENTERTAINMENT RACK                                      | 97.00       | 1.00      | 97             | C     | 9.03     | 876    | 16.02        | A | 1554   | 9.21       | P | 894    |
| 65103  | 3    | MESS OVERHEAD STOWAGE                                   | 90.00       | 1.00      | 90             | C     | 9.19     | 827    | 13.70        | A | 1233   | 6.38       | P | 574    |
| 65201  | 1    | MEDICAL SPACES                                          | 22.40       | 1.00      | 22             |       | 9.31     | 209    | 32.40        | A | 726    | 0.00       | C | 0      |
| 66101  | 1    | CHART TABLE B-4D-6 28W X 36D X 36.75H (DECKHOUSE)       | 125.00      | 1.00      | 125            | C     | 14.39    | 1799   | 27.03        | A | 3378   | 4.93       | S | -617   |
| 66102  | 2    | B-3F-1 BENCH W/ BACKREST, STOWAGE UNDER                 | 301.00      | 1.00      | 301            | C     | 13.63    | 4101   | 30.08        | A | 9055   | 3.26       | P | 982    |
| 66103  | 3    | B-3F-2 BENCHES WITH STOWAGE UNDER, PADDED               | 56.00       | 2.00      | 112            | C     | 13.37    | 1498   | 26.01        | A | 2913   | 3.70       | P | 414    |
| 66104  | 4    | B-4B-2 TABLE, MESS, CREW                                | 90.00       | 1.00      | 90             | C     | 15.07    | 1357   | 27.80        | A | 2502   | 3.96       | P | 357    |
| 66105  | 5    | HELMSMAN CHAIR                                          | 75.00       | 2.00      | 150            | C     | 15.14    | 2271   | 21.34        | A | 3202   | 0.00       | C | 0      |
| 66501  | 1    | WORKSHOP WORKBENCH AND LOCKERS (DECONTAMINATION SPACE)  | 400.00      | 1.00      | 400            | C     | 6.77     | 2709   | 31.20        | A | 12480  | 7.33       | P | 2933   |
| 67101  | 1    | CHAIN STOWAGE BIN & STOWAGE BOX                         | 100.00      | 1.00      | 100            | C     | 6.54     | 654    | 54.29        | A | 5429   | 8.17       | S | -817   |

| SWBS<br># | ITEM | DESCRIPTION                                   | UNIT WEIGHT | NO.<br>UNITS | CURRENT<br>WEIGHT | E<br>C<br>A | VERTICAL |        | LONGITUDINAL |   |        | TRANSVERSE |   |        |
|-----------|------|-----------------------------------------------|-------------|--------------|-------------------|-------------|----------|--------|--------------|---|--------|------------|---|--------|
|           |      |                                               |             |              |                   |             | VCG      | MOMENT | LCG          | S | MOMENT | TCG        | S | MOMENT |
|           |      |                                               | lb          | #            | lb                |             | ft       | ft*lb  | ft           |   | ft*lb  | ft         |   | ft*lb  |
| 67102     | 2    | WC STORAGE                                    | 35.00       | 1.00         | 35                | C           | 6.77     | 237    | 30.76        | A | 1077   | 5.04       | S | -176   |
| 67102     |      |                                               |             |              |                   |             |          |        |              |   |        |            |   |        |
| 67103     | 3    | MAIN DECK STOWAGE FWD                         | 94.00       | 1.00         | 94                | C           | 12.29    | 1155   | 16.75        | A | 1574   | 4.45       | S | -418   |
| 67103     |      |                                               |             |              |                   |             |          |        |              |   |        |            |   |        |
| 67104     | 4    | MAIN DECK STOWAGE AFT PORT                    | 94.00       | 1.00         | 94                | C           | 10.26    | 965    | 40.05        | A | 3764   | 9.24       | P | 868    |
| 67104     |      |                                               |             |              |                   |             |          |        |              |   |        |            |   |        |
| 67105     | 5    | MAIN DECK STOWAGE AFT STBD                    | 94.00       | 1.00         | 94                | C           | 10.24    | 963    | 40.01        | A | 3761   | 9.24       | S | -868   |
| 67105     |      |                                               |             |              |                   |             |          |        |              |   |        |            |   |        |
| 67106     | 6    | HEAD STORAGE                                  | 35.00       | 1.00         | 35                | C           | 6.77     | 237    | 34.44        | A | 1205   | 7.76       | S | -272   |
|           |      |                                               |             |              |                   |             |          |        |              |   |        |            |   |        |
| 69901     |      | OUTFIT AND FURNISHING PARTS AND SPECIAL TOOLS | 784.00      | 1.00         | 784               |             | 6.98     | 5472   | 32.40        | A | 25402  | 0.00       | C | 0      |
|           |      |                                               |             |              |                   |             |          |        |              |   |        |            |   |        |

| Departure W/ AtoN Load |      |                                      |             |           |                |          |       |              |     |        |            |        |              |
|------------------------|------|--------------------------------------|-------------|-----------|----------------|----------|-------|--------------|-----|--------|------------|--------|--------------|
| SWBS #                 | ITEM | DESCRIPTION                          | UNIT WEIGHT | NO. UNITS | CURRENT WEIGHT | Vertical |       | Longitudinal |     |        | Transverse |        |              |
|                        |      |                                      |             |           |                | VCG ABL  | VMOM  | LCG          | F/A | LMOM   | TCG        | P/C/ S | TMOM         |
|                        |      |                                      | lb          | #         | lb             | ft       | ft*lb | ft           |     | ft*lb  | ft         |        | ft*lb        |
| A1301                  | 1    | ENLISTED WEIGHT AND PERSONAL EFFECTS | 220         | 1         | 220            | 6.97     | 1534  | 21.68        | A   | 4770   | 5.48       | P      | 1206.516667  |
| A1301                  | 2    | ENLISTED WEIGHT AND PERSONAL EFFECTS | 220         | 1         | 220            | 6.97     | 1534  | 21.66        | A   | 4764   | 6.34       | S      | -1394.25     |
| A1301                  | 3    | ENLISTED WEIGHT AND PERSONAL EFFECTS | 220         | 4         | 880            | 6.90     | 6071  | 7.38         | A   | 6491   | 0.00       | C      | 0            |
| A2901                  |      |                                      |             |           | 0              |          | 0     |              |     | 0      |            |        | 0            |
| A3101                  | 1    | DRY PROVISIONS                       | 107         | 1         | 107            | 9.603    | 1028  | 15.289       | A   | 1637   | 7.610504   | S      | -814.6283116 |
| A3102                  | 2    | CHILL PROVISIONS                     | 55          | 1         | 55             | 4.500    | 248   | 14.064       | A   | 777    | 2.749858   | S      | -151.8495745 |
| A3103                  | 3    | FREEZE PROVISIONS                    | 54          | 1         | 54             | 6.500    | 354   | 14.064       | A   | 766    | 2.749858   | S      | -149.6802666 |
| A32                    | 1    | GENERAL STORES                       | 1750        | 1         | 1750           | 6.760    | 11830 | 8.500        | A   | 14875  | 0          | C      | 0            |
| A4101                  | 1    | DIESEL TANK                          | 11179       | 1         | 11179          | 1.880    | 21017 | 30.460       | A   | 340512 | 0          | C      | 0            |
| A4101                  | 2    | DIESEL TANK                          | 5027        | 1         | 5027           | 2.400    | 12065 | 30.460       | A   | 153122 | 6.42       | P      | 32273.34     |
| A4101                  | 3    | DIESEL TANK                          | 5001        | 1         | 5001           | 2.400    | 12002 | 30.460       | A   | 152330 | 6.42       | S      | -32106.42    |
| A4301                  | 1    |                                      | 0           | 1         | 0              |          | 0     |              |     | 0      |            |        | 0            |
| A4601                  | 1    |                                      | 0           | 1         | 0              |          | 0     |              |     | 0      |            |        | 0            |
| A51                    | 1    | BALLAST WATER                        | 0           | 1         | 0              |          | 0     |              |     | 0      |            |        | 0            |
| A5201                  | 1    | POTABLE WATER                        | 4080        | 1         | 4080           | 7.460    | 30437 | 30.440       | A   | 124195 | 7.46       | P      | 30436.8      |
| A5201                  | 2    | POTABLE WATER                        | 4080        | 1         | 4080           | 7.460    | 30437 | 30.440       | A   | 124195 | 7.46       | S      | -30436.8     |
| A5401                  | 1    | HYDRAULIC FLUID                      | 1400        | 1         | 1400           | 3.951    | 5532  | 50.926       | A   | 71296  | 0          | C      | 0            |
| A5501                  | 1    | BLACK WATER                          | 44          | 1         | 44             | 0.640    | 28    | 24.570       | A   | 1081   | 0          | C      | 0            |
| A5502                  | 2    | GREY WATER                           | 445         | 1         | 445            | 0.800    | 356   | 18.730       | A   | 8335   | 0          | C      | 0            |
| A62                    |      |                                      |             |           | 0              |          | 0     |              |     | 0      |            |        | 0            |
| A69                    | 1    | 5X11 BUOY                            | 3000        | 2         | 6000           | 12.000   | 72000 | 51.000       | A   | 306000 | 0          | C      | 0            |
| A69                    | 2    | BUOY BRIDLE                          | 120         | 2         | 240            | 9.200    | 2208  | 43.000       | A   | 10320  | 0          | C      | 0            |
| A69                    | 3    | MOORING CHAIN                        | 1050        | 2         | 2100           | 11.300   | 23730 | 43.000       | A   | 90300  | 0          | C      | 0            |
| A69                    | 4    | SINKER                               | 4000        | 2         | 8000           | 9.800    | 78400 | 43.000       | A   | 344000 | 0          | C      | 0            |

| Departure W/O AtoN Load |      |                                      |             |           |                |          |         |              |     |        |            |        |              |
|-------------------------|------|--------------------------------------|-------------|-----------|----------------|----------|---------|--------------|-----|--------|------------|--------|--------------|
| SWBS #                  | ITEM | DESCRIPTION                          | UNIT WEIGHT | NO. UNITS | CURRENT WEIGHT | Vertical |         | Longitudinal |     |        | Transverse |        |              |
|                         |      |                                      | lb          | #         | lb             | VCG ABL  | VMOM    | LCG          | F/A | LMOM   | TCG        | P/C/ S | TMOM         |
|                         |      |                                      |             |           |                | ft       | ft*lb   | ft           |     | ft*lb  | ft         |        | ft*lb        |
| A1301                   | 1    | ENLISTED WEIGHT AND PERSONAL EFFECTS | 220         | 1         | 220            | 6.97     | 1534    | 21.68        | A   | 4770   | 5.48       | P      | 1206.516667  |
| A1301                   | 2    | ENLISTED WEIGHT AND PERSONAL EFFECTS | 220         | 1         | 220            | 6.97     | 1534    | 21.66        | A   | 4764   | 6.34       | S      | -1394.25     |
| A1301                   | 3    | ENLISTED WEIGHT AND PERSONAL EFFECTS | 220         | 4         | 880            | 6.90     | 6071    | 7.38         | A   | 6491   | 0.00       | C      | 0            |
| B2901                   |      |                                      |             |           | 0              |          | 0       |              |     | 0      |            |        | 0            |
| B3101                   | 1    | DRY PROVISIONS                       | 107         | 1         | 107            | 9.603    | 1028    | 15.289       | A   | 1637   | 7.610504   | S      | -814.6283116 |
| B3102                   | 2    | CHILL PROVISIONS                     | 55          | 1         | 55             | 4.500    | 248     | 14.064       | A   | 777    | 2.749858   | S      | -151.8495745 |
| B3103                   | 3    | FREEZE PROVISIONS                    | 54          | 1         | 54             | 6.500    | 354     | 14.064       | A   | 766    | 2.749858   | S      | -149.6802666 |
| B32                     | 1    | GENERAL STORES                       | 1750        | 1         | 1750           | 6.760    | 11830   | 8.500        | A   | 14875  | 0          | C      | 0            |
| B4101                   | 1    | DIESEL TANK                          | 11179       | 1         | 11179          | 1.880    | 21017   | 30.460       | A   | 340512 | 0          | C      | 0            |
| B4101                   | 2    | DIESEL TANK                          | 5027        | 1         | 5027           | 2.400    | 12065   | 30.460       | A   | 153122 | 6.42       | P      | 32273.34     |
| B4101                   | 3    | DIESEL TANK                          | 5001        | 1         | 5001           | 2.400    | 12002   | 30.460       | A   | 152330 | 6.42       | S      | -32106.42    |
| B4301                   | 1    |                                      | 0           | 1         | 0              |          | 0       |              |     | 0      |            |        | 0            |
| B4601                   | 1    |                                      | 0           | 1         | 0              |          | 0       |              |     | 0      |            |        | 0            |
| B51                     | 1    | BALLAST WATER                        | 0           | 1         | 0              |          | 0       |              |     | 0      |            |        | 0            |
| B5201                   | 1    | POTABLE WATER                        | 4080        | 1         | 4080           | 7.46     | 30436.8 | 30.44        | A   | 124195 | 7.46       | P      | 30436.8      |
| B5201                   | 2    | POTABLE WATER                        | 4080        | 1         | 4080           | 7.46     | 30436.8 | 30.44        | A   | 124195 | 7.46       | S      | -30436.8     |
| B5401                   | 1    | HYDRAULIC FLUID                      | 1400        | 1         | 1400           | 3.951    | 5532    | 50.926       | A   | 71296  | 0          | C      | 0            |
| B5501                   | 1    | BLACK WATER                          | 44          | 1         | 44             | 0.640    | 28      | 24.570       | A   | 1081   | 0          | C      | 0            |
| B5502                   | 2    | GREY WATER                           | 445         | 1         | 445            | 0.800    | 356     | 18.730       | A   | 8335   | 0          | C      | 0            |
| B62                     |      |                                      |             |           | 0              |          | 0       |              |     | 0      |            |        | 0            |
| B69                     | 1    | 5X11 BUOY                            | 3000        | 0         | 0              | 12.000   | 0       | 51.000       | A   | 0      |            |        | 0            |
| B69                     | 2    | BUOY BRIDLE                          | 120         | 0         | 0              | 9.200    | 0       | 43.000       | A   | 0      |            |        | 0            |
| B69                     | 3    | MOORING CHAIN                        | 1050        | 0         | 0              | 11.300   | 0       | 43.000       | A   | 0      |            |        | 0            |
| B69                     | 4    | SINKER                               | 4000        | 0         | 0              | 9.800    | 0       | 43.000       | A   | 0      |            |        | 0            |
|                         |      |                                      |             |           | 0              |          | 0       |              |     | 0      |            |        | 0            |

| Mid W/ AtoN |      |                                      |             |           |                |          |          |              |     |          |            |        |              |
|-------------|------|--------------------------------------|-------------|-----------|----------------|----------|----------|--------------|-----|----------|------------|--------|--------------|
| SWBS #      | ITEM | DESCRIPTION                          | UNIT WEIGHT | NO. UNITS | CURRENT WEIGHT | Vertical |          | Longitudinal |     |          | Transverse |        |              |
|             |      |                                      | lb          | #         | lb             | VCG ABL  | VMOM     | LCG          | F/A | LMOM     | TCG        | P/C/ S | TMOM         |
|             |      |                                      |             |           |                | ft       | ft*lb    | ft           |     | ft*lb    | ft         |        | ft*lb        |
| A1301       | 1    | ENLISTED WEIGHT AND PERSONAL EFFECTS | 220         | 1         | 220            | 6.97     | 1534     | 21.68        | A   | 4770     | 5.48       | P      | 1206.516667  |
| A1301       | 2    | ENLISTED WEIGHT AND PERSONAL EFFECTS | 220         | 1         | 220            | 6.97     | 1534     | 21.66        | A   | 4764     | 6.34       | S      | -1394.25     |
| A1301       | 3    | ENLISTED WEIGHT AND PERSONAL EFFECTS | 220         | 4         | 880            | 6.90     | 6071     | 7.38         | A   | 6491     | 0.00       | C      | 0            |
| C2901       |      |                                      |             |           | 0              |          | 0        |              |     | 0        |            |        | 0            |
| C3101       | 1    | DRY PROVISIONS                       | 107         | 1         | 107            | 9.603    | 1028     | 15.289       | A   | 1637     | 7.610504   | S      | -814.6283116 |
| C3102       | 2    | CHILL PROVISIONS                     | 55          | 1         | 55             | 4.500    | 248      | 14.064       | A   | 777      | 2.749858   | S      | -151.8495745 |
| C3103       | 3    | FREEZE PROVISIONS                    | 54          | 1         | 54             | 6.500    | 354      | 14.064       | A   | 766      | 2.749858   | S      | -149.6802666 |
| C32         | 1    | GENERAL STORES                       | 1750        | 1         | 1750           | 6.760    | 11830    | 8.500        | A   | 14875    | 0          | C      | 0            |
| C4101       | 1    | DIESEL TANK                          | 5883        | 1         | 5883           | 1.180    | 6942     | 30.440       | A   | 179079   | 0          | C      | 0            |
| C4101       | 2    | DIESEL TANK                          | 2646        | 0.5       | 1323           | 1.900    | 2514     | 30.450       | A   | 40285    | 6.15       | P      | 8136.45      |
| C4101       | 3    | DIESEL TANK                          | 2632        | 0.5       | 1316           | 1.900    | 2500     | 30.450       | A   | 40072    | 6.15       | S      | -8093.4      |
| C4301       | 1    |                                      | 0           | 1         | 0              |          | 0        |              |     | 0        |            |        | 0            |
| C4601       | 1    |                                      | 0           | 1         | 0              |          | 0        |              |     | 0        |            |        | 0            |
| C51         | 1    | BALLAST WATER                        | 0           | 1         | 0              |          | 0        |              |     | 0        |            |        | 0            |
| C5201       | 1    | POTABLE WATER                        | 2147        | 1         | 2147           | 6.04     | 12967.88 | 30.41        | A   | 65290.27 | 6.04       | P      | 12967.88     |
| C5201       | 2    | POTABLE WATER                        | 2147        | 1         | 2147           | 6.04     | 12967.88 | 30.41        | A   | 65290.27 | 6.04       | S      | -12967.88    |
| C5401       | 1    | HYDRAULIC FLUID                      | 1400        | 1         | 1400           | 3.951    | 5532     | 50.926       | A   | 71296    | 0          | C      | 0            |
| C5501       | 1    | BLACK WATER                          | 436         | 1         | 436            | 1.240    | 541      | 24.580       | A   | 10717    | 0          | C      | 0            |
| C5502       | 2    | GREY WATER                           | 4447        | 1         | 4447           | 1.540    | 6848     | 18.670       | A   | 83025    | 0          | C      | 0            |
| C62         |      |                                      |             |           | 0              |          | 0        |              |     | 0        |            |        | 0            |
| C69         | 1    | 5X11 BUOY                            | 3000        | 2         | 6000           | 12.000   | 72000    | 51.000       | A   | 306000   |            |        | 0            |
| C69         | 2    | BUOY BRIDLE                          | 120         | 2         | 240            | 9.200    | 2208     | 43.000       | A   | 10320    |            |        | 0            |
| C69         | 3    | MOORING CHAIN                        | 1050        | 2         | 2100           | 11.300   | 23730    | 43.000       | A   | 90300    |            |        | 0            |
| C69         | 4    | SINKER                               | 4000        | 2         | 8000           | 9.800    | 78400    | 43.000       | A   | 344000   |            |        | 0            |
|             |      |                                      |             |           | 0              |          | 0        |              |     | 0        |            |        | 0            |

| Mid W/O AtoN |      |                                      |             |           |                |          |          |              |     |          |            |       |              |
|--------------|------|--------------------------------------|-------------|-----------|----------------|----------|----------|--------------|-----|----------|------------|-------|--------------|
| SWBS #       | ITEM | DESCRIPTION                          | UNIT WEIGHT | NO. UNITS | CURRENT WEIGHT | Vertical |          | Longitudinal |     |          | Transverse |       |              |
|              |      |                                      | lb          | #         | lb             | VCG ABL  | VMOM     | LCG          | F/A | LMOM     | TCG        | P/C/S | TMOM         |
|              |      |                                      |             |           |                | ft       | ft*lb    | ft           |     | ft*lb    | ft         |       | ft*lb        |
| A1301        | 1    | ENLISTED WEIGHT AND PERSONAL EFFECTS | 220         | 1         | 220            | 6.97     | 1534     | 21.68        | A   | 4770     | 5.48       | P     | 1206.516667  |
| A1301        | 2    | ENLISTED WEIGHT AND PERSONAL EFFECTS | 220         | 1         | 220            | 6.97     | 1534     | 21.66        | A   | 4764     | 6.34       | S     | -1394.25     |
| A1301        | 3    | ENLISTED WEIGHT AND PERSONAL EFFECTS | 220         | 4         | 880            | 6.90     | 6071     | 7.38         | A   | 6491     | 0.00       | C     | 0            |
| D2901        |      |                                      |             |           | 0              |          | 0        |              |     | 0        |            |       | 0            |
| D3101        | 1    | DRY PROVISIONS                       | 107         | 1         | 107            | 9.603    | 1028     | 15.289       | A   | 1637     | 7.610504   | S     | -814.6283116 |
| D3102        | 2    | CHILL PROVISIONS                     | 55          | 1         | 55             | 4.500    | 248      | 14.064       | A   | 777      | 2.749858   | S     | -151.8495745 |
| D3103        | 3    | FREEZE PROVISIONS                    | 54          | 1         | 54             | 6.500    | 354      | 14.064       | A   | 766      | 2.749858   | S     | -149.6802666 |
| D32          | 1    | GENERAL STORES                       | 1750        | 1         | 1750           | 6.760    | 11830    | 8.500        | A   | 14875    | 0          | C     | 0            |
| D4101        | 1    | DIESEL TANK                          | 5883        | 1         | 5883           | 1.180    | 6942     | 30.440       | A   | 179079   | 0          | C     | 0            |
| D4101        | 2    | DIESEL TANK                          | 2646        | 0.5       | 1323           | 1.900    | 2514     | 30.450       | A   | 40285    | 6.15       | P     | 8136.45      |
| D4101        | 3    | DIESEL TANK                          | 2632        | 0.5       | 1316           | 1.900    | 2500     | 30.450       | A   | 40072    | 6.15       | S     | -8093.4      |
| D4301        | 1    |                                      | 0           | 1         | 0              |          | 0        |              |     | 0        |            |       | 0            |
| D4601        | 1    |                                      | 0           | 1         | 0              |          | 0        |              |     | 0        |            |       | 0            |
| D51          | 1    | BALLAST WATER                        | 0           | 1         | 0              |          | 0        |              |     | 0        |            |       | 0            |
| D5201        | 1    | POTABLE WATER                        | 2147        | 1         | 2147           | 6.04     | 12967.88 | 30.41        | A   | 65290.27 | 6.04       | P     | 12967.88     |
| D5201        | 2    | POTABLE WATER                        | 2147        | 1         | 2147           | 6.04     | 12967.88 | 30.41        | A   | 65290.27 | 6.04       | S     | -12967.88    |
| D5401        | 1    | HYDRAULIC FLUID                      | 1400        | 1         | 1400           | 3.951    | 5532     | 50.926       | A   | 71296    | 0          | C     | 0            |
| D5501        | 1    | BLACK WATER                          | 436         | 1         | 436            | 1.240    | 541      | 24.580       | A   | 10717    | 0          | C     | 0            |
| D5502        | 2    | GREY WATER                           | 4447        | 1         | 4447           | 1.540    | 6848     | 18.670       | A   | 83025    | 0          | C     | 0            |
| D62          |      |                                      |             |           | 0              |          | 0        |              |     | 0        |            |       | 0            |
| D69          | 1    | 5X11 BUOY                            | 3000        | 0         | 0              | 12.000   | 0        | 51.000       | A   | 0        |            |       | 0            |
| D69          | 2    | BUOY BRIDLE                          | 120         | 0         | 0              | 9.200    | 0        | 43.000       | A   | 0        |            |       | 0            |
| D69          | 3    | MOORING CHAIN                        | 1050        | 0         | 0              | 11.300   | 0        | 43.000       | A   | 0        |            |       | 0            |
| D69          | 4    | SINKER                               | 4000        | 0         | 0              | 9.800    | 0        | 43.000       | A   | 0        |            |       | 0            |
|              |      |                                      |             |           | 0              |          | 0        |              |     | 0        |            |       | 0            |

| Min-Op W/ AtoN |      |                                      |             |           |                |          |        |              |     |         |            |        |              |
|----------------|------|--------------------------------------|-------------|-----------|----------------|----------|--------|--------------|-----|---------|------------|--------|--------------|
| SWBS #         | ITEM | DESCRIPTION                          | UNIT WEIGHT | NO. UNITS | CURRENT WEIGHT | Vertical |        | Longitudinal |     |         | Transverse |        |              |
|                |      |                                      | lb          | #         | lb             | VCG ABL  | VMOM   | LCG          | F/A | LMOM    | TCG        | P/C/ S | TMOM         |
|                |      |                                      |             |           |                | ft       | ft*lb  | ft           |     | ft*lb   | ft         |        | ft*lb        |
| A1301          | 1    | ENLISTED WEIGHT AND PERSONAL EFFECTS | 220         | 1         | 220            | 6.97     | 1534   | 21.68        | A   | 4770    | 5.48       | P      | 1206.516667  |
| A1301          | 2    | ENLISTED WEIGHT AND PERSONAL EFFECTS | 220         | 1         | 220            | 6.97     | 1534   | 21.66        | A   | 4764    | 6.34       | S      | -1394.25     |
| A1301          | 3    | ENLISTED WEIGHT AND PERSONAL EFFECTS | 220         | 4         | 880            | 6.90     | 6071   | 7.38         | A   | 6491    | 0.00       | C      | 0            |
| E2901          |      |                                      |             |           | 0              |          | 0      |              |     | 0       |            |        | 0            |
| E3101          | 1    | DRY PROVISIONS                       | 107         | 1         | 107            | 9.603    | 1028   | 15.289       | A   | 1637    | 7.610504   | S      | -814.6283116 |
| E3102          | 2    | CHILL PROVISIONS                     | 55          | 1         | 55             | 4.500    | 248    | 14.064       | A   | 777     | 2.749858   | S      | -151.8495745 |
| E3103          | 3    | FREEZE PROVISIONS                    | 54          | 1         | 54             | 6.500    | 354    | 14.064       | A   | 766     | 2.749858   | S      | -149.6802666 |
| E32            | 1    | GENERAL STORES                       | 1750        | 1         | 1750           | 6.760    | 11830  | 8.500        | A   | 14875   | 0          | C      | 0            |
| E4101          | 1    | DIESEL TANK                          | 1177        | 1         | 1177           | 0.490    | 577    | 30.220       | A   | 35569   | 0          | C      | 0            |
| E4101          | 2    | DIESEL TANK                          | 530         | 1         | 530            | 1.320    | 700    | 30.330       | A   | 16075   | 5.09       | P      | 2697.7       |
| E4101          | 3    | DIESEL TANK                          | 527         | 1         | 527            | 1.320    | 696    | 30.320       | A   | 30      | 5.09       | S      | -2682.43     |
| E4301          | 1    |                                      | 0           | 1         | 0              |          | 0      |              |     | 0       |            |        | 0            |
| E4601          | 1    |                                      | 0           | 1         | 0              |          | 0      |              |     | 0       |            |        | 0            |
| E51            | 1    | BALLAST WATER                        | 0           | 1         | 0              |          | 0      |              |     | 0       |            |        | 0            |
| E5201          | 1    | POTABLE WATER                        | 215         | 1         | 215            | 4.33     | 930.95 | 30.15        | A   | 6482.25 | 8.94       | P      | 1922.1       |
| E5201          | 2    | POTABLE WATER                        | 215         | 1         | 215            | 4.33     | 930.95 | 30.15        | A   | 6482.25 | 8.94       | S      | -1922.1      |
| E5401          | 1    | HYDRAULIC FLUID                      | 1400        | 1         | 1400           | 3.951    | 5532   | 50.926       | A   | 71296   | 0          | C      | 0            |
| E5501          | 1    | BLACK WATER                          | 828         | 1         | 828            | 1.790    | 1482   | 24.590       | A   | 20361   | 0          | C      | 0            |
| E5502          | 2    | GREY WATER                           | 8448        | 1         | 8448           | 2.020    | 17065  | 18.500       | A   | 156288  | 0          | C      | 0            |
| E62            |      |                                      |             |           | 0              |          | 0      |              |     | 0       |            |        | 0            |
| E69            | 1    | 5X11 BUOY                            | 3000        | 2         | 6000           | 12.000   | 72000  | 51.000       | A   | 306000  |            |        | 0            |
| E69            | 2    | BUOY BRIDLE                          | 120         | 2         | 240            | 9.200    | 2208   | 43.000       | A   | 10320   |            |        | 0            |
| E69            | 3    | MOORING CHAIN                        | 1050        | 2         | 2100           | 11.300   | 23730  | 43.000       | A   | 90300   |            |        | 0            |
| E69            | 4    | SINKER                               | 4000        | 2         | 8000           | 9.800    | 78400  | 43.000       | A   | 344000  |            |        | 0            |
|                |      |                                      |             |           | 0              |          | 0      |              |     | 0       |            |        | 0            |

| Min-Op W/O AtoN |      |                                      |             |           |                |          |        |              |     |         |            |       |              |
|-----------------|------|--------------------------------------|-------------|-----------|----------------|----------|--------|--------------|-----|---------|------------|-------|--------------|
| SWBS #          | ITEM | DESCRIPTION                          | UNIT WEIGHT | NO. UNITS | CURRENT WEIGHT | Vertical |        | Longitudinal |     |         | Transverse |       |              |
|                 |      |                                      | lb          | #         | lb             | VCG ABL  | VMOM   | LCG          | F/A | LMOM    | TCG        | P/C/S | TMOM         |
|                 |      |                                      |             |           |                | ft       | ft*lb  | ft           |     | ft*lb   | ft         |       | ft*lb        |
| A1301           | 1    | ENLISTED WEIGHT AND PERSONAL EFFECTS | 220         | 1         | 220            | 6.97     | 1534   | 21.68        | A   | 4770    | 5.48       | P     | 1206.516667  |
| A1301           | 2    | ENLISTED WEIGHT AND PERSONAL EFFECTS | 220         | 1         | 220            | 6.97     | 1534   | 21.66        | A   | 4764    | 6.34       | S     | -1394.25     |
| A1301           | 3    | ENLISTED WEIGHT AND PERSONAL EFFECTS | 220         | 4         | 880            | 6.90     | 6071   | 7.38         | A   | 6491    | 0.00       | C     | 0            |
| F2901           |      |                                      |             |           | 0              |          | 0      |              |     | 0       |            |       | 0            |
| F3101           | 1    | DRY PROVISIONS                       | 107         | 1         | 107            | 9.603    | 1028   | 15.289       | A   | 1637    | 7.610504   | S     | -814.6283116 |
| F3102           | 2    | CHILL PROVISIONS                     | 55          | 1         | 55             | 4.500    | 248    | 14.064       | A   | 777     | 2.749858   | S     | -151.8495745 |
| F3103           | 3    | FREEZE PROVISIONS                    | 54          | 1         | 54             | 6.500    | 354    | 14.064       | A   | 766     | 2.749858   | S     | -149.6802666 |
| F32             | 1    | GENERAL STORES                       | 1750        | 1         | 1750           | 6.760    | 11830  | 8.500        | A   | 14875   | 0          | C     | 0            |
| F4101           | 1    | DIESEL TANK                          | 1177        | 1         | 1177           | 0.490    | 577    | 30.220       | A   | 35569   | 0          | C     | 0            |
| F4101           | 2    | DIESEL TANK                          | 530         | 1         | 530            | 1.320    | 700    | 30.330       | A   | 16075   | 5.09       | P     | 2697.7       |
| F4101           | 3    | DIESEL TANK                          | 527         | 1         | 527            | 1.320    | 696    | 30.320       | A   | 30      | 5.09       | S     | -2682.43     |
| F4301           | 1    |                                      | 0           | 1         | 0              |          | 0      |              |     | 0       |            |       | 0            |
| F4601           | 1    |                                      | 0           | 1         | 0              |          | 0      |              |     | 0       |            |       | 0            |
| F51             | 1    | BALLAST WATER                        | 0           | 1         | 0              |          | 0      |              |     | 0       |            |       | 0            |
| F5201           | 1    | POTABLE WATER                        | 215         | 1         | 215            | 4.33     | 930.95 | 30.15        | A   | 6482.25 | 8.94       | P     | 1922.1       |
| F5201           | 2    | POTABLE WATER                        | 215         | 1         | 215            | 4.33     | 930.95 | 30.15        | A   | 6482.25 | 8.94       | S     | -1922.1      |
| F5401           | 1    | HYDRAULIC FLUID                      | 1400        | 1         | 1400           | 3.951    | 5532   | 50.926       | A   | 71296   | 0          | C     | 0            |
| F5501           | 1    | BLACK WATER                          | 828         | 1         | 828            | 1.790    | 1482   | 24.590       | A   | 20361   | 0          | C     | 0            |
| F5502           | 2    | GREY WATER                           | 8448        | 1         | 8448           | 2.020    | 17065  | 18.500       | A   | 156288  | 0          | C     | 0            |
| F62             |      |                                      |             |           | 0              |          | 0      |              |     | 0       |            |       | 0            |
| F69             | 1    | 5X11 BUOY                            | 3000        | 0         | 0              | 12.000   | 0      | 51.000       | A   | 0       |            |       | 0            |
| F69             | 2    | BUOY BRIDLE                          | 120         | 0         | 0              | 9.200    | 0      | 43.000       | A   | 0       |            |       | 0            |
| F69             | 3    | MOORING CHAIN                        | 1050        | 0         | 0              | 11.300   | 0      | 43.000       | A   | 0       |            |       | 0            |
| F69             | 4    | SINKER                               | 4000        | 0         | 0              | 9.800    | 0      | 43.000       | A   | 0       |            |       | 0            |
|                 |      |                                      |             |           | 0              |          | 0      |              |     | 0       |            |       | 0            |

| SWBS #   | SWBS Name                                    | Applied Weight | Vertical |                | Longitudinal |     |                | Transverse |      |                 |
|----------|----------------------------------------------|----------------|----------|----------------|--------------|-----|----------------|------------|------|-----------------|
|          |                                              |                | VCG ABL  | VMOM           | LCG          | F/A | LMOM           | TCG        | P/C/ | TMOM            |
|          |                                              | lb             | ft       | ft*lb          | ft           |     | ft*lb          | ft         | S    | ft*lb           |
|          |                                              |                |          |                |              |     |                |            |      |                 |
|          | <b>Light Ship, No Margins</b>                | <b>182643</b>  | 7.46     | <b>1362512</b> | 34.56        | A   | <b>6311343</b> | 0.01       | P    | <b>2206.899</b> |
|          |                                              |                |          |                |              |     |                |            |      |                 |
|          | <b>Design Margin</b>                         |                |          |                |              |     |                |            |      |                 |
|          |                                              |                |          |                |              |     |                |            |      |                 |
| M11      | Design Margin Weight (10.3%)                 | <b>18810</b>   | 7.46     | <b>140325</b>  | 34.56        | A   | <b>650005</b>  | 0.01       | S    | <b>227.289</b>  |
| M11      | Design Margin KG (5.0%)                      |                | 0.37     |                |              |     |                |            |      |                 |
|          |                                              |                |          |                |              |     |                |            |      |                 |
|          | <b>Construction Margin</b>                   |                |          |                |              |     |                |            |      |                 |
|          |                                              |                |          |                |              |     |                |            |      |                 |
| M21      | Construction Margin Weight (5.0%)            | <b>9132</b>    | 7.46     | <b>68126</b>   | 34.56        | A   | <b>315567</b>  | 0.01       | S    | <b>110.345</b>  |
| M21      | Construction Margin KG (3.0%)                |                | 0.22     |                |              |     |                |            |      |                 |
|          |                                              |                |          |                |              |     |                |            |      |                 |
|          | <b>Contract Modification</b>                 |                |          |                |              |     |                |            |      |                 |
|          |                                              |                |          |                |              |     |                |            |      |                 |
| M31      | Contract Modification Weight (0.7%)          | <b>1279</b>    | 7.46     | <b>9538</b>    | 34.56        | A   | <b>44179</b>   | 0.01       | S    | <b>15.448</b>   |
| M31      | Contract Modification KG (0.5%)              |                | 0.04     |                |              |     |                |            |      |                 |
|          |                                              |                |          |                |              |     |                |            |      |                 |
|          | <b>Government Furnished Equipment</b>        |                |          |                |              |     |                |            |      |                 |
|          |                                              |                |          |                |              |     |                |            |      |                 |
| M41      | Government Furnished Equipment Weight (0.3%) | <b>548</b>     | 7.46     | <b>4088</b>    | 34.56        | A   | <b>18934</b>   | 0.01       | S    | <b>6.621</b>    |
| M41      | Government Furnished Equipment KG (0.2%)     |                | 0.01     |                |              |     |                |            |      |                 |
|          |                                              |                |          |                |              |     |                |            |      |                 |
|          |                                              |                |          |                |              |     |                |            |      |                 |
| <b>M</b> | <b>Margins</b>                               | <b>29769</b>   | 12.09    | <b>359935</b>  | 34.56        | A   | <b>1028686</b> | 0.01       | S    | <b>359.702</b>  |
|          |                                              |                |          |                |              |     |                |            |      |                 |
|          | <b>Light Ship Condition</b>                  | <b>212412</b>  | 8.11     | <b>1722447</b> | 34.56        | A   | <b>7340029</b> | 0.01       | S    | <b>2566.601</b> |
|          |                                              |                |          |                |              |     |                |            |      |                 |